SEMNEST: proofs of the nesting and equivalence relations

all 118 items of the capsule

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1 Setting and definitions

Each item compares two models, A (left) and B (right), for the same p observed variables. A model is written in RAM form

$$v = \mathbf{A}v + u, \qquad \mathrm{Cov}(u) = \mathbf{S}, \qquad \Sigma_{\mathrm{full}} = (I - \mathbf{A})^{-1}\mathbf{S}(I - \mathbf{A})^{-\top},$$

with v the observed and latent variables and Σ the observed block of Σ_{full}. $\mathbf{A}$ holds the drawn directed effects (loadings λ, regressions β, second-order loadings γ); $\mathbf{S}$ holds a free variance for every variable (the residual variance θ_{ii} or ψ of an endogenous variable) and the drawn covariances. For variables in a *network block*, the block of $\mathbf{S}$ is K^{-1}, where the precision matrix K has a free diagonal and free off-diagonal elements only for the drawn undirected edges: a Gaussian graphical model (GGM) on observed variables, a residual network model (RNM) on the residuals of indicators, or a latent network model (LNM) on latent variables (Epskamp, Rhemtulla & Borsboom, 2017, *Psychometrika*).

Parameters. The free parameters $\theta \in \mathbb{R}^q$ are unrestricted real numbers: variances may be negative (Heywood solutions), as in lavaan's default estimation. The scale of each latent variable is fixed by its first loading (marker variable); q_X counts the free parameters of model X. Let $\Theta_X = \{\theta : \Sigma_X(\theta) \text{ exists and is positive definite}\}$ (an open set) and

$$\mathcal{M}_X = \{\Sigma_X(\theta) : \theta \in \Theta_X\},$$

the set of covariance matrices model X can produce. $\overline{\mathcal{M}}$ denotes the closure within the positive definite matrices.

Definition. X is *nested* in Y, written $X \preceq Y$, if $\mathcal{M}_X \subseteq \overline{\mathcal{M}_Y}$: every covariance matrix that X can produce, Y can produce or approximate arbitrarily closely, so fitting Y to it gives a perfect fit ($F_{\mathrm{ML}} = 0$, $\chi^2 = 0$). This is exactly what the numerical check of Bentler & Satorra (2010) and of the PL5225 assignment tests. X and Y are *equivalent* ($X \equiv Y$) if $X \preceq Y$ and $Y \preceq X$. *A is nested in B* in the game means $A \preceq B$ and $B \not\preceq A$; *neither* means $A \not\preceq B$ and $B \not\preceq A$.

The closure makes the definition insensitive to the measure-zero exceptions that are typical for latent variable models (e.g. a mapping that divides by a covariance that happens to be zero). Allowing negative variances is what makes the textbook statements ("a one-factor model with three indicators is saturated", "a second-order factor on three factors is equivalent to three correlated factors") literally true.

2 Lemmas

Lemma (R, restriction)**.** *If X is obtained from Y by fixing free parameters of Y to constants or by equating them, then $X \preceq Y$.*

Proof. Θ_X embeds in Θ_Y with the same implied Σ. □

Lemma (G, generic maps)**.** *If there are a dense subset $D \subseteq \Theta_X$ and a map $\varphi : D \to \Theta_Y$ with $\Sigma_Y(\varphi(\theta)) = \Sigma_X(\theta)$ for all $\theta \in D$, then $X \preceq Y$.*

Proof. Σ_X is continuous, so $\mathcal{M}_X = \Sigma_X(\Theta_X) \subseteq \overline{\Sigma_X(D)} \subseteq \overline{\mathcal{M}_Y}$. In the items D is Θ_X minus the zero set of a polynomial that is not identically zero ("generic" parameter values), which is dense. □

Markers. Fixing the marker loading of a latent variable to 1 only removes a scale indeterminacy: a parameter value in which that loading is $c \neq 0$ can be rescaled ($F \to cF$: loadings divided by c, variances and covariances of F multiplied by c^2 and c) without changing Σ. So the maps in the proofs may treat loadings and factor variances as free; the values with a zero marker loading form a set of measure zero, covered by Lemma G.

Lemma (S, separating constraint)**.** *If a polynomial f in the entries of Σ vanishes on $\mathcal{M}_Y$ and $f(\Sigma_X(\theta^*)) \neq 0$ for some $\theta^* \in \Theta_X$, then $X \not\preceq Y$.*

Proof. f is continuous, so it vanishes on $\overline{\mathcal{M}_Y}$, which therefore does not contain $\Sigma_X(\theta^*)$. □

In every application below, f vanishing on $\mathcal{M}_Y$ is shown algebraically in the text (and was also checked in exact rational arithmetic at random rational parameter values), and the point θ^* with rational coordinates is listed in Appendix B together with the exact value of f.

Lemma (D, dimension)**.** *Let $J_X(\theta)$ be the Jacobian of $\theta \mapsto \operatorname{vech}\Sigma_X(\theta)$. If $\operatorname{rank} J_X(\theta^*) > q_Y$ for some $\theta^* \in \Theta_X$, then $X \not\preceq Y$.*

Proof. $\mathcal{M}_Y$ is the image of a semialgebraic set of dimension q_Y under a rational map, so $\overline{\mathcal{M}_Y}$ is semialgebraic with dimension at most q_Y. The rank of J_X is lower semicontinuous, so its maximum $r \geq \operatorname{rank} J_X(\theta^*) > q_Y$ is attained on a nonempty open subset of Θ_X, where the rank is constant; by the constant rank theorem $\mathcal{M}_X$ contains an r-dimensional embedded submanifold, which cannot lie in a semialgebraic set of dimension $< r$. □

The ranks were computed exactly (rational arithmetic) at the rational points listed in Appendix B.

Lemma (B, saturated blocks)**.** *(a) The following are saturated ($\overline{\mathcal{M}}$ = all positive definite matrices): the model with all covariances free; a complete recursive path model (every pair connected by a directed path following a fixed order); a one-factor model with three indicators. (b) If two models differ only in how they model the covariance matrix of a block of variables that is connected to the rest of the model in the same way, and both can produce every covariance matrix of that block (up to closure), the models are equivalent; if only one of them can, the other is nested in it.*

Proof. (a) Every $\Sigma \succ 0$ has a Cholesky factorization $\Sigma = (I-B)^{-1}D(I-B)^{-\top}$ with B strictly lower triangular and D diagonal; that is a complete recursive path model. For one factor with three indicators and $\sigma_{12}\sigma_{13}\sigma_{23} \neq 0$: $\lambda_1 = 1$, $\psi = \sigma_{12}\sigma_{13}/\sigma_{23}$, $\lambda_2 = \sigma_{23}/\sigma_{13}$, $\lambda_3 = \sigma_{23}/\sigma_{12}$, $\theta_{ii} = \sigma_{ii} - \lambda_i^2\psi$ reproduce Σ; apply Lemma G. (b) The implied Σ depends on the block only through the block's covariance matrix (e.g. $\Sigma = \Lambda\Psi\Lambda^\top + \Theta$ for the latent block). □

Lemma (M, merging factors)**.** *In a factor model where the covariance matrix Ψ of the factors is free, setting the variances and covariances of factors F_k and F_l equal ($\psi_{kk} = \psi_{ll} = \psi_{kl}$ and $\psi_{km} = \psi_{lm}$ for all other m) makes $F_k = F_l$ almost surely. Hence the model in which F_k and F_l are replaced by one factor loading on the union of their indicators is nested in it.*

Proof. With these values $\operatorname{Var}(F_k - F_l) = 0$; the implied Σ is that of the merged model (Lemma R on the parameters). □

Lemma (P, pairs)**.** *Let F be a latent variable of model X and y_i, y_j two observed variables whose only paths are loadings from F. Let X^+ be X plus the residual covariance $y_i \leftrightarrow y_j$, and X^G be X in which y_i, y_j instead load on a new latent variable G whose only path is $F \to G$. Then $X^+ \equiv X^G$.*

Proof. In X^G write $G = \gamma F + u$ with u independent of all other variables. Then $y_i = \lambda_i^G\gamma F + \lambda_i^G u + e_i$, and likewise for y_j: this is X^+ with loadings $\lambda_i = \lambda_i^G\gamma$, $\lambda_j = \lambda_j^G\gamma$, residual covariance $\theta_{ij} = \lambda_i^G\lambda_j^G\operatorname{Var}(u)$ and residual variances $\theta_{ii} + (\lambda_i^G)^2\operatorname{Var}(u)$; nothing else changes. Conversely, given X^+ with $\lambda_i\lambda_j \neq 0$, take $\gamma = 1$, $\lambda_i^G = \lambda_i$, $\lambda_j^G = \lambda_j$, $\operatorname{Var}(u) = \theta_{ij}/(\lambda_i\lambda_j)$ (possibly negative) and residual variances $\theta_{ii} - \lambda_i^2\operatorname{Var}(u)$. Lemma G gives both directions. □

Note that if F is exogenous and G has no other paths, $F \to G$ and $F \leftrightarrow G$ give the same (saturated) 2×2 latent block, so by Lemma B(b) G may also simply be a factor correlated with F.

Lemma (I, inverses)**.** *For a positive definite Σ, $(\Sigma^{-1})_{ij} = 0$ iff the partial covariance of i and j given all other variables is 0, iff the minor of Σ obtained by deleting row i and column j vanishes. Σ is block-diagonal iff Σ^{-1} is block-diagonal (with the same blocks). For three variables $\{i,j,k\}$, $(\Sigma^{-1})_{ij} = 0$ iff $\sigma_{ij}\sigma_{kk} - \sigma_{ik}\sigma_{jk} = 0$.*

Proof. Standard: $(\Sigma^{-1})_{ij} = C_{ji}/\det\Sigma$ with C the cofactor matrix (for symmetric Σ, $C_{ji} = \pm$ the minor without row i and column j); the inverse of a block-diagonal matrix is block-diagonal; for 3×3, $C_{ji} = \pm(\sigma_{ij}\sigma_{kk} - \sigma_{ik}\sigma_{jk})$. □

Lemma (C, trees)**.** *Let a path model be a DAG whose skeleton is a tree and in which no variable has two parents (no colliders: chains and forks). Then its set of covariance matrices equals the GGM on the skeleton, $\{\Sigma \succ 0 : (\Sigma^{-1})_{ij} = 0$ for every non-adjacent pair$\}$.*

Proof. For a DAG, $\Sigma^{-1} = (I-B)^\top D^{-1}(I-B)$ with B supported on the arrows. If every node has at most one parent, $\big((I-B)^\top D^{-1}(I-B)\big)_{ij}$ for $i \neq j$ is nonzero only if i is the parent of j or vice versa (a product term $b_{ki}b_{kj}$ would need a common child k with two parents). So the DAG is contained in the GGM. Conversely, a $\Sigma \succ 0$ with tree-shaped inverse factorizes along any orientation of the tree away from a root (the Gaussian density factorizes over a tree as $\prod_k p(x_k \mid x_{\mathrm{pa}(k)})$; see Lauritzen, 1996), and every collider-free orientation of a tree points away from some root. □

Lemma (T, perfect indicators)**.** *In a factor model, an indicator y_1 of F with $\theta_{11} = 0$ and $\lambda_1 = 1$ equals F. Hence the path model in which y_1 is an exogenous cause of the other indicators of F ($y_k = \lambda_k y_1 + e_k$) is nested in the factor model.*

Proof. Take $\theta_{11} = 0$ (allowed: parameters are unrestricted) and $\psi = \mathrm{Var}(y_1)$; Lemma R. □

Lemma (H, higher-order and bifactor; Yung, Thissen & McLeod, 1999)**.** *A higher-order model (first-order factors F_k with disjoint indicator sets, $F_k = \gamma_k G + \zeta_k$) is nested in the bifactor model with a general factor on all indicators and one orthogonal group factor per first-order factor.*

Proof. For an indicator y_i of F_k: $y_i = \lambda_i\gamma_k G + \lambda_i\zeta_k + e_i$. The general factor G and the disturbances ζ_k are mutually uncorrelated, so this is a bifactor model with general loadings $\lambda_i\gamma_k$, group factor $S_k = \zeta_k$ with loadings λ_i and variance $\mathrm{Var}(\zeta_k)$ (possibly negative). Rescale G to its marker loading (generic: $\lambda_1\gamma_1 \neq 0$). □

3 How to read an item page

Each page shows the two path diagrams exactly as in the game (observed variables as squares, latent variables as circles, residuals as small circles, covariances as curved double-headed arrows, network edges as undirected lines; variances are free and not drawn), the model syntax (∼∼ is a covariance, `-` a network edge, `network` lists a network block, `a*` an equality label), the claim and its proof. (i) concerns $A \preceq B$, (ii) concerns $B \preceq A$. σ_{ij} is the covariance of y_i and y_j.

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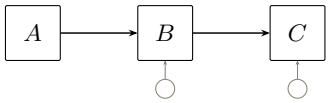
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E01 Chain versus fork

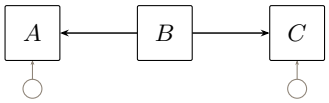
Level: Easy. Model families: Path analysis. Observed variables: A, B, C ($p = 3$, 6 moments).

Model A $q_A = 5$




```
B ~ A; C ~ B
```

Model B $q_B = 5$




```
A ~ B; C ~ B
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Both graphs have the skeleton $A - B - C$ and no collider, so by Lemma C both models equal the GGM with $(\Sigma^{-1})_{AC} = 0$, i.e. $\mathcal{M} = \{\Sigma \succ 0 : \sigma_{AC}\sigma_{BB} = \sigma_{AB}\sigma_{BC}\}$.

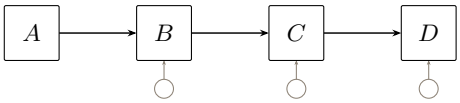
(ii) $B \preceq A$. See the previous line: $\mathcal{M}_A = \mathcal{M}_B$. □

In the game: Both models say that A and C are unrelated once you know B. Turning the first arrow around does not change that, so both reproduce exactly the same covariance matrices.

E02 A chain and its reverse

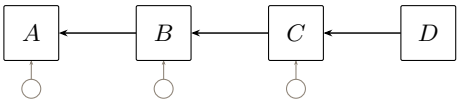
Level: Easy. Model families: Path analysis. Observed variables: A, B, C, D ($p = 4$, 10 moments).

Model A $q_A = 7$




```
B ~ A; C ~ B; D ~ C
```

Model B $q_B = 7$




```
C ~ D; B ~ C; A ~ B
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Both are chains on the path $A - B - C - D$ without colliders. By Lemma C both equal the GGM on that path: all $\Sigma \succ 0$ whose inverse is tridiagonal.

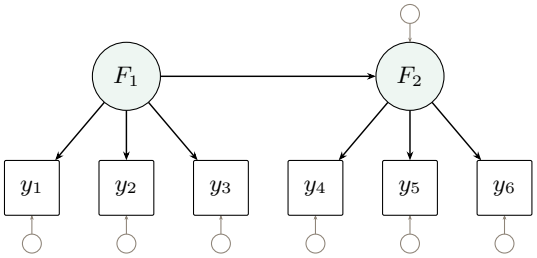
(ii) $B \preceq A$. Same argument: $\mathcal{M}_A = \mathcal{M}_B$. □

In the game: A chain read backwards is still a chain: in both models every variable only "talks" to its neighbours. Same conditional independencies, same fit on every dataset.

E03 Regression or correlation between two factors

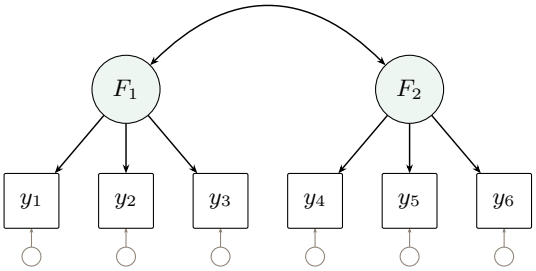
Level: Easy. Model families: SEM. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 13$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F2 ~ F1
```

Model B $q_B = 13$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. The measurement parts are identical; the models differ only in the latent block (F_1, F_2). In A, $\Psi = \begin{pmatrix} \psi_{11} & \beta\psi_{11} \\ \beta\psi_{11} & \beta^2\psi_{11} + \zeta \end{pmatrix}$; in B, Ψ is free. Both reach every 2×2 Ψ with $\psi_{11} \neq 0$ ($\beta = \psi_{12}/\psi_{11}$, $\zeta = \psi_{22} - \psi_{12}^2/\psi_{11}$), so by Lemmas B and G the models are equivalent.

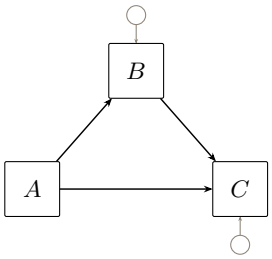
(ii) $B \preceq A$. Lemma B: in B the block is free, so $\mathcal{M}_A \subseteq \mathcal{M}_B$ trivially; with the map above $\mathcal{M}_B \subseteq \overline{\mathcal{M}_A}$. □

In the game: With only two factors, a regression F1 → F2 and a correlation F1 ↔ F2 both spend one parameter on the single covariance between the factors. The fit is identical.

E04 Two saturated models

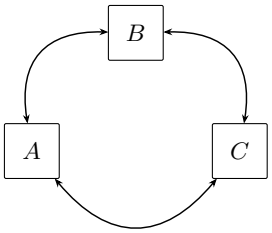
Level: Easy. Model families: Path analysis. Observed variables: A, B, C ($p = 3$, 6 moments).

Model A $q_A = 6$




```
B ~ A; C ~ A + B
```

Model B $q_B = 6$




```
A ~~ B + C; B ~~ C
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. A is a complete recursive path model and B the model with all covariances free; both are saturated (Lemma B(a)): $\overline{\mathcal{M}_A} = \overline{\mathcal{M}_B} =$ all positive definite 3×3 matrices.

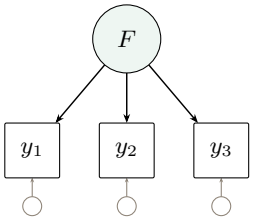
(ii) $B \preceq A$. As above. □

In the game: Both models are saturated: they have as many parameters (6) as there are variances and covariances. Every saturated model reproduces any covariance matrix, so all saturated models are equivalent.

E05 One factor with three indicators is saturated

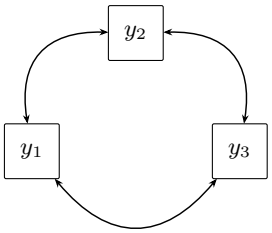
Level: Easy. Model families: CFA, Path analysis. Observed variables: y_1, y_2, y_3 ($p = 3$, 6 moments).

Model A $q_A = 6$




```
F =~ y1 + y2 + y3
```

Model B $q_B = 6$




```
y1 ~~ y2 + y3; y2 ~~ y3
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. B (all covariances free) is saturated, so $A \preceq B$.

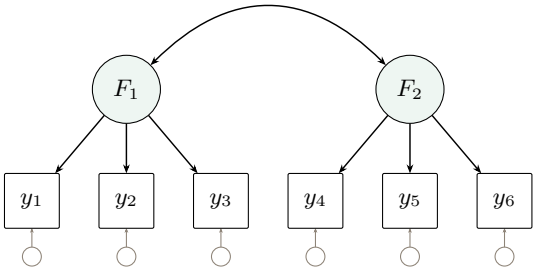
(ii) $B \preceq A$. By Lemma B(a) the one-factor model with three indicators reproduces every $\Sigma \succ 0$ with $\sigma_{12}\sigma_{13}\sigma_{23} \neq 0$ (a dense set): $\lambda_1 = 1$, $\psi = \sigma_{12}\sigma_{13}/\sigma_{23}$, $\lambda_2 = \sigma_{23}/\sigma_{13}$, $\lambda_3 = \sigma_{23}/\sigma_{12}$, $\theta_{ii} = \sigma_{ii} - \lambda_i^2\psi$. ($\psi < 0$ when the product of the covariances is negative: a Heywood solution.) Hence $B \preceq A$. □

In the game: A one-factor model with three indicators has 3 loadings and 3 residual variances: 6 parameters for 6 (co)variances. It is just-identified (saturated), like the model with all covariances free.

E06 A cross-loading

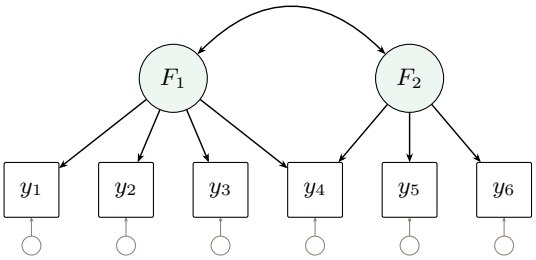
Level: Easy. Model families: CFA. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 13$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2
```

Model B $q_B = 14$




```
F1 =~ y1 + y2 + y3 + y4; F2 =~ y4 + y5 + y6; F1 ~~ F2
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Fix the cross-loading of y_4 on F_1 in B to 0 (Lemma R).

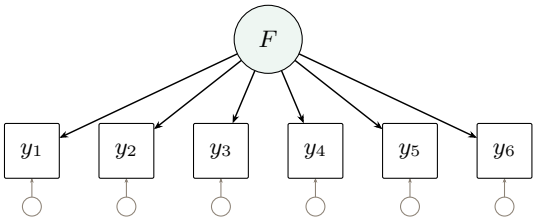
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 14 at the rational point listed in Appendix B, while model A has only $q_A = 13$ free parameters. □

In the game: B is A plus one cross-loading (F1 → y4). Fix that loading to zero and you get A back, so A is nested in B.

E07 One factor versus two

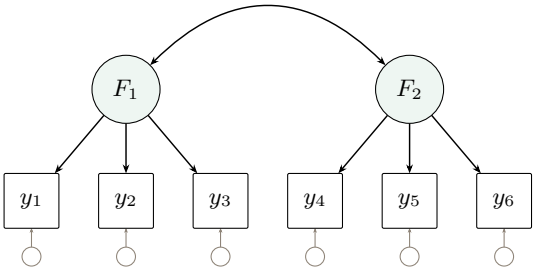
Level: Easy. Model families: CFA. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 12$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6
```

Model B $q_B = 13$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Lemma M: in B take $\psi_{11} = \psi_{22} = \psi_{12}$; then F_1 and F_2 act as one factor and B reproduces A.

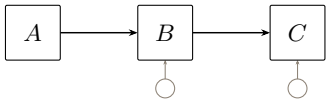
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 13 at the rational point listed in Appendix B, while model A has only $q_A = 12$ free parameters. □

In the game: A one-factor model is a two-factor model whose factors correlate perfectly (r = 1). So the one-factor model is nested in the two-factor model.

E08 Adding a direct effect

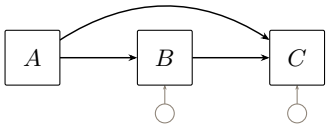
Level: Easy. Model families: Path analysis. Observed variables: A, B, C ($p = 3$, 6 moments).

Model A $q_A = 5$




```
B ~ A; C ~ B
```

Model B $q_B = 6$




```
B ~ A; C ~ A + B
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Fix $\beta_{CA} = 0$ in B (Lemma R).

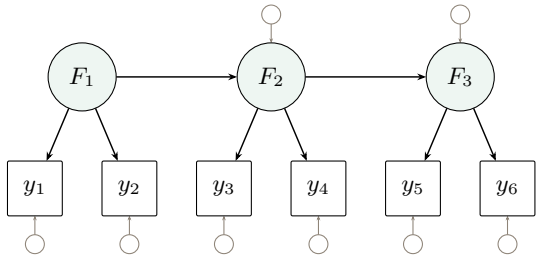
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 6 at the rational point listed in Appendix B, while model A has only $q_A = 5$ free parameters. □

In the game: B adds the direct effect A → C (and is saturated). Setting that effect to zero gives A: A is nested in B.

E09 A latent chain versus correlated factors

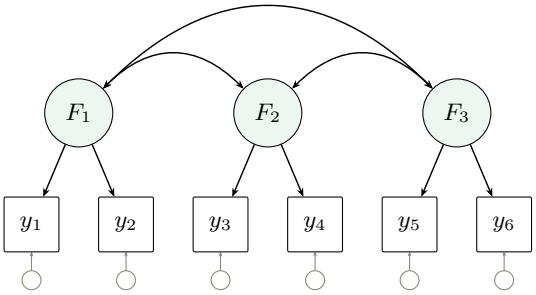
Level: Easy. Model families: SEM. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F2 ~ F1; F3 ~ F2
```

Model B $q_B = 15$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F1 ~~ F2 + F3; F2
~~ F3
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Same measurement part; in B the latent covariance matrix Ψ is free (a saturated latent block), and the chain in A produces one particular Ψ. By Lemma B(b), $A \preceq B$.

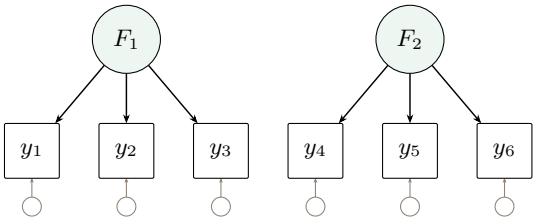
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 15 at the rational point listed in Appendix B, while model A has only $q_A = 14$ free parameters. □

In the game: B leaves the three factor covariances free (a saturated latent block). The chain F1 → F2 → F3 restricts them (no direct F1–F3 link), so A is nested in B.

E10 Orthogonal versus correlated factors

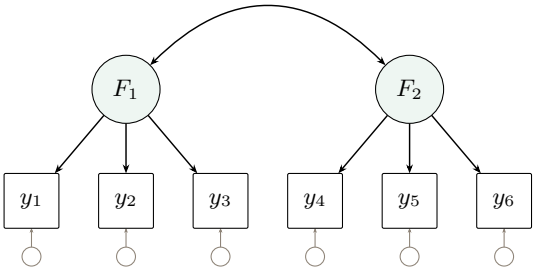
Level: Easy. Model families: CFA. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 12$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6
```

Model B $q_B = 13$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Fix $\psi_{12} = 0$ in B (Lemma R).

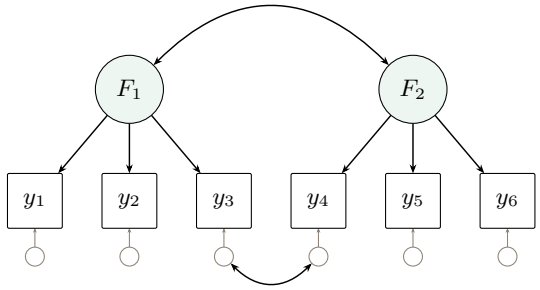
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 13 at the rational point listed in Appendix B, while model A has only $q_A = 12$ free parameters. □

In the game: Orthogonal factors are correlated factors with the correlation fixed to zero: A is nested in B.

E11 One residual covariance more

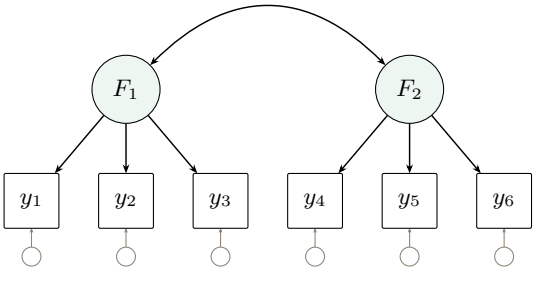
Level: Easy. Model families: CFA. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2; y3 ~~ y4
```

Model B $q_B = 13$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 14 at the rational point listed in Appendix B, while model B has only $q_B = 13$ free parameters.

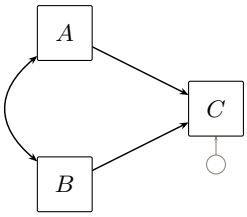
(ii) $B \preceq A$. Fix $\theta_{34} = 0$ in A (Lemma R). □

In the game: A is B plus one residual covariance (y3 ↔ y4). Fix it to zero and you get B, so B is nested in A.

E12 Correlated or independent predictors

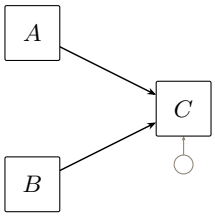
Level: Easy. Model families: Path analysis. Observed variables: A, B, C ($p = 3$, 6 moments).

Model A $q_A = 6$




```
C ~ A + B; A ~~ B
```

Model B $q_B = 5$




```
C ~ A + B
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 6 at the rational point listed in Appendix B, while model B has only $q_B = 5$ free parameters.

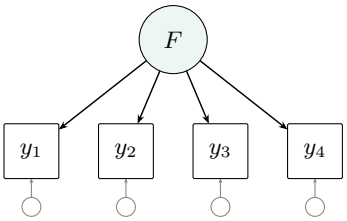
(ii) $B \preceq A$. Fix $\mathrm{Cov}(A, B) = 0$ in A (Lemma R). □

In the game: B is A with the covariance between the predictors fixed to zero (A and B independent). B is nested in A.

E13 Equal loadings

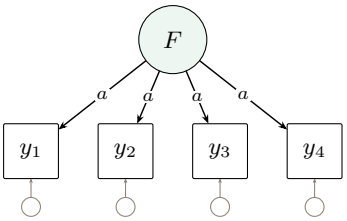
Level: Easy. Model families: CFA. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 8$




```
F =~ y1 + y2 + y3 + y4
```

Model B $q_B = 6$




```
F =~ a*y1 + a*y2 + a*y3 + a*y4
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 8 at the rational point listed in Appendix B, while model B has only $q_B = 6$ free parameters.

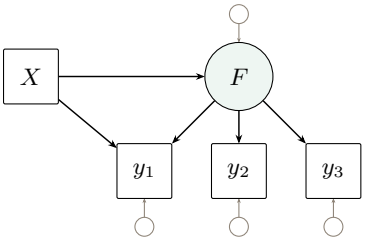
(ii) $B \preceq A$. B is A with the equality constraints $\lambda_1 = \lambda_2 = \lambda_3 = \lambda_4$ (Lemma R; with free factor variance the common loading can be set to 1). □

In the game: B forces all loadings to be equal (label a). That is a special case of A, where the loadings are free: B is nested in A.

E14 A direct effect of the covariate

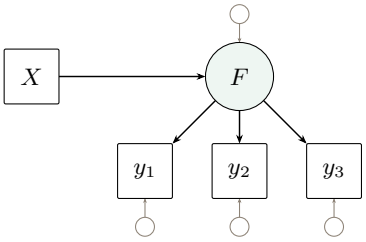
Level: Easy. Model families: SEM. Observed variables: X, y_1, y_2, y_3 ($p = 4$, 10 moments).

Model A $q_A = 9$




```
F =~ y1 + y2 + y3; F ~ X; y1 ~ X
```

Model B $q_B = 8$




```
F =~ y1 + y2 + y3; F ~ X
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 9 at the rational point listed in Appendix B, while model B has only $q_B = 8$ free parameters.

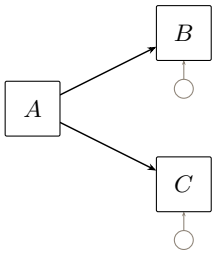
(ii) $B \preceq A$. Fix the direct effect $X \to y_1$ in A to 0 (Lemma R). □

In the game: A adds a direct effect of the covariate X on item y1. Fixing it to zero gives B, so B is nested in A.

E15 The independence model

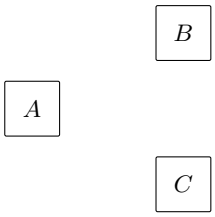
Level: Easy. Model families: Path analysis. Observed variables: A, B, C ($p = 3$, 6 moments).

Model A $q_A = 5$




```
B ~ A; C ~ A
```

Model B $q_B = 3$




```
(no paths:  variances only)
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 5 at the rational point listed in Appendix B, while model B has only $q_B = 3$ free parameters.

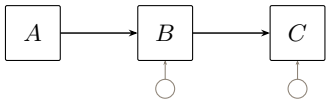
(ii) $B \preceq A$. Fix both regression coefficients in A to 0 (Lemma R): B only has variances. □

In the game: B has no paths at all: only variances (the independence or "baseline" model). It is nested in every model with the same variables, including A.

E16 Chain versus collider

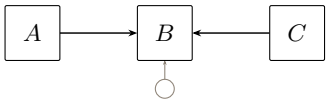
Level: Easy. Model families: Path analysis. Observed variables: A, B, C ($p = 3$, 6 moments).

Model A $q_A = 5$




```
B ~ A; C ~ B
```

Model B $q_B = 5$




```
B ~ A + C
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{AC}$. In B, A and C are exogenous and not connected, so $\sigma_{AC} = 0$. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \beta_{BA}\beta_{CB}\,\mathrm{Var}(A) \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.3000 \neq 0$.

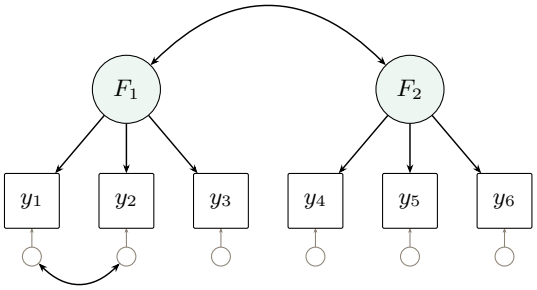
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{AC}\sigma_{BB} - \sigma_{AB}\sigma_{BC}$. In A, $\sigma_{AC} = \beta_{CB}\sigma_{AB}$ and $\sigma_{BC} = \beta_{CB}\sigma_{BB}$. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B, $f = -\sigma_{AB}\sigma_{BC} \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = -0.2700 \neq 0$. □

In the game: The chain says A and C are independent given B; the collider says A and C are independent (period), but dependent given B. Each restriction is violated by the other model.

E17 Two different residual covariances

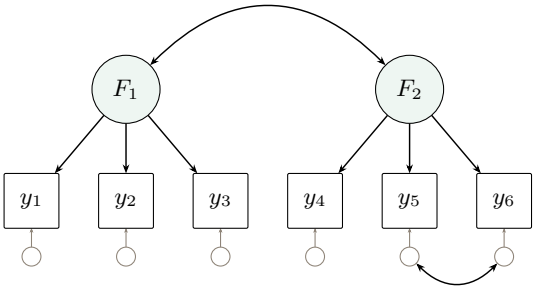
Level: Easy. Model families: CFA. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2; y1 ~~ y2
```

Model B $q_B = 14$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2; y5 ~~ y6
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{12}\sigma_{35} - \sigma_{13}\sigma_{25}$. In B these four covariances are products $\lambda_i\lambda_j\psi_{f(i)f(j)}$ (no residual covariance involved), and the two products are equal. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \theta_{12}\lambda_3\lambda_5\psi_{12} \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.06480 \neq 0$.

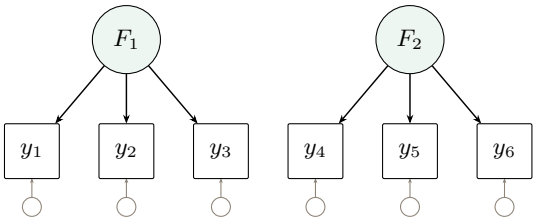
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{56}\sigma_{24} - \sigma_{45}\sigma_{26}$. In A these covariances are products of loadings and factor covariances (no θ_{56}). So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B, $f = \theta_{56}\lambda_2\lambda_4\psi_{12} \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.06000 \neq 0$. □

In the game: Same number of parameters, but each model frees a residual covariance that the other fixes to zero. Neither can mimic the other.

E18 Two orthogonal factors versus one factor

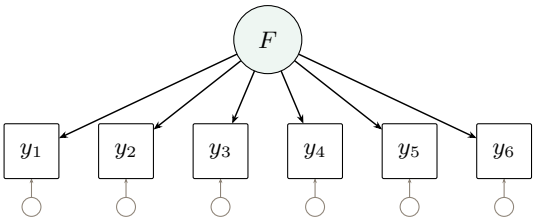
Level: Easy. Model families: CFA. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 12$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6
```

Model B $q_B = 12$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{12}\sigma_{45} - \sigma_{14}\sigma_{25}$. One-factor model: every off-diagonal $\sigma_{ij} = \lambda_i \lambda_j \psi$, so all tetrads vanish. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \lambda_1\lambda_2\lambda_4\lambda_5\psi_{11}\psi_{22} \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 1.440 \neq 0$.

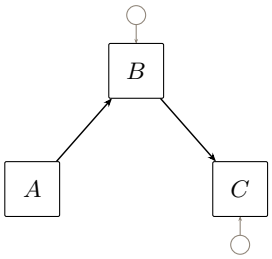
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{14}$. Orthogonal factors: the two clusters are uncorrelated. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B, $f = \lambda_1\lambda_4\psi \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 1.200 \neq 0$. □

In the game: A forces the two clusters to be uncorrelated; B forces all items through one factor, so the clusters must correlate. Neither restriction holds in the other model.

E19 Two different mediators

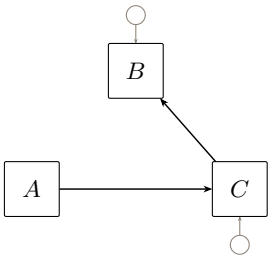
Level: Easy. Model families: Path analysis. Observed variables: A, B, C ($p = 3$, 6 moments).

Model A $q_A = 5$




```
B ~ A; C ~ B
```

Model B $q_B = 5$




```
C ~ A; B ~ C
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{AB}\sigma_{CC} - \sigma_{AC}\sigma_{BC}$. In B ($A \to C \to B$), A and B are independent given C. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A at the point of Appendix B, $f = 0.4500 \neq 0$.

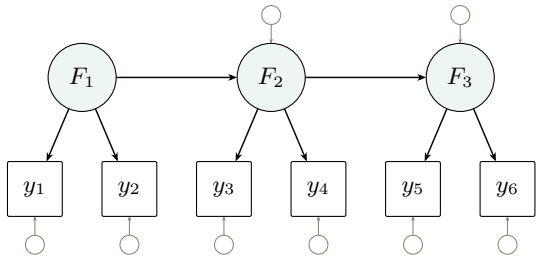
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{AC}\sigma_{BB} - \sigma_{AB}\sigma_{BC}$. In A ($A \to B \to C$), A and C are independent given B. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = 0.6000 \neq 0$. □

In the game: Different mediators: A says A and C are independent given B, B says A and B are independent given C. Both are chains, but they make different claims.

E20 Latent chain versus latent collider

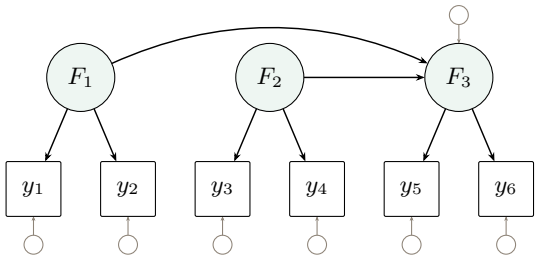
Level: Easy. Model families: SEM. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F2 ~ F1; F3 ~ F2
```

Model B $q_B = 14$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F3 ~ F1 + F2
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{13}$. In B, F_1 and F_2 are exogenous and uncorrelated, so indicators of F_1 and F_2 are uncorrelated. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \lambda_1\lambda_3\psi_{12} \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.5000 \neq 0$.

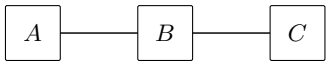
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{15}\sigma_{34} - \sigma_{13}\sigma_{45}$. In A this equals $\lambda_1\lambda_3\lambda_4\lambda_5(\psi_{13}\psi_{22} - \psi_{12}\psi_{23})$, and the chain implies $\psi_{13}\psi_{22} = \psi_{12}\psi_{23}$. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = 0.5400 \neq 0$.□

In the game: A chain (F1 and F3 independent given F2) against a collider (F1 and F2 independent). Each model has a restriction the other violates.

E21 A network path equals a chain

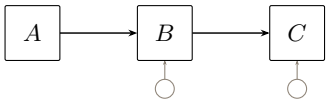
Level: Easy. Model families: GGM. Observed variables: A, B, C ($p = 3$, 6 moments).

Model A $q_A = 5$




```
network A B C; A - B; B - C
```

Model B $q_B = 5$




```
B ~ A; C ~ B
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. By Lemma C the chain $A \to B \to C$ (skeleton $A - B - C$, no collider) equals the GGM on its skeleton, which is model A.

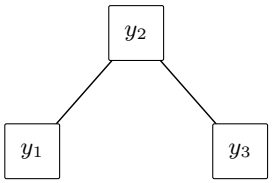
(ii) $B \preceq A$. See above: $\mathcal{M}_A = \mathcal{M}_B$. □

In the game: The network has no edge A — C: A and C are independent once you know B (a zero partial correlation). The chain A → B → C makes exactly the same claim, and no other.

E22 One network edge more

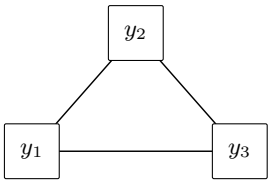
Level: Easy. Model families: GGM. Observed variables: y_1, y_2, y_3 ($p = 3$, 6 moments).

Model A $q_A = 5$




```
network y1 y2 y3; y1 - y2; y2 - y3
```

Model B $q_B = 6$




```
network y1 y2 y3; y1 - y2; y2 - y3; y1 - y3
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Fix the precision element k_{13} of B to 0 (Lemma R).

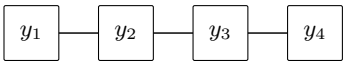
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 6 at the rational point listed in Appendix B, while model A has only $q_A = 5$ free parameters. □

In the game: B adds the edge y1 — y3 (a complete network, saturated). Fix that partial correlation to zero and you get A.

E23 Cutting a network in two

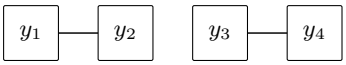
Level: Easy. Model families: GGM. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 7$




```
network y1 y2 y3 y4; y1 - y2; y2 - y3; y3 - y4
```

Model B $q_B = 6$




```
network y1 y2 y3 y4; y1 - y2; y3 - y4
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 7 at the rational point listed in Appendix B, while model B has only $q_B = 6$ free parameters.

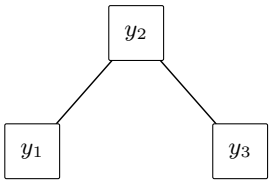
(ii) $B \preceq A$. Fix $k_{23} = 0$ in A (Lemma R). □

In the game: B drops the edge y2 — y3, which splits the network into two separate pairs. That is A with one partial correlation fixed to zero.

E24 Two networks with a different centre

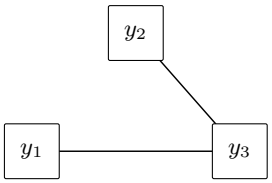
Level: Easy. Model families: GGM. Observed variables: y_1, y_2, y_3 ($p = 3$, 6 moments).

Model A $q_A = 5$




```
network y1 y2 y3; y1 - y2; y2 - y3
```

Model B $q_B = 5$




```
network y1 y2 y3; y1 - y3; y3 - y2
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{12}\sigma_{33} - \sigma_{13}\sigma_{23}$. In B there is no edge $y_1 - y_2$, so $k_{12} = 0$, which by Lemma I means $\sigma_{12}\sigma_{33} = \sigma_{13}\sigma_{23}$. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A at the point of Appendix B, $f = 0.04852 \neq 0$.

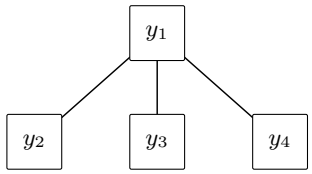
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{13}\sigma_{22} - \sigma_{12}\sigma_{23}$. In A there is no edge $y_1 - y_3$: $k_{13} = 0$ (Lemma I). So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = 0.04823 \neq 0$. □

In the game: A says y1 and y3 are independent given y2; B says y1 and y2 are independent given y3. Each network has an edge the other one lacks.

E25 The empty network

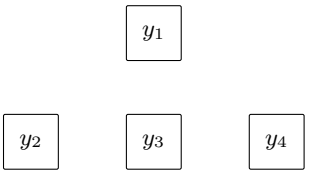
Level: Easy. Model families: GGM. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 7$




```
network y1 y2 y3 y4; y1 - y2 + y3 + y4
```

Model B $q_B = 4$




```
network y1 y2 y3 y4
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 7 at the rational point listed in Appendix B, while model B has only $q_B = 4$ free parameters.

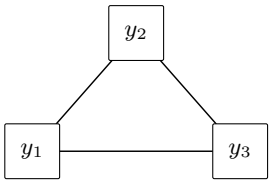
(ii) $B \preceq A$. Fix $k_{12} = k_{13} = k_{14} = 0$ in A (Lemma R): K and hence Σ become diagonal, which is B. □

In the game: A network without edges says that all variables are independent. Fix all partial correlations of A to zero and you get B.

E26 A complete network is saturated

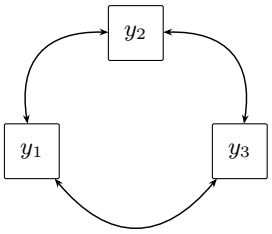
Level: Easy. Model families: GGM. Observed variables: y_1, y_2, y_3 ($p = 3$, 6 moments).

Model A $q_A = 6$




```
network y1 y2 y3; y1 - y2 + y3; y2 - y3
```

Model B $q_B = 6$




```
y1 ~~ y2 + y3; y2 ~~ y3
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. B (all covariances free) is saturated (Lemma B(a)).

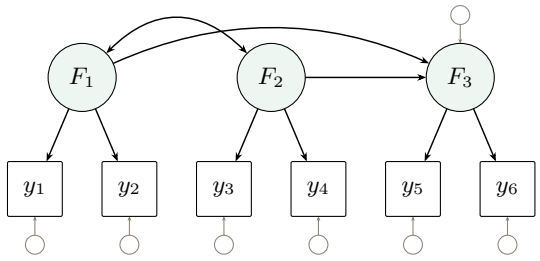
(ii) $B \preceq A$. In A every element of K is free, so $\Sigma = K^{-1}$ is any positive definite matrix: A is saturated too. □

In the game: Both models are saturated: the complete network has a free partial correlation for every pair, B a free covariance for every pair. Both reproduce every covariance matrix.

E27 Correlated or uncorrelated latent predictors

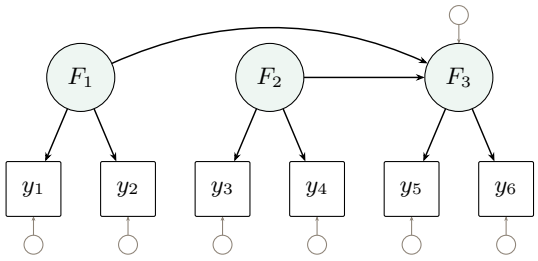
Level: Easy. Model families: SEM. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 15$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F3 ~ F1 + F2; F1
~~ F2
```

Model B $q_B = 14$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F3 ~ F1 + F2
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 15 at the rational point listed in Appendix B, while model B has only $q_B = 14$ free parameters.

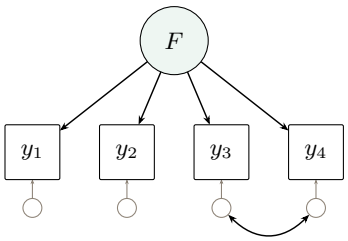
(ii) $B \preceq A$. Fix $\psi_{12} = \mathrm{Cov}(F_1, F_2) = 0$ in A (Lemma R). □

In the game: B is A with the covariance between the two predicting factors (F1 and F2) fixed to zero.

M01 Residual covariance versus a second factor

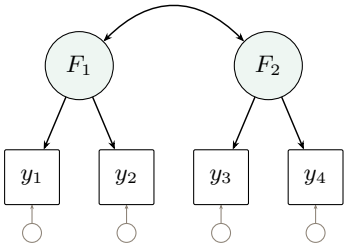
Level: Medium. Model families: CFA. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 9$




```
F =~ y1 + y2 + y3 + y4; y3 ~~ y4
```

Model B $q_B = 9$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F1 ~~ F2
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Lemma P with the pair (y_3, y_4): $A \equiv F\{y_1, y_2\} \to G\{y_3, y_4\}$. The latent block (F, G) is saturated in both that model and B (Lemma B(b), as in E03), so $A \equiv B$. Both equal $\{\Sigma \succ 0 : \sigma_{13}\sigma_{24} = \sigma_{14}\sigma_{23}\}$.

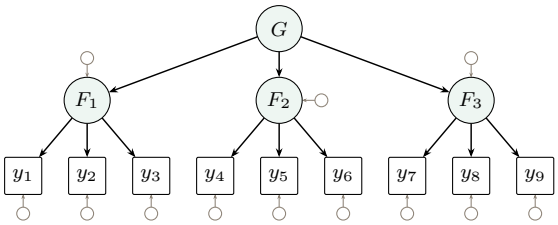
(ii) $B \preceq A$. See above. □

In the game: Both models imply a single restriction, the tetrad $\sigma 13{\cdot}\sigma 24 = \sigma 14{\cdot}\sigma 23$, and nothing else (9 parameters each). The residual covariance does the same job as splitting off a second factor.

M02 Higher-order model with three first-order factors

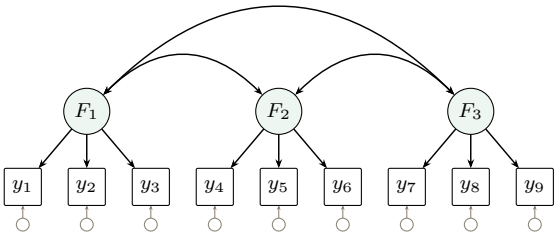
Level: Medium. Model families: Higher-order. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9$ ($p = 9$, 45 moments).

Model A $q_A = 21$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F3 =~ y7 + y8 + y9; G
=~ F1 + F2 + F3
```

Model B $q_B = 21$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F3 =~ y7 + y8 + y9; F1
~~ F2 + F3; F2 ~~ F3
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Same measurement part. In B, Ψ is free. In A, $\Psi = \gamma\gamma^\top \psi_G + \mathrm{diag}(\zeta)$ is a one-factor model on three variables, which by Lemma B(a) reproduces every Ψ with $\psi_{12}\psi_{13}\psi_{23} \neq 0$. Lemmas B(b) and G give $A \equiv B$.

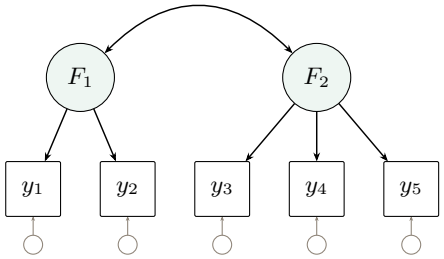
(ii) $B \preceq A$. See above (with $\gamma_1 = 1$, $\psi_G = \psi_{12}\psi_{13}/\psi_{23}$, ...). □

In the game: A second-order factor on three first-order factors has three loadings for three factor covariances: it is just-identified, so it reproduces any correlation pattern among three factors.

M03 A two-indicator factor

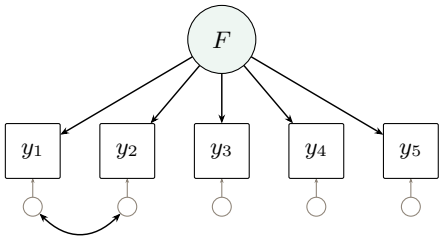
Level: Medium. Model families: CFA. Observed variables: y_1, y_2, y_3, y_4, y_5 ($p = 5$, 15 moments).

Model A $q_A = 11$




```
F1 =~ y1 + y2; F2 =~ y3 + y4 + y5; F1 ~~ F2
```

Model B $q_B = 11$




```
F =~ y1 + y2 + y3 + y4 + y5; y1 ~~ y2
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. A has a factor F_1 with only y_1, y_2 and a free covariance with F_2. By Lemma P (pair (y_1, y_2)), $B \equiv F\{y_3, y_4, y_5\} \to G\{y_1, y_2\}$, and $F \to G$ and $F \leftrightarrow G$ give the same (saturated) 2×2 latent block (Lemma B(b)).

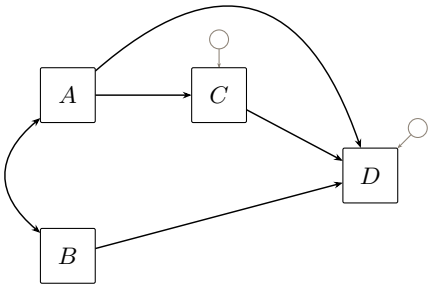
(ii) $B \preceq A$. See above. □

In the game: A factor with only two indicators adds exactly one free covariance (between those two items) on top of a single common factor. That is what the residual covariance y1 ↔ y2 does in B.

M04 The replacing rule

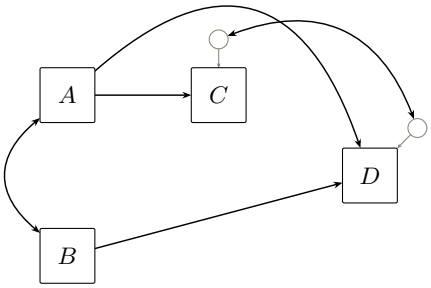
Level: Medium. Model families: Path analysis. Observed variables: A, B, C, D ($p = 4$, 10 moments).

Model A $q_A = 9$




```
A ~~ B; C ~ A; D ~ C + A + B
```

Model B $q_B = 9$




```
A ~~ B; C ~ A; D ~ A + B; C ~~ D
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Substitute $C = \beta_{CA}A + \zeta_C$ in D's equation of A: $D = (\beta_{DA} + \beta_{DC}\beta_{CA})A + \beta_{DB}B + (\beta_{DC}\zeta_C + \zeta_D)$. This is model B with residual $\zeta'_D = \beta_{DC}\zeta_C + \zeta_D$, $\mathrm{Var}(\zeta'_D) = \beta^2_{DC}\psi_C + \psi_D$ and $\mathrm{Cov}(\zeta_C, \zeta'_D) = \beta_{DC}\psi_C$.

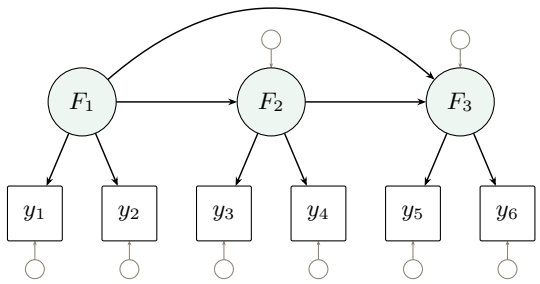
(ii) $B \preceq A$. Conversely, for $\psi_C \neq 0$ set $\beta_{DC} = \omega/\psi_C$ ($\omega = \mathrm{Cov}(\zeta_C, \zeta'_D)$), $\beta_{DA} = \beta'_{DA} - \beta_{DC}\beta_{CA}$, $\psi_D = \psi'_D - \omega^2/\psi_C$ (Lemma G). □

In the game: Replacing rule (Lee & Hershberger): C → D may be replaced by a residual covariance C ↔ D because the predictors of D include all predictors of C. Both models only claim that C and B are independent given A.

M05 A saturated structural block

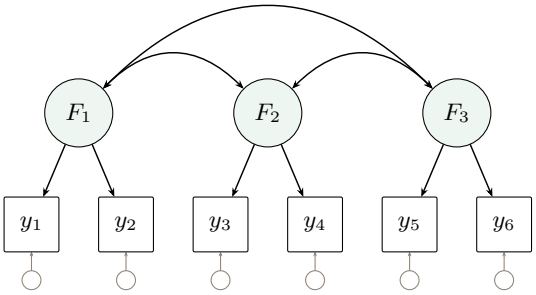
Level: Medium. Model families: SEM. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 15$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F2 ~ F1; F3 ~ F1 +
F2
```

Model B $q_B = 15$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F1 ~~ F2 + F3; F2
~~ F3
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. The structural part of A is a complete recursive model on (F_1, F_2, F_3), which reproduces every $\Psi \succ 0$ (Cholesky, Lemma B(a)); in B, Ψ is free. Lemma B(b).

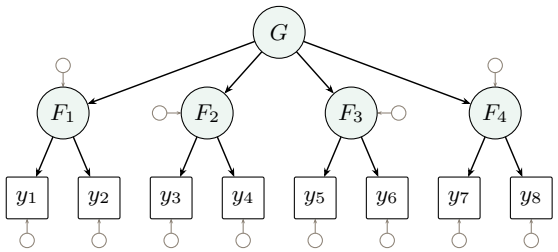
(ii) $B \preceq A$. See above. □

In the game: Three structural paths among three factors make a saturated latent block, just like three free factor covariances. Same measurement part, same fit.

M06 Higher-order model with four first-order factors

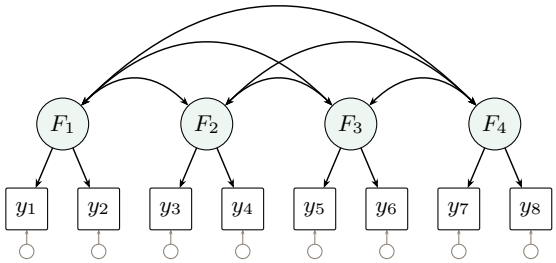
Level: Medium. Model families: Higher-order. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 20$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; G
=~ F1 + F2 + F3 + F4
```

Model B $q_B = 22$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; F1
~~ F2 + F3 + F4; F2 ~~ F3 + F4; F3 ~~ F4
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Same measurement part; B has a free Ψ and A's second-order factor gives one particular Ψ (Lemma B(b)).

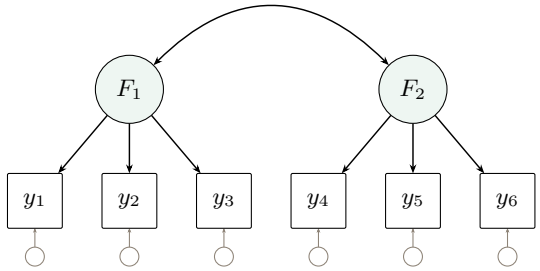
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 22 at the rational point listed in Appendix B, while model A has only $q_A = 20$ free parameters. □

In the game: With four first-order factors the second-order factor explains six covariances with four loadings, imposing two restrictions. B leaves the six covariances free: A is nested in B.

M07 Two factors inside one factor with correlated residuals

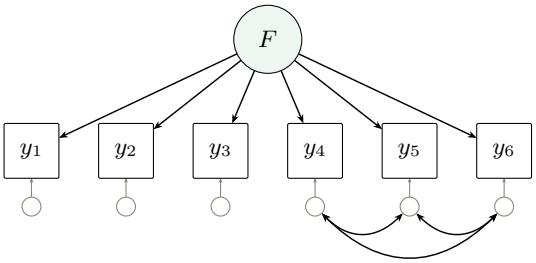
Level: Medium. Model families: CFA. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 13$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2
```

Model B $q_B = 15$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6; y4 ~~ y5 + y6; y5 ~~ y6
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Given A ($\Psi = (\psi_{kl})$, $\psi_{11} \neq 0$), take in B: $\psi = \psi_{11}$, loadings λ_i for y_1, y_2, y_3 and $\lambda_j \psi_{12}/\psi_{11}$ for $j = 4, 5, 6$, $\theta_{jk} = \lambda_j \lambda_k (\psi_{22} - \psi_{12}^2/\psi_{11})$ for $j \neq k \in \{4, 5, 6\}$ and $\theta_{jj} = \theta_{jj}^A + \lambda_j^2 (\psi_{22} - \psi_{12}^2/\psi_{11})$. Then $\Sigma_B = \Sigma_A$ (Lemma G).

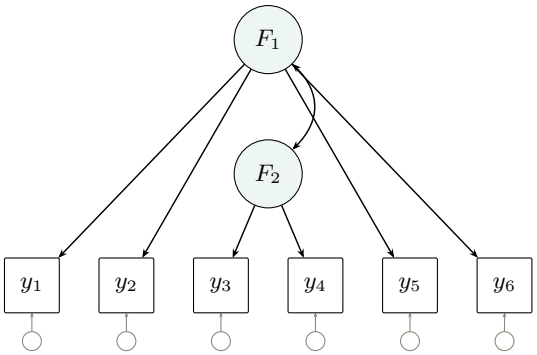
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 15 at the rational point listed in Appendix B, while model A has only $q_A = 13$ free parameters. □

In the game: Surprise: one factor plus free residual covariances among y4–y6 can mimic the second factor. The part of y4–y6 that F1 does not explain goes into those residual covariances. B has two more parameters, so A is nested in B.

M08 Merging two factors

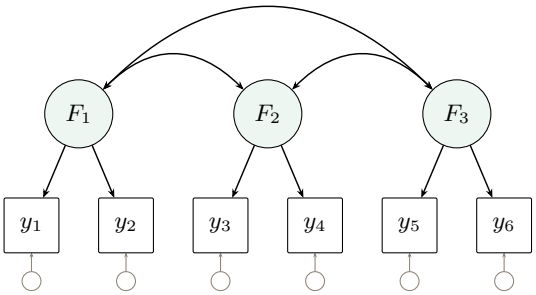
Level: Medium. Model families: CFA. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 13$




```
F1 =~ y1 + y2 + y5 + y6; F2 =~ y3 + y4; F1 ~~ F2
```

Model B $q_B = 15$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F1 ~~ F2 + F3; F2
~~ F3
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Lemma M: merge F_1 and F_3 of B ($\psi_{11} = \psi_{33} = \psi_{13}$, $\psi_{12} = \psi_{23}$).

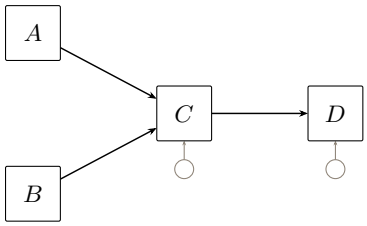
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 15 at the rational point listed in Appendix B, while model A has only $q_A = 13$ free parameters. □

In the game: Merge the first and third factor of B (correlation 1) and you get A. So A, with its non-adjacent indicators, is nested in the three-factor model.

M09 Every model is nested in a saturated model

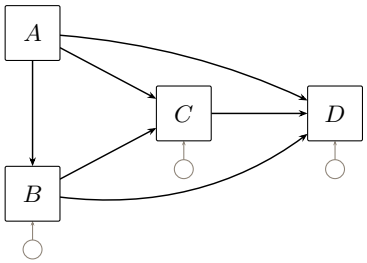
Level: Medium. Model families: Path analysis. Observed variables: A, B, C, D ($p = 4$, 10 moments).

Model A $q_A = 7$




```
C ~ A + B; D ~ C
```

Model B $q_B = 10$




```
B ~ A; C ~ A + B; D ~ A + B + C
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. B is a complete recursive path model on four variables, hence saturated (Lemma B(a)).

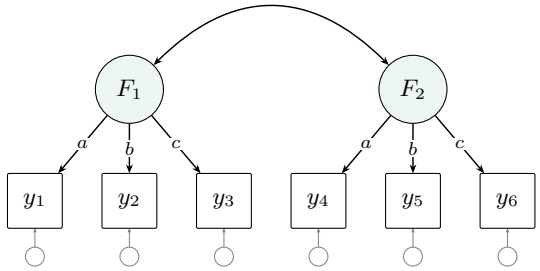
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 10 at the rational point listed in Appendix B, while model A has only $q_A = 7$ free parameters. □

In the game: B has every possible path between the four variables: it is saturated. Any model on the same variables is nested in a saturated model.

M10 Equal loading patterns

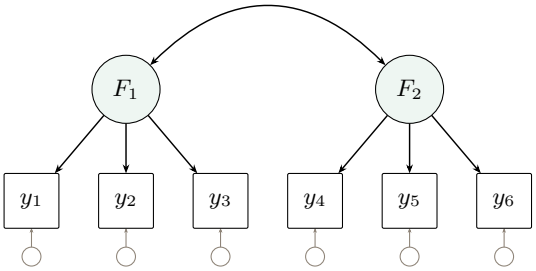
Level: Medium. Model families: CFA. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 12$




```
F1 =~ a*y1 + b*y2 + c*y3; F2 =~ a*y4 + b*y5 + c*y6; F1 ~~ F2
```

Model B $q_B = 13$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. A is B with the constraints $\lambda_1 = \lambda_4$, $\lambda_2 = \lambda_5$, $\lambda_3 = \lambda_6$ (Lemma R).

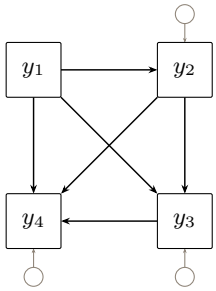
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 13 at the rational point listed in Appendix B, while model A has only $q_A = 12$ free parameters. □

In the game: A forces the same loading pattern (a, b, c) on both factors; B leaves all loadings free. A is a constrained version of B.

M11 A factor model inside a saturated path model

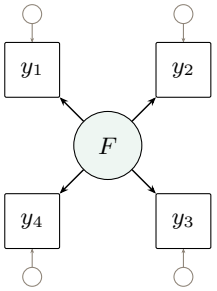
Level: Medium. Model families: CFA, Path analysis. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 10$




```
y2 ~ y1; y3 ~ y1 + y2; y4 ~ y1 + y2 + y3
```

Model B $q_B = 8$




```
F =~ y1 + y2 + y3 + y4
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 10 at the rational point listed in Appendix B, while model B has only $q_B = 8$ free parameters.

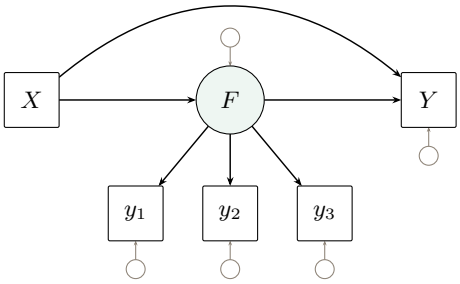
(ii) $B \preceq A$. A is a complete recursive path model on $y_1, \ldots, y_4$: saturated (Lemma B(a)). □

In the game: A is a saturated path model (all six paths). Every model on these four variables, also a factor model, is nested in it.

M12 Full versus partial mediation

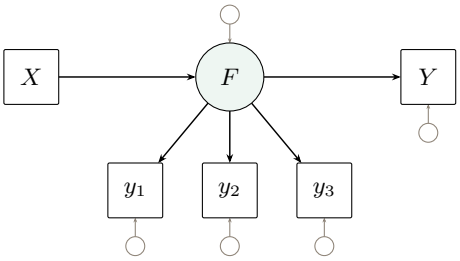
Level: Medium. Model families: SEM. Observed variables: X, y_1, y_2, y_3, Y ($p = 5$, 15 moments).

Model A $q_A = 11$




```
F =~ y1 + y2 + y3; F ~ X; Y ~ F + X
```

Model B $q_B = 10$




```
F =~ y1 + y2 + y3; F ~ X; Y ~ F
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 11 at the rational point listed in Appendix B, while model B has only $q_B = 10$ free parameters.

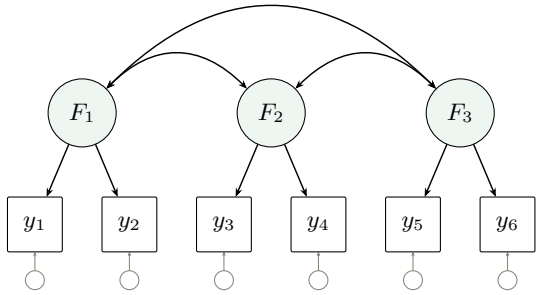
(ii) $B \preceq A$. Fix the direct effect $X \to Y$ in A to 0 (Lemma R). □

In the game: Partial mediation (A) against full mediation (B). Full mediation is partial mediation with the direct effect X → Y fixed to zero.

M13 A residual covariance as a factor

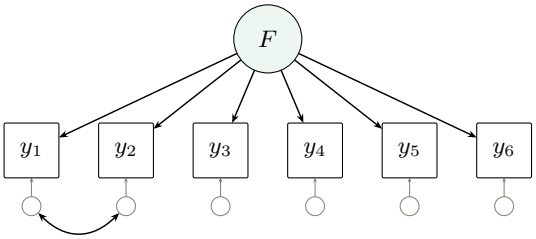
Level: Medium. Model families: CFA. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 15$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F1 ~~ F2 + F3; F2
~~ F3
```

Model B $q_B = 13$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6; y1 ~~ y2
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 15 at the rational point listed in Appendix B, while model B has only $q_B = 13$ free parameters.

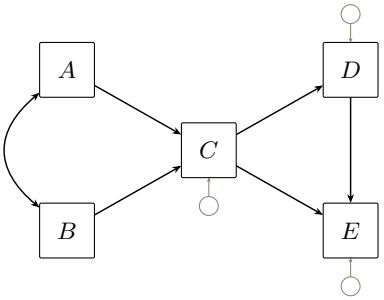
(ii) $B \preceq A$. By Lemma P, $B \equiv F\{y_3, \ldots, y_6\} \to G\{y_1, y_2\}$, which by Lemma B(b) is the two-factor model $\{y_1, y_2\} \,|\, \{y_3, \ldots, y_6\}$ with correlated factors. Merging F_2 and F_3 of A gives that model (Lemma M). Hence $B \preceq A$. □

In the game: A residual covariance between y1 and y2 acts like a separate two-indicator factor for them. So B is a two-factor model, which in turn is the three-factor model with F2 and F3 merged.

M14 Dropping one path

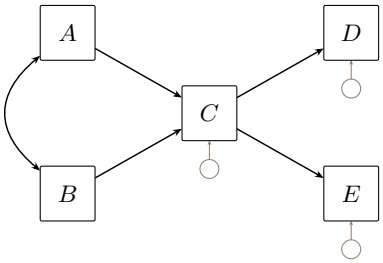
Level: Medium. Model families: Path analysis. Observed variables: A, B, C, D, E ($p = 5$, 15 moments).

Model A $q_A = 11$




```
A ~~ B; C ~ A + B; D ~ C; E ~ C + D
```

Model B $q_B = 10$




```
A ~~ B; C ~ A + B; D ~ C; E ~ C
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 11 at the rational point listed in Appendix B, while model B has only $q_B = 10$ free parameters.

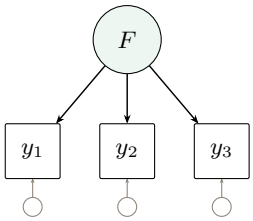
(ii) $B \preceq A$. Fix $\beta_{ED} = 0$ in A (Lemma R). □

In the game: B is A without the path D → E. Fix that coefficient to zero and A becomes B.

M15 A chain inside a saturated factor model

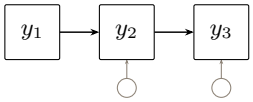
Level: Medium. Model families: CFA, Path analysis. Observed variables: y_1, y_2, y_3 ($p = 3$, 6 moments).

Model A $q_A = 6$




```
F =~ y1 + y2 + y3
```

Model B $q_B = 5$




```
y2 ~ y1; y3 ~ y2
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 6 at the rational point listed in Appendix B, while model B has only $q_B = 5$ free parameters.

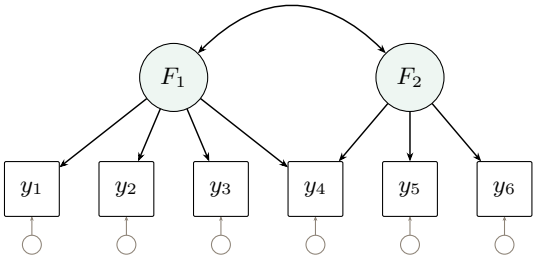
(ii) $B \preceq A$. The one-factor model with three indicators is saturated (Lemma B(a), see E05). □

In the game: The one-factor model with three indicators is saturated, so it contains every model for these three variables, including the chain y1 → y2 → y3.

M16 Cross-loadings on different items

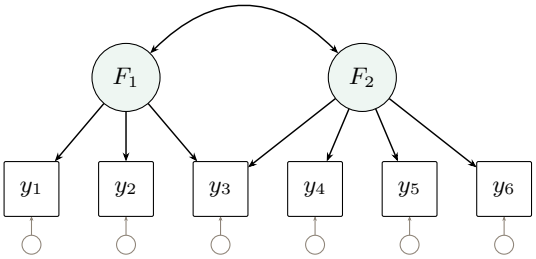
Level: Medium. Model families: CFA. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F1 =~ y1 + y2 + y3 + y4; F2 =~ y4 + y5 + y6; F1 ~~ F2
```

Model B $q_B = 14$




```
F1 =~ y1 + y2 + y3; F2 =~ y3 + y4 + y5 + y6; F1 ~~ F2
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{56}\sigma_{24} - \sigma_{46}\sigma_{25}$. In B, y_4, y_5, y_6 load only on F_2 and y_2 only on F_1, so the four covariances are $\lambda\lambda\psi$ products that cancel. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \lambda_2\lambda_4\lambda_5\lambda_6(\psi_{11}\psi_{22} - \psi_{12}^2) \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 1.172 \neq 0$.

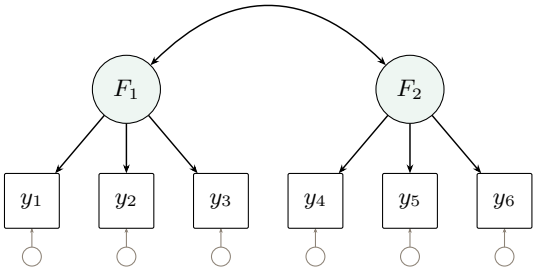
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{12}\sigma_{35} - \sigma_{13}\sigma_{25}$. In A, y_1, y_2, y_3 load only on F_1 and y_5 only on F_2. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = 0.8880 \neq 0$. □

In the game: The cross-loading sits on a different item (y4 in A, y3 in B). Each model has a loading that the other fixes to zero.

M17 Different splits of the items

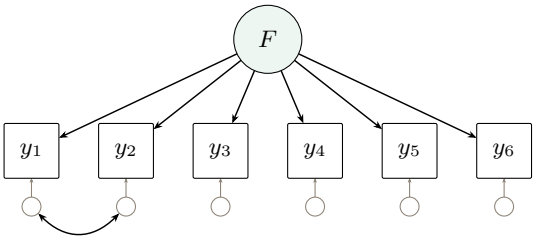
Level: Medium. Model families: CFA. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 13$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2
```

Model B $q_B = 13$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6; y1 ~~ y2
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{13}\sigma_{45} - \sigma_{14}\sigma_{35}$. In B these covariances do not involve θ_{12}, so they are $\lambda_i\lambda_j\psi$ and the tetrad vanishes. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \lambda_1\lambda_3\lambda_4\lambda_5(\psi_{11}\psi_{22} - \psi_{12}^2) \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 1.199 \neq 0$.

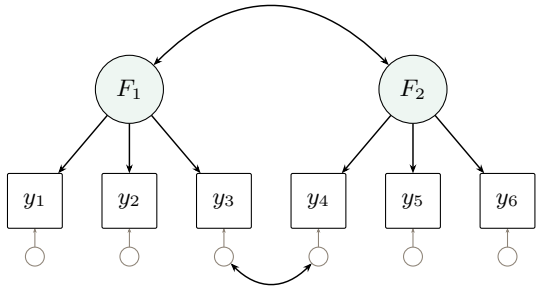
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{12}\sigma_{34} - \sigma_{14}\sigma_{23}$. In A: $\lambda_1\lambda_2\psi_{11} \cdot \lambda_3\lambda_4\psi_{12} - \lambda_1\lambda_4\psi_{12} \cdot \lambda_2\lambda_3\psi_{11} = 0$. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B, $f = \theta_{12}\lambda_3\lambda_4\psi \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.3240 \neq 0$. □

In the game: B behaves like a two-factor model with y1, y2 as a separate factor; A splits the items as y1, y2, y3 and y4, y5, y6. Different splits: neither is nested.

M18 Residual covariance versus cross-loading

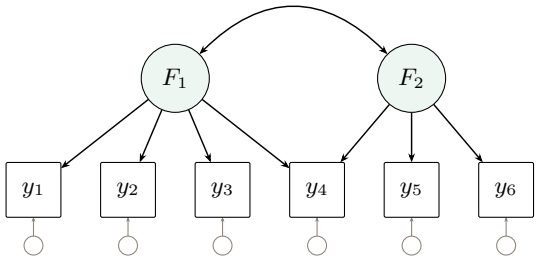
Level: Medium. Model families: CFA. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2; y3 ~~ y4
```

Model B $q_B = 14$




```
F1 =~ y1 + y2 + y3 + y4; F2 =~ y4 + y5 + y6; F1 ~~ F2
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{34}\sigma_{15} - \sigma_{14}\sigma_{35}$. In B, y_4's covariances with y_1 and y_3 are both $\lambda_i(\lambda_4\psi_{12} + \kappa\psi_{11})$, so the rows of y_1 and y_3 stay proportional. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \theta_{34}\lambda_1\lambda_5\psi_{12} \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.07200 \neq 0$.

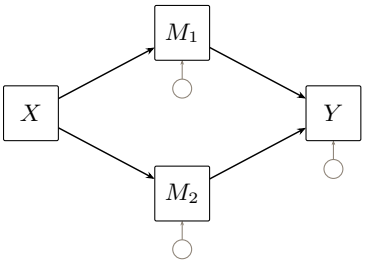
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{14}\sigma_{56} - \sigma_{15}\sigma_{46}$. In A, y_4 loads only on F_2 (the residual covariance only touches σ_{34}). So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B, $f = \lambda_1\lambda_5\lambda_6\kappa(\psi_{11}\psi_{22} - \psi_{12}^2) \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 1.172 \neq 0$. □

In the game: A residual covariance only adds covariance between y3 and y4. A cross-loading changes the covariances of y4 with all items. Same number of parameters, different restrictions.

M19 Parallel versus serial mediation

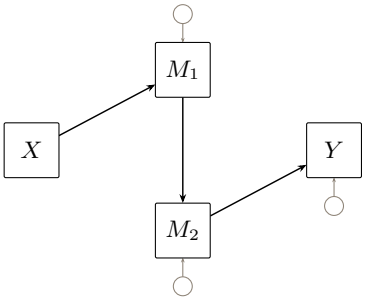
Level: Medium. Model families: Path analysis. Observed variables: X, M_1, M_2, Y ($p = 4$, 10 moments).

Model A $q_A = 8$




```
M1 ~ X; M2 ~ X; Y ~ M1 + M2
```

Model B $q_B = 7$




```
M1 ~ X; M2 ~ M1; Y ~ M2
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 8 at the rational point listed in Appendix B, while model B has only $q_B = 7$ free parameters.

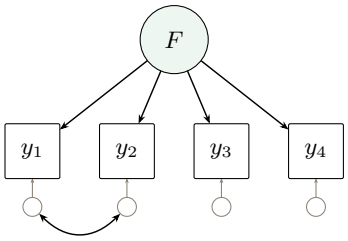
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{12}\sigma_{XX} - \sigma_{X1}\sigma_{X2}$. In A, M_1 and M_2 are independent given X (their only common cause). So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = 0.7200 \neq 0$. □

In the game: Parallel mediators (A: M1 and M2 independent given X) against serial mediation (B: M1 → M2). A has more parameters but still cannot produce B.

M20 Residual covariance on different pairs

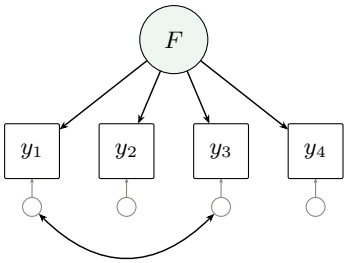
Level: Medium. Model families: CFA. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 9$




```
F =~ y1 + y2 + y3 + y4; y1 ~~ y2
```

Model B $q_B = 9$




```
F =~ y1 + y2 + y3 + y4; y1 ~~ y3
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{12}\sigma_{34} - \sigma_{14}\sigma_{23}$. In B only σ_{13} contains a residual covariance; this tetrad avoids it. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \theta_{12}\lambda_3\lambda_4\psi \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.3240 \neq 0$.

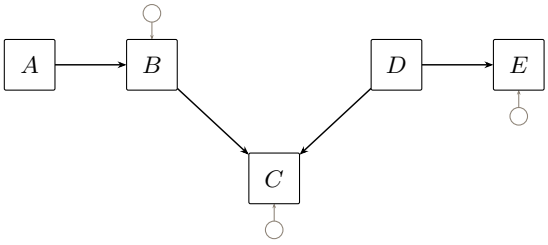
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{13}\sigma_{24} - \sigma_{14}\sigma_{23}$. In A only σ_{12} contains a residual covariance; this tetrad avoids it. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B, $f = \theta_{13}\lambda_2\lambda_4\psi \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.3600 \neq 0$. □

In the game: Each model keeps a different tetrad: A implies $\sigma 13{\cdot}\sigma 24 = \sigma 14{\cdot}\sigma 23$, B implies $\sigma 12{\cdot}\sigma 34 = \sigma 14{\cdot}\sigma 23$. Neither model satisfies the other's restriction.

M21 Same skeleton, same collider

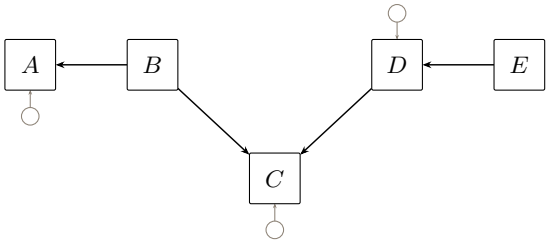
Level: Medium. Model families: Path analysis. Observed variables: A, B, C, D, E ($p = 5$, 15 moments).

Model A $q_A = 9$




```
B ~ A; C ~ B + D; E ~ D
```

Model B $q_B = 9$




```
A ~ B; C ~ B + D; D ~ E
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. In both models $C = \beta_{CB}B + \beta_{CD}D + \zeta_C$, the pairs (A, B) and (D, E) are not connected to each other, and (A, B) is connected to the rest only through $B \to C$, (D, E) only through $D \to C$. The models differ only in the direction of $A - B$ and $D - E$, and either direction reproduces every 2×2 covariance matrix of the pair (as in E03). Hence Lemma B(b), applied to the block (A, B) and then to (D, E), gives $A \equiv B$. (This is the rule of Verma & Pearl, 1990: same skeleton, same colliders.)

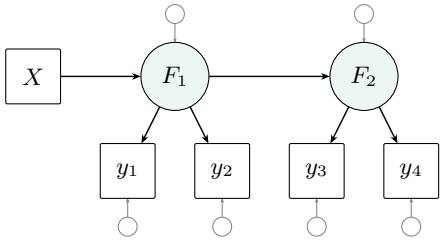
(ii) $B \preceq A$. See above. □

In the game: Same skeleton, same collider (B → C ← D). Turning A — B or D — E around changes no independence claim: A depends on the rest only through B, E only through D. Equivalent (the rule of Verma & Pearl).

M22 Mediator or common cause

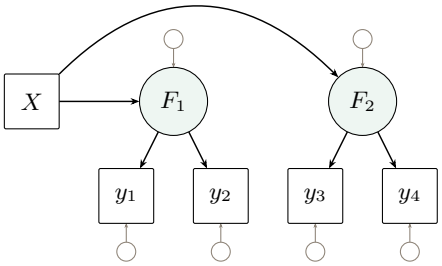
Level: Medium. Model families: SEM. Observed variables: X, y_1, y_2, y_3, y_4 ($p = 5$, 15 moments).

Model A $q_A = 11$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F1 ~ X; F2 ~ F1
```

Model B $q_B = 11$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F1 ~ X; F2 ~ X
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{13}\sigma_{XX} - \sigma_{X1}\sigma_{X3}$. In B, F_1 and F_2 are related only through X, so $\mathrm{Cov}(F_1, F_2)\mathrm{Var}(X) = \mathrm{Cov}(X, F_1)\mathrm{Cov}(X, F_2)$; with $\sigma_{y_1 y_3} = \lambda_1\lambda_3\mathrm{Cov}(F_1, F_2)$ and $\sigma_{X y_k} = \lambda_k \mathrm{Cov}(X, F_{f(k)})$ this is $f = 0$. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A at the point of Appendix B, $f = 0.6000 \neq 0$.

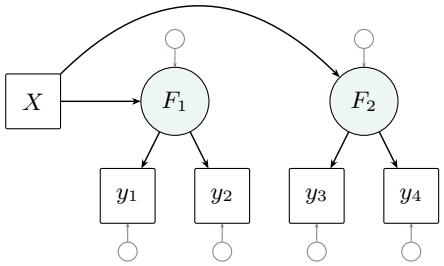
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{X3}\sigma_{12} - \sigma_{X1}\sigma_{23}$. In A ($X \to F_1 \to F_2$), X and F_2 are independent given F_1: $\mathrm{Cov}(X, F_2)\mathrm{Var}(F_1) = \mathrm{Cov}(X, F_1)\mathrm{Cov}(F_1, F_2)$, and f is $\lambda_1\lambda_2\lambda_3$ times this difference. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = 0.6000 \neq 0$. □

In the game: A: X affects F2 only through F1 (X and F2 independent given F1). B: X is a common cause of F1 and F2 (F1 and F2 independent given X). Different claims, so neither is nested.

M23 A disturbance covariance

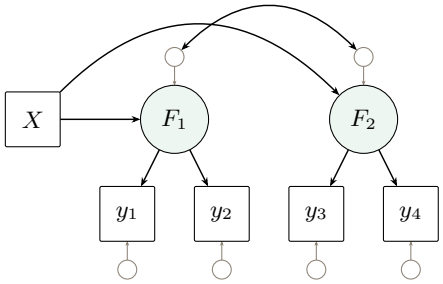
Level: Medium. Model families: SEM. Observed variables: X, y_1, y_2, y_3, y_4 ($p = 5$, 15 moments).

Model A $q_A = 11$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F1 ~ X; F2 ~ X
```

Model B $q_B = 12$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F1 ~ X; F2 ~ X; F1 ~~ F2
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Fix the covariance of the disturbances of F_1 and F_2 in B to 0 (Lemma R).

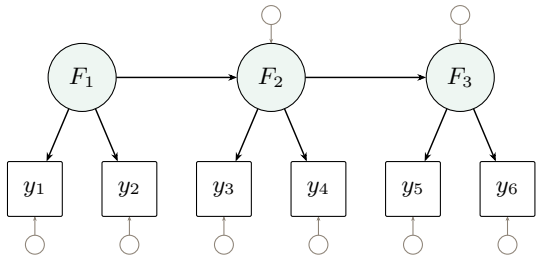
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 12 at the rational point listed in Appendix B, while model A has only $q_A = 11$ free parameters. □

In the game: B adds a covariance between the disturbances of F1 and F2 (something they share besides X). Fix it to zero and you get A.

M24 Latent chain or latent fork

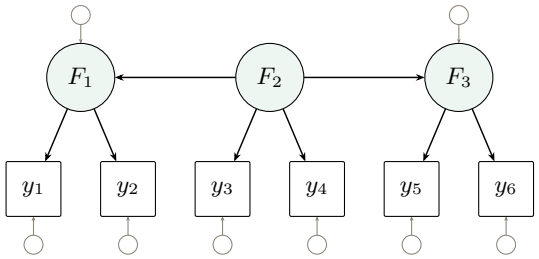
Level: Medium. Model families: SEM. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F2 ~ F1; F3 ~ F2
```

Model B $q_B = 14$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F1 ~ F2; F3 ~ F2
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Same measurement part. The structural parts are the chain $F_1 \to F_2 \to F_3$ and the fork $F_1 \leftarrow F_2 \to F_3$: same skeleton, no collider, so by Lemma C (applied to the latent variables) both produce exactly the Ψ with $(\Psi^{-1})_{13} = 0$. Lemma B(b).

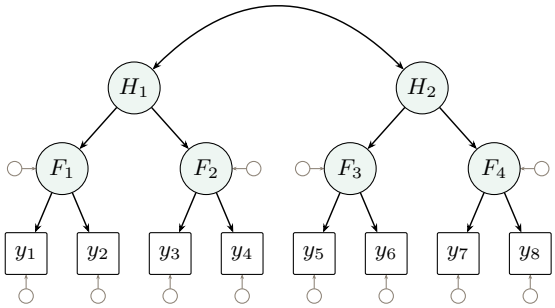
(ii) $B \preceq A$. See above. □

In the game: A latent chain F1 → F2 → F3 and a fork F1 ← F2 → F3 both say only that F1 and F3 are independent given F2. Same claim, so the models are equivalent (the chain-or-fork rule, one level up).

M25 One or two second-order factors

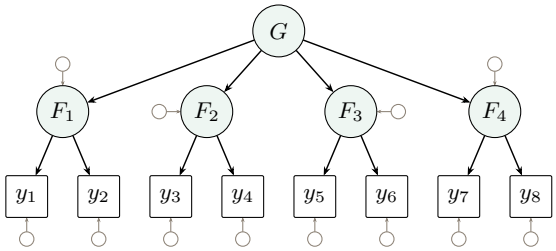
Level: Medium. Model families: Higher-order. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 21$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; H1
=~ F1 + F2; H2 =~ F3 + F4; H1 ~~ H2
```

Model B $q_B = 20$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; G
=~ F1 + F2 + F3 + F4
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 21 at the rational point listed in Appendix B, while model B has only $q_B = 20$ free parameters.

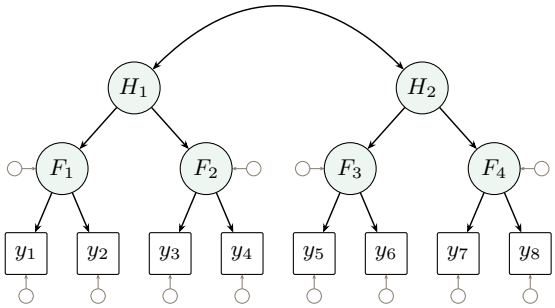
(ii) $B \preceq A$. In A the covariance matrix of (H_1, H_2) is free. Lemma M one level up (the first-order factors play the role of indicators): with $\psi_{H_1} = \psi_{H_2} = \psi_{H_1 H_2}$, $H_1 = H_2$ and A becomes the model with one second-order factor on $F_1, \ldots, F_4$, which is B. □

In the game: Two second-order factors that correlate perfectly act as one: merge H1 and H2 and you get B. So one second-order factor is nested in two correlated ones.

M26 Two second-order groupings

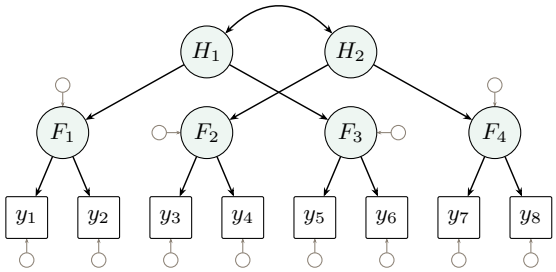
Level: Medium. Model families: Higher-order. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 21$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; H1
=~ F1 + F2; H2 =~ F3 + F4; H1 ~~ H2
```

Model B $q_B = 21$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; H1
=~ F1 + F3; H2 =~ F2 + F4; H1 ~~ H2
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{13}\sigma_{57} - \sigma_{17}\sigma_{35}$. With the marker indicators y_1, y_3, y_5, y_7 of $F_1, \ldots, F_4$, $\sigma_{y_{2k-1}y_{2l-1}} = \psi_{kl}$. In B the factors are grouped $\{F_1, F_3\}$, $\{F_2, F_4\}$, so the covariances between the groups have rank one: $\psi_{12}\psi_{34} = \psi_{14}\psi_{32}$. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \gamma_1\gamma_2\gamma_3\gamma_4(\phi_{11}\phi_{22} - \phi_{12}^2) \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.2430 \neq 0$.

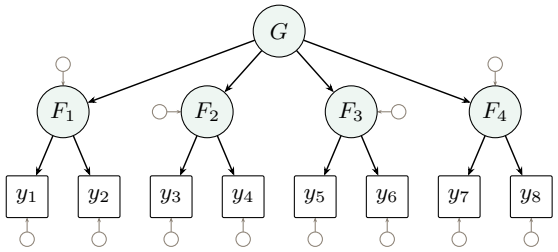
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{15}\sigma_{37} - \sigma_{17}\sigma_{35}$. In A the groups are $\{F_1, F_2\}$, $\{F_3, F_4\}$: $\psi_{13}\psi_{24} = \psi_{14}\psi_{23}$. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = 0.2430 \neq 0$. □

In the game: Same number of parameters, different grouping of the first-order factors. Each grouping implies its own tetrad among the factor covariances, which the other model violates.

M27 A disturbance covariance between first-order factors

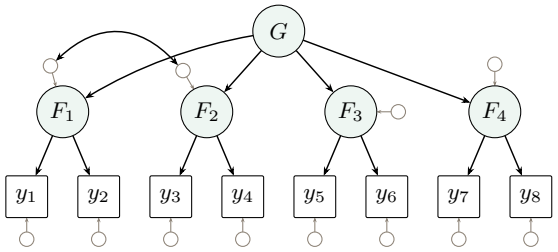
Level: Medium. Model families: Higher-order. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 20$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; G
=~ F1 + F2 + F3 + F4
```

Model B $q_B = 21$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; G
=~ F1 + F2 + F3 + F4; F1 ~~ F2
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Fix the covariance of the disturbances of F_1 and F_2 in B to 0 (Lemma R).

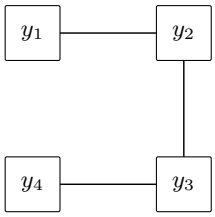
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 21 at the rational point listed in Appendix B, while model A has only $q_A = 20$ free parameters. □

In the game: B also frees the covariance between the disturbances of F1 and F2. Fix it to zero and you are back at A.

M28 Path network versus star network

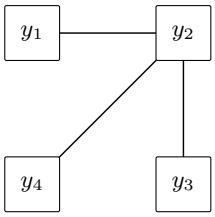
Level: Medium. Model families: GGM. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 7$




```
network y1 y2 y3 y4; y1 - y2; y2 - y3; y3 - y4
```

Model B $q_B = 7$




```
network y1 y2 y3 y4; y2 - y1 + y3 + y4
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \det \begin{pmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{41} & \sigma_{42} & \sigma_{43} \end{pmatrix}$. This is the minor of Σ without row 3 and column 4. By Lemma I it vanishes iff $(\Sigma^{-1})_{34} = 0$, which holds in B (no edge $y_3 - y_4$). So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A at the point of Appendix B, $f = 0.01346 \neq 0$.

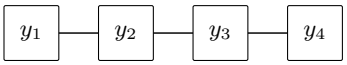
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \det \begin{pmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \\ \sigma_{41} & \sigma_{42} & \sigma_{43} \end{pmatrix}$. The minor without row 2 and column 4: zero iff $(\Sigma^{-1})_{24} = 0$, which holds in A (no edge $y_2 - y_4$). So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = -0.01350 \neq 0$. □

In the game: Both networks have three edges. A has no edge y2 — y4 (independent given y1 and y3), B has no edge y3 — y4. Each sets a partial correlation to zero that the other leaves free.

M29 A path network is a chain

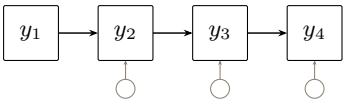
Level: Medium. Model families: GGM. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 7$




```
network y1 y2 y3 y4; y1 - y2; y2 - y3; y3 - y4
```

Model B $q_B = 7$




```
y2 ~ y1; y3 ~ y2; y4 ~ y3
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Lemma C: the directed chain has skeleton $y_1 - y_2 - y_3 - y_4$ and no collider, so it equals the GGM on that path.

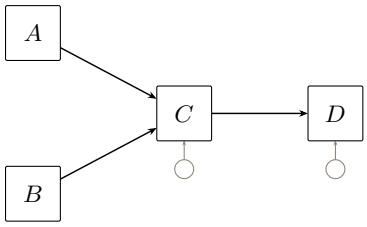
(ii) $B \preceq A$. See above. □

In the game: A Gaussian graphical model (GGM) whose graph is a path has the same conditional independencies as the directed chain: both are exactly the covariance matrices with a tridiagonal inverse.

M30 A DAG inside its moral graph

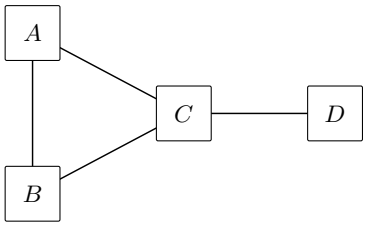
Level: Medium. Model families: GGM. Observed variables: A, B, C, D ($p = 4$, 10 moments).

Model A $q_A = 7$




```
C ~ A + B; D ~ C
```

Model B $q_B = 8$




```
network A B C D; A - B + C; B - C; C - D
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. In A, D is independent of $\{A, B\}$ given C (local Markov property), so $(\Sigma^{-1})_{AD} = (\Sigma^{-1})_{BD} = 0$: $\Sigma \in \mathcal{M}_B$.

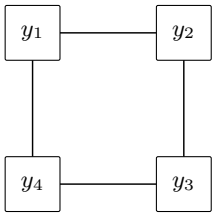
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 8 at the rational point listed in Appendix B, while model A has only $q_A = 7$ free parameters. □

In the game: B is the moral graph of the DAG in A (marry the parents of C). Every conditional independence in the GGM also holds in the DAG, but the DAG adds one: A and B are marginally independent.

M31 A chain inside a cycle network

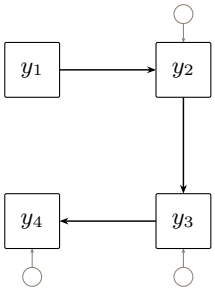
Level: Medium. Model families: GGM. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 8$




```
network y1 y2 y3 y4; y1 - y2; y2 - y3; y3 - y4; y4 - y1
```

Model B $q_B = 7$




```
y2 ~ y1; y3 ~ y2; y4 ~ y3
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 8 at the rational point listed in Appendix B, while model B has only $q_B = 7$ free parameters.

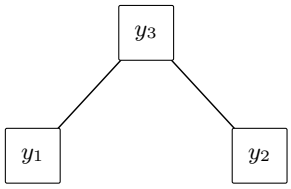
(ii) $B \preceq A$. By Lemma C, B is the GGM on the path $y_1 - y_2 - y_3 - y_4$, which is A with the edge $y_4 - y_1$ fixed to 0 (Lemma R). □

In the game: The directed chain equals the GGM with the path y1 — y2 — y3 — y4. The cycle network adds the edge y4 — y1, so it contains the chain.

M32 Undirected path versus collider

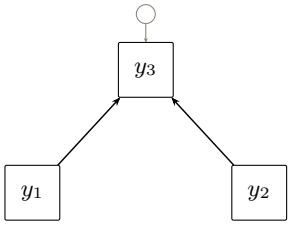
Level: Medium. Model families: GGM. Observed variables: y_1, y_2, y_3 ($p = 3$, 6 moments).

Model A $q_A = 5$




```
network y1 y2 y3; y1 - y3; y3 - y2
```

Model B $q_B = 5$




```
y3 ~ y1 + y2
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{12}$. In B, y_1 and y_2 are exogenous and unconnected. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A at the point of Appendix B, $f = 0.01929 \neq 0$.

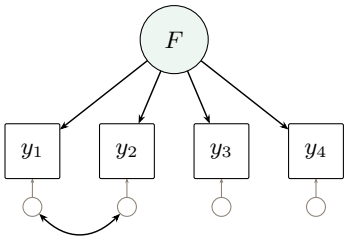
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{12}\sigma_{33} - \sigma_{13}\sigma_{23}$. In A, $k_{12} = 0$ (Lemma I). So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B, $f = -\sigma_{13}\sigma_{23} \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = -0.3600 \neq 0$. □

In the game: The network says y1 and y2 are independent given y3; the collider says they are independent without conditioning (and dependent given y3).

H01 Equivalent residual covariances

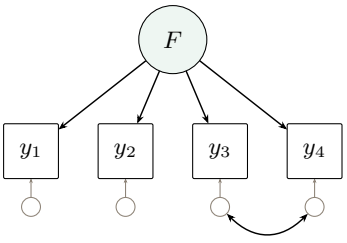
Level: Hard. Model families: CFA. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 9$




```
F =~ y1 + y2 + y3 + y4; y1 ~~ y2
```

Model B $q_B = 9$




```
F =~ y1 + y2 + y3 + y4; y3 ~~ y4
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. By Lemma P, $A \equiv F\{y_3, y_4\} \to G\{y_1, y_2\}$ and $B \equiv F\{y_1, y_2\} \to G\{y_3, y_4\}$; by Lemma B(b) both are the two-factor model $\{y_1, y_2\}|\{y_3, y_4\}$ with correlated factors. Hence $A \equiv B$; both equal $\{\Sigma \succ 0 : \sigma_{13}\sigma_{24} = \sigma_{14}\sigma_{23}\}$.

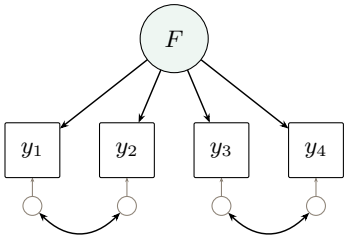
(ii) $B \preceq A$. See above. □

In the game: Both are the same as a two-factor model y1, y2 | y3, y4: either residual covariance frees the covariance within one pair, and the factor variance takes care of the other pair. Both imply only $\sigma 13 \cdot \sigma 24 = \sigma 14 \cdot \sigma 23$.

H02 A counting trap

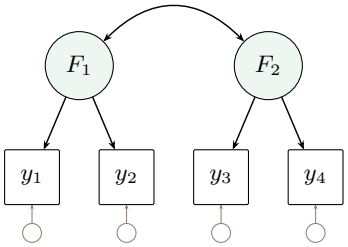
Level: Hard. Model families: CFA. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 10$




```
F =~ y1 + y2 + y3 + y4; y1 ~~ y2; y3 ~~ y4
```

Model B $q_B = 9$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F1 ~~ F2
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Apply Lemma P to both pairs: $A \equiv G_1\{y_1, y_2\} \leftarrow F \rightarrow G_2\{y_3, y_4\}$ with F without indicators. The observed covariances depend on the latent block (G_1, G_2) only, which is saturated ($\mathrm{Cov}(G_1, G_2) = \gamma_1\gamma_2\psi_F$, variances free). By Lemma B(b), $A \equiv B$. Note $q_A = 10$ but $\dim \mathcal{M}_A = 9$: A is not identified.

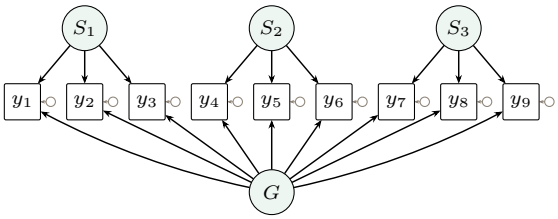
(ii) $B \preceq A$. See above. □

In the game: Counting trap: A has 10 parameters for 10 moments but is not identified, and still implies $\sigma 13 \cdot \sigma 24 = \sigma 14 \cdot \sigma 23$. It is exactly as restrictive as the two-factor model B.

H03 A three-indicator group factor equals a free residual block

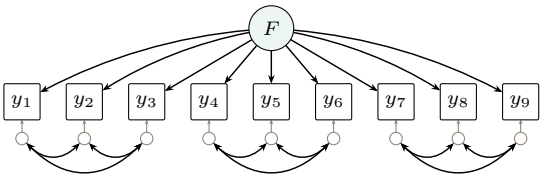
Level: Hard. Model families: Bifactor. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9$ ($p = 9$, 45 moments).

Model A $q_A = 27$




```
S1 =~ y1 + y2 + y3; S2 =~ y4 + y5 + y6; S3 =~ y7 + y8 + y9; G
=~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8 + y9
```

Model B $q_B = 27$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8 + y9; y1 ~~ y2 + y3;
y2 ~~ y3; y4 ~~ y5 + y6; y5 ~~ y6; y7 ~~ y8 + y9; y8 ~~ y9
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. In A, group factor S_k adds $s_i s_j \psi_{S_k}$ to σ_{ij} for $i \neq j$ in group k, and $s_i^2 \psi_{S_k}$ to σ_{ii}. Map to B: same general loadings, $\theta_{ij} = s_i s_j \psi_{S_k}$, $\theta_{ii} = \theta_{ii}^A + s_i^2 \psi_{S_k}$.

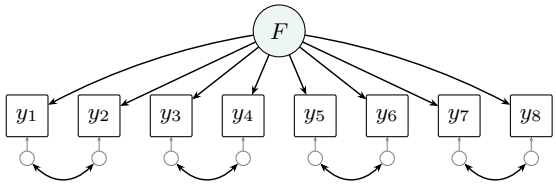
(ii) $B \preceq A$. Conversely, in group $k = \{a, b, c\}$ with $\theta_{ab}\theta_{ac}\theta_{bc} \neq 0$: $s_a = 1$, $\psi_{S_k} = \theta_{ab}\theta_{ac}/\theta_{bc}$, $s_b = \theta_{bc}/\theta_{ac}$, $s_c = \theta_{bc}/\theta_{ab}$ (Lemma B(a) on the residual block), residual variances adjusted. Lemma G. □

In the game: A group factor with three indicators has three loadings for the three covariances within its group: it can produce any pattern there. That is exactly what three free residual covariances do.

H04 One factor with paired residuals equals a higher-order model

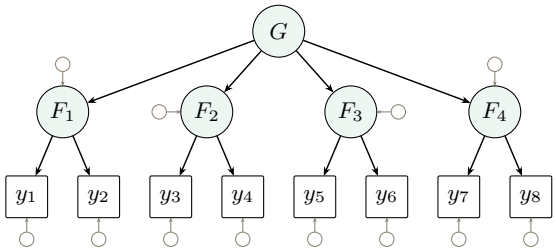
Level: Hard. Model families: Higher-order. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 20$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8; y1 ~~ y2; y3 ~~ y4;
y5 ~~ y6; y7 ~~ y8
```

Model B $q_B = 20$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; G
=~ F1 + F2 + F3 + F4
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Lemma P applied to the four pairs: $A \equiv G_k\{y_{2k-1}, y_{2k}\} \leftarrow F$ ($k = 1, \ldots, 4$) with F without indicators. That is exactly the higher-order model B with second-order factor F.

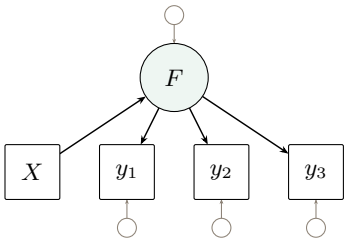
(ii) $B \preceq A$. See above. □

In the game: In both models all covariances between pairs follow one general factor, and the covariance within each pair is free (a residual covariance in A, a first-order factor variance in B). Same 20 free parameters, same restrictions.

H05 Covariate or indicator

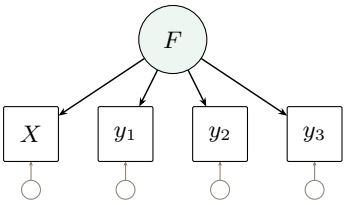
Level: Hard. Model families: SEM. Observed variables: X, y_1, y_2, y_3 ($p = 4$, 10 moments).

Model A $q_A = 8$




```
F =~ y1 + y2 + y3; F ~ X
```

Model B $q_B = 8$




```
F =~ X + y1 + y2 + y3
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Given A ($F = \gamma X + \zeta$, $v = \mathrm{Var}(X)$): put $\psi = \gamma^2 v + \psi_\zeta$, keep λ_i, and $\lambda_X = \gamma v/\psi$, $\theta_X = v - \lambda_X^2 \psi$. Then $\sigma_{X,i} = \lambda_X \lambda_i \psi = \gamma v \lambda_i$ and $\sigma_{ij} = \lambda_i \lambda_j \psi$ as in A.

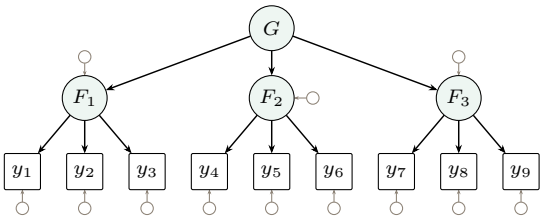
(ii) $B \preceq A$. Given B: $v = \sigma_{XX}$, $\gamma = \lambda_X \psi / v$, $\psi_\zeta = \psi - \gamma^2 v$ (Lemma G; $v \neq 0$). □

In the game: A covariate that predicts the factor (A) cannot be told apart from one more indicator of that factor (B). Both say that X relates to the items only through F.

H06 Higher-order inside bifactor

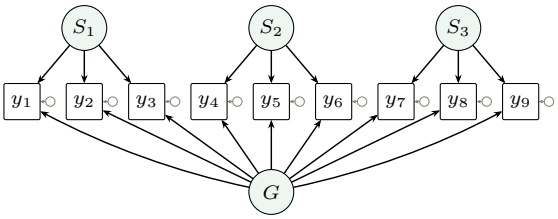
Level: Hard. Model families: Bifactor. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9$ ($p = 9$, 45 moments).

Model A $q_A = 21$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F3 =~ y7 + y8 + y9; G
=~ F1 + F2 + F3
```

Model B $q_B = 27$




```
S1 =~ y1 + y2 + y3; S2 =~ y4 + y5 + y6; S3 =~ y7 + y8 + y9; G
=~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8 + y9
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Lemma H (Schmid–Leiman; Yung, Thissen & McLeod, 1999).

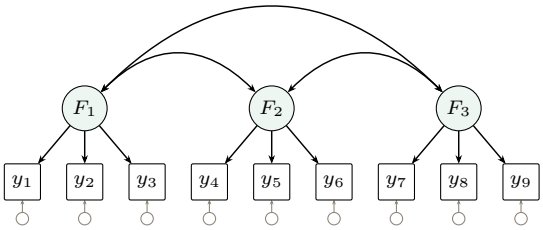
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 27 at the rational point listed in Appendix B, while model A has only $q_A = 21$ free parameters. □

In the game: Schmid–Leiman: a higher-order model is a bifactor model whose general and group loadings are proportional within each group (Yung, Thissen & McLeod, 1999). The bifactor model drops those proportionality constraints.

H07 Correlated factors inside bifactor

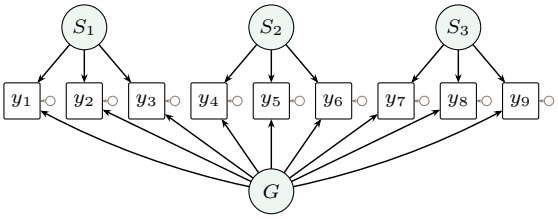
Level: Hard. Model families: Bifactor. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9$ ($p = 9$, 45 moments).

Model A $q_A = 21$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F3 =~ y7 + y8 + y9; F1
~~ F2 + F3; F2 ~~ F3
```

Model B $q_B = 27$




```
S1 =~ y1 + y2 + y3; S2 =~ y4 + y5 + y6; S3 =~ y7 + y8 + y9; G
=~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8 + y9
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. By M02, three correlated factors are equivalent to the higher-order model, which is nested in the bifactor model (Lemma H).

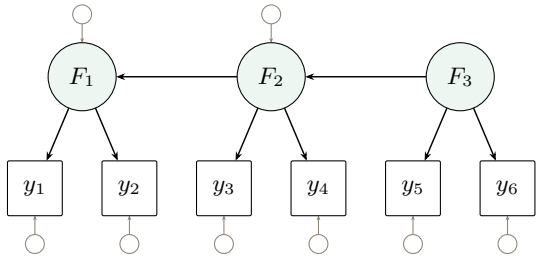
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 27 at the rational point listed in Appendix B, while model A has only $q_A = 21$ free parameters. □

In the game: Three correlated factors are equivalent to a higher-order model (just-identified), and a higher-order model is nested in the bifactor model. So the correlated-factors model is nested in the bifactor model.

H08 Arrows the other way, still nested

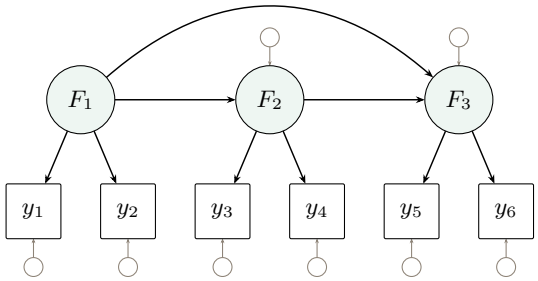
Level: Hard. Model families: SEM. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F2 ~ F3; F1 ~ F2
```

Model B $q_B = 15$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F2 ~ F1; F3 ~ F1 +
F2
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. The structural part of B is a complete recursive model on (F_1, F_2, F_3), which reproduces every $\Psi \succ 0$ (Lemma B(a)), including the Ψ of A's chain $F_3 \to F_2 \to F_1$. Lemma B(b).

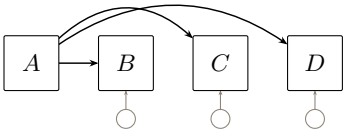
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 15 at the rational point listed in Appendix B, while model A has only $q_A = 14$ free parameters. □

In the game: The arrows in B point the other way, but B's latent block is saturated: it reproduces any covariance matrix of the three factors, including the one implied by A's chain.

H09 A perfect indicator

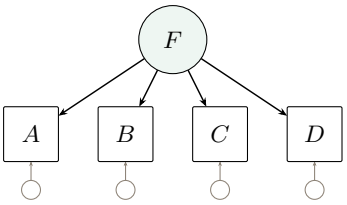
Level: Hard. Model families: CFA, Path analysis. Observed variables: A, B, C, D ($p = 4$, 10 moments).

Model A $q_A = 7$




```
B ~ A; C ~ A; D ~ A
```

Model B $q_B = 8$




```
F =~ A + B + C + D
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Lemma T with A as the perfect indicator: set $\lambda_A = 1$, $\theta_A = 0$, $\psi = \mathrm{Var}(A)$ and $\lambda_k = \beta_{kA}$, $\theta_k = \psi_k$ $(k = B, C, D)$.

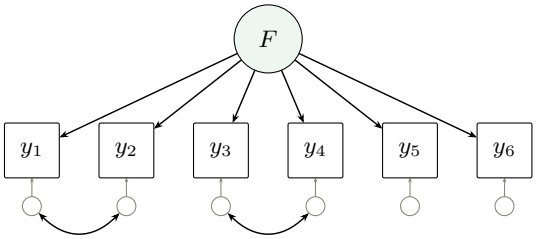
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 8 at the rational point listed in Appendix B, while model A has only $q_A = 7$ free parameters. □

In the game: Let A be a perfect indicator of F (residual variance 0): then F is A, and the factor model becomes the path model with A as the common cause. A is nested in B.

H10 Paired residuals inside three factors

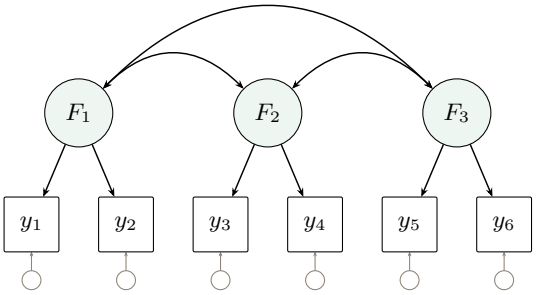
Level: Hard. Model families: CFA. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6; y1 ~~ y2; y3 ~~ y4
```

Model B $q_B = 15$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F1 ~~ F2 + F3; F2
~~ F3
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Lemma P twice: $A \equiv G_1\{y_1, y_2\} \leftarrow F\{y_5, y_6\} \rightarrow G_2\{y_3, y_4\}$, a three-factor model with a particular latent covariance matrix. In B, Ψ is free (Lemma B(b)).

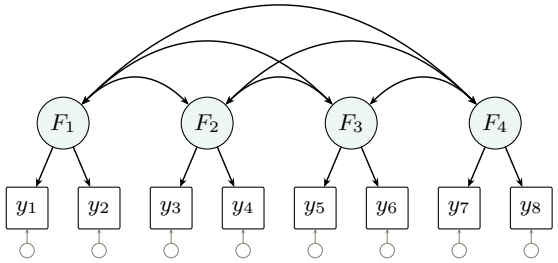
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 15 at the rational point listed in Appendix B, while model A has only $q_A = 14$ free parameters. □

In the game: Each residual covariance within a pair acts like a two-indicator factor. A is a one-factor model with two such pairs; the three-factor model B also frees the third pair.

H11 Paired residuals inside four correlated factors

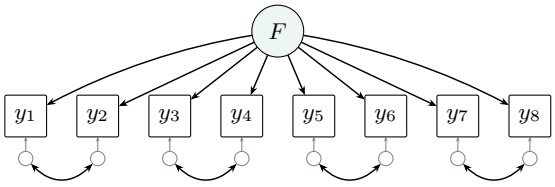
Level: Hard. Model families: CFA. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 22$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; F1
~~ F2 + F3 + F4; F2 ~~ F3 + F4; F3 ~~ F4
```

Model B $q_B = 20$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8; y1 ~~ y2; y3 ~~ y4;
y5 ~~ y6; y7 ~~ y8
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 22 at the rational point listed in Appendix B, while model B has only $q_B = 20$ free parameters.

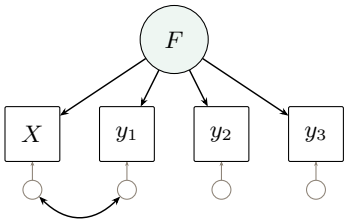
(ii) $B \preceq A$. By H04, B is equivalent to the higher-order model over the four pair factors, and that model is nested in A, where Ψ is free (Lemma B(b)). □

In the game: B is equivalent to a second-order model over four two-indicator factors (one general factor plus a free covariance within each pair), and that is nested in four freely correlated factors.

H12 A covariate inside a factor model with one residual covariance

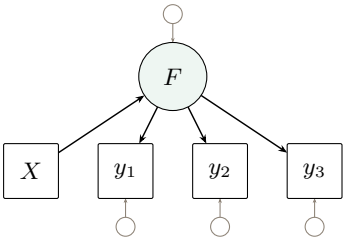
Level: Hard. Model families: SEM. Observed variables: X, y_1, y_2, y_3 ($p = 4$, 10 moments).

Model A $q_A = 9$




```
F =~ X + y1 + y2 + y3; X ~~ y1
```

Model B $q_B = 8$




```
F =~ y1 + y2 + y3; F ~ X
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 9 at the rational point listed in Appendix B, while model B has only $q_B = 8$ free parameters.

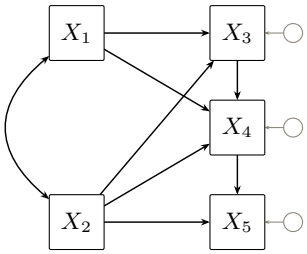
(ii) $B \preceq A$. By H05, B is equivalent to the one-factor model with indicators X, y_1, y_2, y_3, which is A with $\theta_{X1} = 0$ (Lemma R). □

In the game: B (X predicts F) is equivalent to the one-factor model with X as fourth indicator. A is that same model plus a residual covariance, so B is nested in A.

H13 Replacing rule plus one restriction

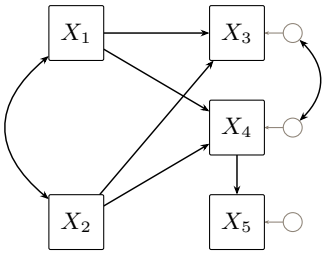
Level: Hard. Model families: Path analysis. Observed variables: X_1, X_2, X_3, X_4, X_5 ($p = 5$, 15 moments).

Model A $q_A = 13$




```
X3 ~ X1 + X2; X4 ~ X1 + X2 + X3; X5 ~ X4 + X2; X1 ~~ X2
```

Model B $q_B = 12$




```
X3 ~ X1 + X2; X4 ~ X1 + X2; X5 ~ X4; X1 ~~ X2; X3 ~~ X4
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 13 at the rational point listed in Appendix B, while model B has only $q_B = 12$ free parameters.

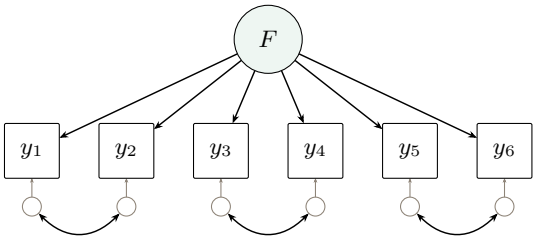
(ii) $B \preceq A$. As in M04 (with $C = X_3$, $D = X_4$; the equation of X_5 is unchanged), replacing $X_3 \to X_4$ by the residual covariance $X_3 \leftrightarrow X_4$ gives a model equivalent to A without $X_2 \to X_5$, which is nested in A (Lemma R). □

In the game: Swapping X3 → X4 for the residual covariance X3 ↔ X4 is allowed (same predictors), so that change alone gives an equivalent model. B also drops X2 → X5, which makes it nested in A.

H14 A two-indicator factor inside residual pairs

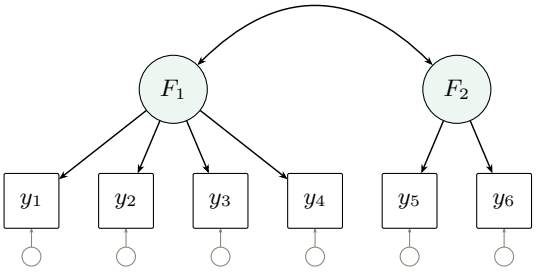
Level: Hard. Model families: CFA. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 15$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6; y1 ~~ y2; y3 ~~ y4; y5 ~~ y6
```

Model B $q_B = 13$




```
F1 =~ y1 + y2 + y3 + y4; F2 =~ y5 + y6; F1 ~~ F2
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 15 at the rational point listed in Appendix B, while model B has only $q_B = 13$ free parameters.

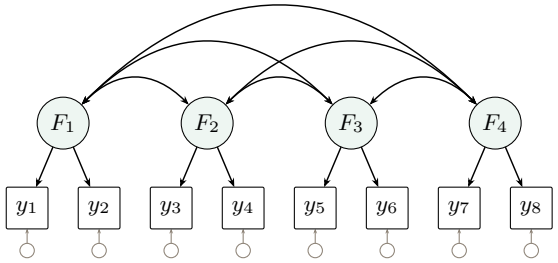
(ii) $B \preceq A$. By Lemma P and Lemma B(b), $B \equiv$ one factor plus θ_{56}. That is A with $\theta_{12} = \theta_{34} = 0$ (Lemma R). □

In the game: A two-indicator factor (F2 in B) is equivalent to one residual covariance on top of one factor. So B is one factor plus y5 ↔ y6, which is A with two of its residual covariances fixed to zero.

H15 Merging non-adjacent factors

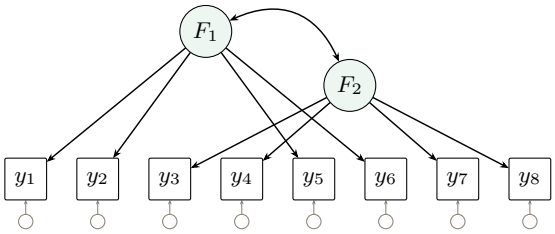
Level: Hard. Model families: CFA. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 22$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; F1
~~ F2 + F3 + F4; F2 ~~ F3 + F4; F3 ~~ F4
```

Model B $q_B = 17$




```
F1 =~ y1 + y2 + y5 + y6; F2 =~ y3 + y4 + y7 + y8; F1 ~~ F2
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 22 at the rational point listed in Appendix B, while model B has only $q_B = 17$ free parameters.

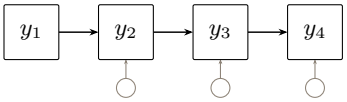
(ii) $B \preceq A$. Lemma M: in A merge F_1 with F_3 and F_2 with F_4. □

In the game: Merge F1 with F3 and F2 with F4 (correlations of 1) in the four-factor model and you get B. Non-adjacent indicators do not matter for nesting.

H16 Markov chain versus one factor

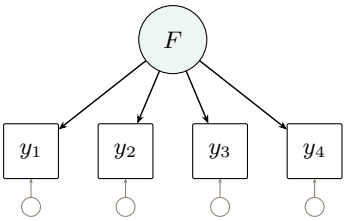
Level: Hard. Model families: CFA, Path analysis. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 7$




```
y2 ~ y1; y3 ~ y2; y4 ~ y3
```

Model B $q_B = 8$




```
F =~ y1 + y2 + y3 + y4
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{12}\sigma_{34} - \sigma_{13}\sigma_{24}$. A tetrad, which vanishes in every one-factor model. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \sigma_{12}\sigma_{34}\big(1 - \sigma_{23}^2/(\sigma_{22}\sigma_{33})\big) \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.1800 \neq 0$.

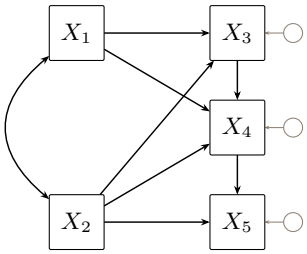
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 8 at the rational point listed in Appendix B, while model A has only $q_A = 7$ free parameters. □

In the game: The chain makes y1 and y3 independent given y2; the factor model never makes items conditionally independent given another item. The factor model implies tetrads, which the chain violates.

H17 An illegal replacement

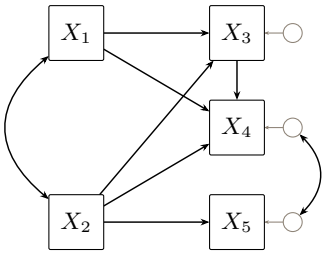
Level: Hard. Model families: Path analysis. Observed variables: X_1, X_2, X_3, X_4, X_5 ($p = 5$, 15 moments).

Model A $q_A = 13$




```
X3 ~ X1 + X2; X4 ~ X1 + X2 + X3; X5 ~ X4 + X2; X1 ~~ X2
```

Model B $q_B = 13$




```
X3 ~ X1 + X2; X4 ~ X1 + X2 + X3; X5 ~ X2; X1 ~~ X2; X4 ~~ X5
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{51}\sigma_{22} - \sigma_{52}\sigma_{12}$. In B, $X_5 = \beta X_2 + \zeta_5$ and ζ_5 is correlated only with ζ_4, so X_5 and X_1 are independent given X_2. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A at the point of Appendix B, $f = 0.2747 \neq 0$.

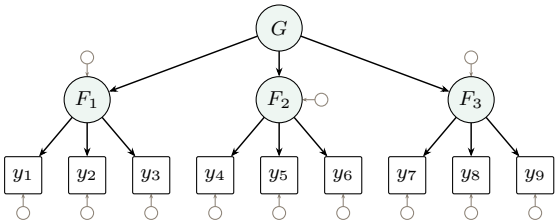
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \det \begin{pmatrix} \sigma_{51} & \sigma_{52} & \sigma_{54} \\ \sigma_{21} & \sigma_{22} & \sigma_{24} \\ \sigma_{41} & \sigma_{42} & \sigma_{44} \end{pmatrix}$. In A, X_5 is independent of its non-descendant X_1 given its parents X_2, X_4 (local Markov property); this determinant is the corresponding partial covariance times $\det \Sigma_{\{2,4\}}$. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = -0.1221 \neq 0$. □

In the game: An illegal use of the replacing rule: X4 → X5 cannot become X4 ↔ X5, because X5's predictors (X2) do not include those of X4. The models make different independence claims about X5.

H18 Higher-order versus one factor with one residual block

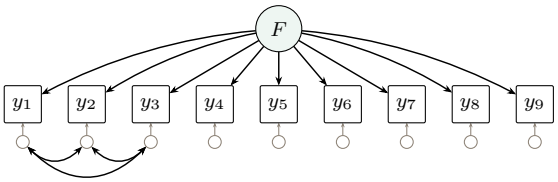
Level: Hard. Model families: Higher-order. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9$ ($p = 9$, 45 moments).

Model A $q_A = 21$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F3 =~ y7 + y8 + y9; G
=~ F1 + F2 + F3
```

Model B $q_B = 21$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8 + y9; y1 ~~ y2 + y3;
y2 ~~ y3
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{45}\sigma_{17} - \sigma_{14}\sigma_{57}$. In B these four covariances do not involve residual covariances (those are only within y_1, y_2, y_3), so the tetrad vanishes. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \lambda_1\lambda_4\lambda_5\lambda_7\gamma_1\gamma_3\psi_G\zeta_2 \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.9504 \neq 0$.

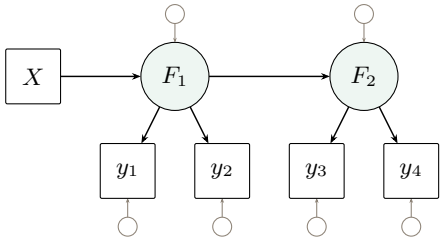
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{12}\sigma_{34} - \sigma_{13}\sigma_{24}$. In A, y_1, y_2, y_3 load only on F_1 and y_4 only on F_2. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B, $f = \lambda_4\psi(\theta_{12}\lambda_3 - \theta_{13}\lambda_2) \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.08400 \neq 0$. □

In the game: B forces the items of the second and third cluster to follow one general factor exactly; A gives every cluster its own factor. But A restricts the first cluster, which B leaves free. Neither contains the other.

H19 Two mediation chains

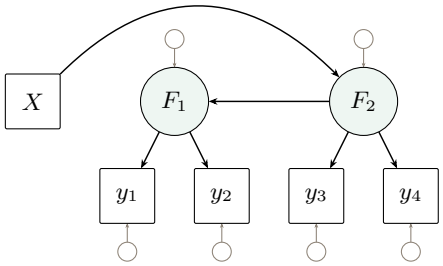
Level: Hard. Model families: SEM. Observed variables: X, y_1, y_2, y_3, y_4 ($p = 5$, 15 moments).

Model A $q_A = 11$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F1 ~ X; F2 ~ F1
```

Model B $q_B = 11$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F2 ~ X; F1 ~ F2
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{X1}\sigma_{34} - \sigma_{X3}\sigma_{14}$. In B ($X \to F_2 \to F_1$), X and F_1 are independent given F_2. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \lambda_1\lambda_3\lambda_4\,\gamma\,\mathrm{Var}(X)\,\zeta_2 \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.5400 \neq 0$.

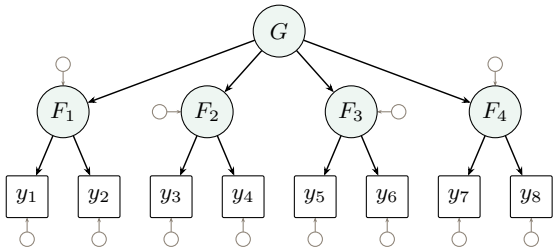
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{X3}\sigma_{12} - \sigma_{X1}\sigma_{23}$. In A ($X \to F_1 \to F_2$), X and F_2 are independent given F_1. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = 0.5000 \neq 0$. □

In the game: Both are mediation chains starting at X, but with a different mediator. A says X and F2 are independent given F1; B says X and F1 are independent given F2.

H20 General factor versus latent chain

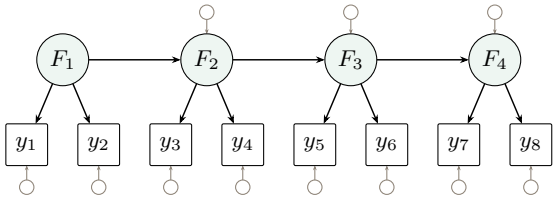
Level: Hard. Model families: SEM, Higher-order. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 20$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; G
=~ F1 + F2 + F3 + F4
```

Model B $q_B = 19$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; F2
~ F1; F3 ~ F2; F4 ~ F3
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 20 at the rational point listed in Appendix B, while model B has only $q_B = 19$ free parameters.

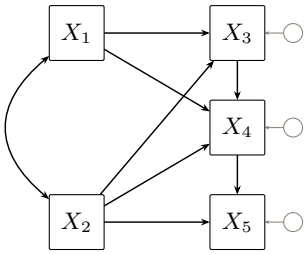
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{13}\sigma_{57} - \sigma_{15}\sigma_{37}$. This equals $\lambda\lambda\lambda\lambda(\psi_{12}\psi_{34} - \psi_{13}\psi_{24})$, which vanishes when Ψ follows one second-order factor. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = 0.1600 \neq 0$. □

In the game: Two ways to explain the correlations among four factors: one common cause (tetrad restrictions) or a chain (conditional independencies). Neither set of restrictions implies the other.

H21 A legal replacement

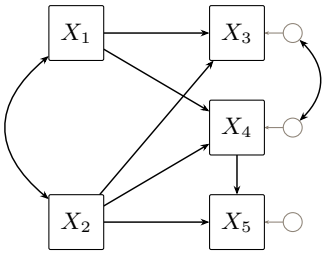
Level: Hard. Model families: Path analysis. Observed variables: X_1, X_2, X_3, X_4, X_5 ($p = 5$, 15 moments).

Model A $q_A = 13$




```
X3 ~ X1 + X2; X4 ~ X1 + X2 + X3; X5 ~ X4 + X2; X1 ~~ X2
```

Model B $q_B = 13$




```
X3 ~ X1 + X2; X4 ~ X1 + X2; X5 ~ X4 + X2; X1 ~~ X2; X3 ~~ X4
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Substitute $X_3 = \beta_{31}X_1 + \beta_{32}X_2 + \zeta_3$ in the equation of X_4 in A: $X_4 = (\beta_{41} + \beta_{43}\beta_{31})X_1 + (\beta_{42} + \beta_{43}\beta_{32})X_2 + \zeta_4'$ with $\zeta_4' = \beta_{43}\zeta_3 + \zeta_4$, $\mathrm{Var}(\zeta_4') = \beta_{43}^2\psi_3 + \psi_4$ and $\mathrm{Cov}(\zeta_3, \zeta_4') = \beta_{43}\psi_3$. The variable X_4 itself is unchanged, so the equation of X_5 (on X_4 and X_2, with ζ_5 uncorrelated with everything) is the same in both models: this is model B.

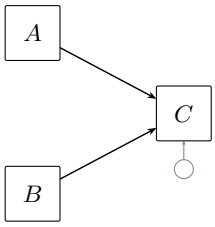
(ii) $B \preceq A$. Conversely, given B with $\psi_3 \neq 0$ and $\omega = \mathrm{Cov}(\zeta_3, \zeta_4')$: $\beta_{43} = \omega/\psi_3$, $\beta_{4k} = \beta_{4k}' - \beta_{43}\beta_{3k}$ ($k = 1, 2$), $\psi_4 = \psi_4' - \omega^2/\psi_3$ (Lemma G). □

In the game: The replacing rule (Lee & Hershberger): X3 → X4 may become the residual covariance X3 ↔ X4, because the other predictors of X4 (X1, X2) include all predictors of X3.

H22 A collider equals two covariances

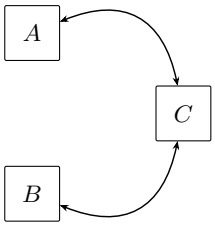
Level: Hard. Model families: Path analysis. Observed variables: A, B, C ($p = 3$, 6 moments).

Model A $q_A = 5$




```
C ~ A + B
```

Model B $q_B = 5$




```
A ~~ C; B ~~ C
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Both models equal $\{\Sigma \succ 0 : \sigma_{AB} = 0\}$. In A, A and B are exogenous and uncorrelated, so $\sigma_{AB} = 0$, and every other entry of Σ is free in B: $A \preceq B$.

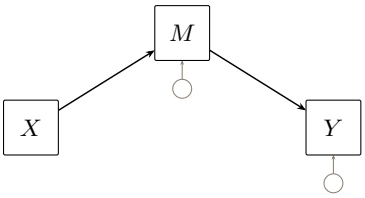
(ii) $B \preceq A$. Given $\Sigma \succ 0$ with $\sigma_{AB} = 0$: $\beta_{CA} = \sigma_{AC}/\sigma_{AA}$, $\beta_{CB} = \sigma_{BC}/\sigma_{BB}$, $\psi_C = \sigma_{CC} - \beta_{CA}^2\sigma_{AA} - \beta_{CB}^2\sigma_{BB}$ reproduce it in A. □

In the game: Both models say only one thing: A and B are uncorrelated. In the collider C may depend on both, in B it may covary with both. Five parameters each and the same single restriction.

H23 Full mediation inside a model without the mediated path

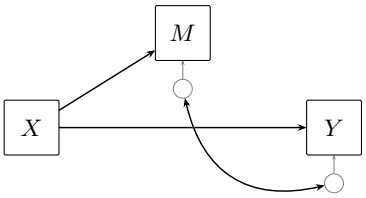
Level: Hard. Model families: Path analysis. Observed variables: X, M, Y ($p = 3$, 6 moments).

Model A $q_A = 5$




```
M ~ X; Y ~ M
```

Model B $q_B = 6$




```
M ~ X; Y ~ X; M ~~ Y
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. B is saturated: for $\Sigma \succ 0$, $\beta_{MX} = \sigma_{XM}/\sigma_{XX}$, $\beta_{YX} = \sigma_{XY}/\sigma_{XX}$, $\psi_{MY} = \sigma_{MY} - \beta_{MX}\beta_{YX}\sigma_{XX}$, $\psi_M = \sigma_{MM} - \beta_{MX}^2\sigma_{XX}$, $\psi_Y = \sigma_{YY} - \beta_{YX}^2\sigma_{XX}$ reproduce it. Hence $A \preceq B$.

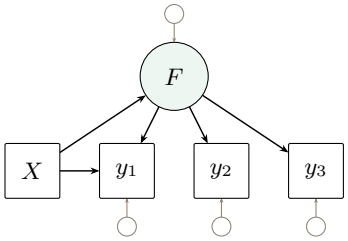
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 6 at the rational point listed in Appendix B, while model A has only $q_A = 5$ free parameters. □

In the game: B has no arrow M → Y and still contains A: its six parameters are identified (σXM and σXY give the two effects, the disturbance covariance takes the rest), so B is saturated.

H24 Direct effect of a covariate or residual covariance of an indicator

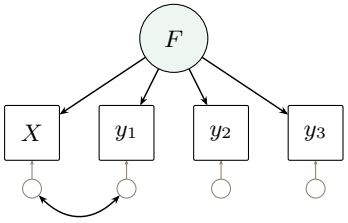
Level: Hard. Model families: SEM. Observed variables: X, y_1, y_2, y_3 ($p = 4$, 10 moments).

Model A $q_A = 9$




```
F =~ y1 + y2 + y3; F ~ X; y1 ~ X
```

Model B $q_B = 9$




```
F =~ X + y1 + y2 + y3; X ~~ y1
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. In A let $v = \mathrm{Var}(X)$, $F = \gamma X + \zeta$, $\Phi = \mathrm{Var}(F) = \gamma^2 v + \psi_\zeta$, $y_1 = \lambda_1 F + \delta X + e_1$, $y_j = \lambda_j F + e_j$ ($j = 2, 3$). In B take $\psi = \Phi$, the same λ_2, λ_3, $\lambda_X = \gamma v/\Phi$, $\theta_X = v - \lambda_X^2 \Phi$, $\lambda_1^B = \lambda_1 + \delta\gamma v/\Phi$, $\theta_{X1} = \delta v \psi_\zeta/\Phi$ and adjust θ_{11}. Then $\sigma_{X y_j} = \lambda_X \lambda_j \Phi = \gamma v \lambda_j$, $\sigma_{y_1 y_j} = \lambda_1^B \lambda_j \Phi = \lambda_j(\lambda_1 \Phi + \delta\gamma v)$ and $\sigma_{X y_1} = \lambda_X \lambda_1^B \Phi + \theta_{X1} = \gamma v \lambda_1 + \delta v$, all as in A. (Loadings and factor variances are rescaled to the markers as explained after Lemma G.)

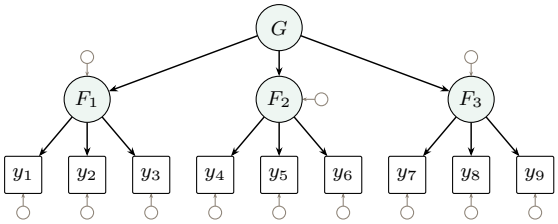
(ii) $B \preceq A$. Conversely, given B: $v = \lambda_X^2 \psi + \theta_X$, $\Phi = \psi$, $\gamma = \lambda_X \psi / v$, $\psi_\zeta = \psi - \gamma^2 v$, $\delta = \theta_{X1} \Phi/(v \psi_\zeta)$, $\lambda_1 = \lambda_1^B - \delta\gamma v/\Phi$ (Lemma G; $v\psi_\zeta \neq 0$ generically). □

In the game: Covariate or indicator, one step further: a direct effect X → y1 on top of X → F does the same as a residual covariance X ↔ y1 on top of X as an indicator. Both models imply one tetrad, σ(y1,y2)·σ(X,y3) = σ(y1,y3)·σ(X,y2), and nothing else.

H25 Higher-order model inside a cross-loading model

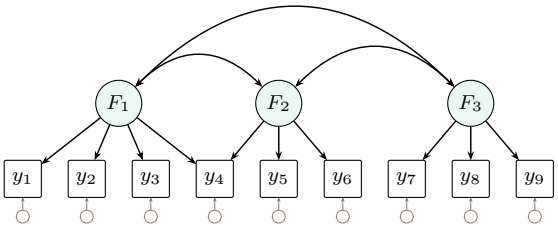
Level: Hard. Model families: Higher-order. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9$ ($p = 9$, 45 moments).

Model A $q_A = 21$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F3 =~ y7 + y8 + y9; G
=~ F1 + F2 + F3
```

Model B $q_B = 22$




```
F1 =~ y1 + y2 + y3 + y4; F2 =~ y4 + y5 + y6; F3 =~ y7 + y8 +
y9; F1 ~~ F2 + F3; F2 ~~ F3
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. By M02, A is equivalent to three correlated factors, which is B with the cross-loading of y_4 on F_1 fixed to 0 (Lemma R).

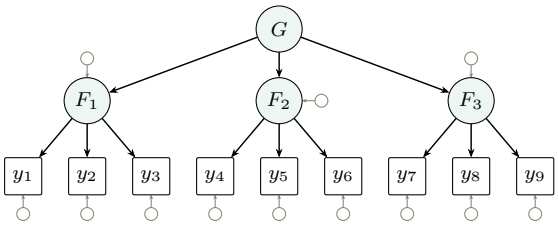
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 22 at the rational point listed in Appendix B, while model A has only $q_A = 21$ free parameters. □

In the game: A second-order factor on three factors is just-identified: A is the same as three correlated factors. And those are B with its cross-loading fixed to zero.

H26 One factor inside a higher-order model

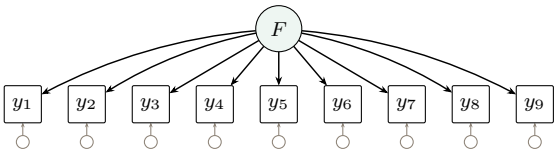
Level: Hard. Model families: Higher-order. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9$ ($p = 9$, 45 moments).

Model A $q_A = 21$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F3 =~ y7 + y8 + y9; G
   =~ F1 + F2 + F3
```

Model B $q_B = 18$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8 + y9
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 21 at the rational point listed in Appendix B, while model B has only $q_B = 18$ free parameters.

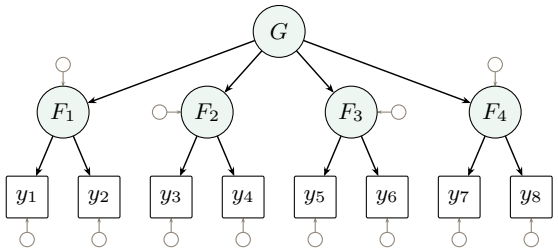
(ii) $B \preceq A$. In A fix the disturbance variances $\zeta_1 = \zeta_2 = \zeta_3 = 0$: then $F_k = \gamma_k G$ and $y_i = \lambda_i \gamma_{k(i)} G + e_i$, a one-factor model whose loadings $\lambda_i \gamma_{k(i)}$ can take any values (choose γ_k equal to the loading of the marker item of group k, then λ_i). Lemmas R and G. □

In the game: Fix the disturbance variances of F1, F2 and F3 to zero: every first-order factor becomes a multiple of G, so all items follow one factor. The one-factor model is nested in the higher-order model.

H27 General factor versus a missing factor covariance

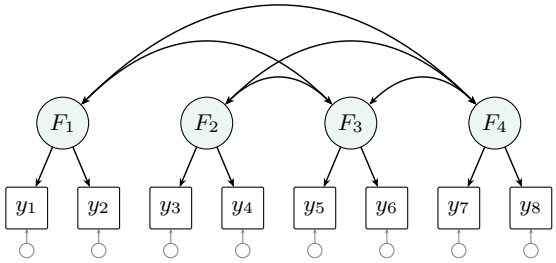
Level: Hard. Model families: Higher-order. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 20$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; G
=~ F1 + F2 + F3 + F4
```

Model B $q_B = 21$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; F1
~~ F3 + F4; F2 ~~ F3 + F4; F3 ~~ F4
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{13}$. In B, $\psi_{12} = 0$, so the marker indicators of F_1 and F_2 are uncorrelated. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \gamma_1\gamma_2\phi \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.4500 \neq 0$.

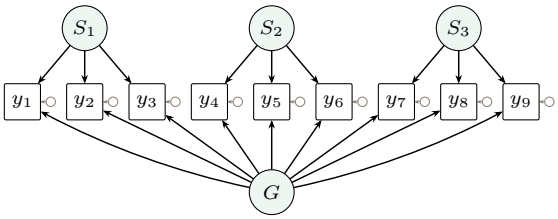
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 21 at the rational point listed in Appendix B, while model A has only $q_A = 20$ free parameters. □

In the game: B has more parameters but fixes the covariance of F1 and F2 to zero, which a general factor never does. And B's five free covariances need not follow A's tetrads. Neither is nested.

H28 One residual block inside a bifactor model

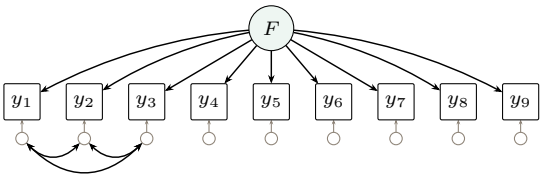
Level: Hard. Model families: Bifactor. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9$ ($p = 9$, 45 moments).

Model A $q_A = 27$




```
S1 =~ y1 + y2 + y3; S2 =~ y4 + y5 + y6; S3 =~ y7 + y8 + y9; G
=~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8 + y9
```

Model B $q_B = 21$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8 + y9; y1 ~~ y2 + y3;
y2 ~~ y3
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 27 at the rational point listed in Appendix B, while model B has only $q_B = 21$ free parameters.

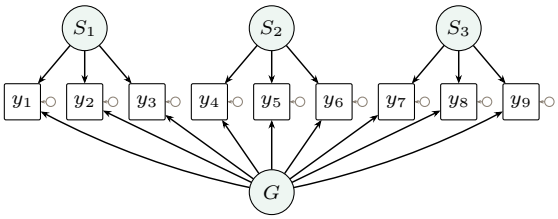
(ii) $B \preceq A$. By H03, A is equivalent to one factor plus free residual covariances within $\{y_1, y_2, y_3\}$, $\{y_4, y_5, y_6\}$ and $\{y_7, y_8, y_9\}$. B is that model with the second and third block fixed to 0 (Lemma R). □

In the game: A group factor with three indicators can produce any covariances within its group, so A is one factor plus three free residual blocks. B keeps only the first block: nested in A.

H29 Two bifactor groupings

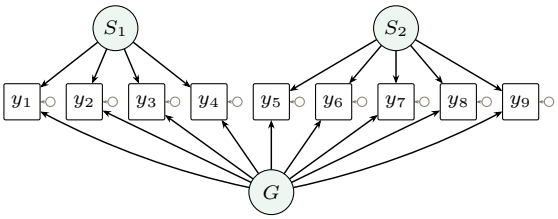
Level: Hard. Model families: Bifactor. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9$ ($p = 9$, 45 moments).

Model A $q_A = 27$




```
S1 =~ y1 + y2 + y3; S2 =~ y4 + y5 + y6; S3 =~ y7 + y8 + y9; G
=~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8 + y9
```

Model B $q_B = 27$




```
S1 =~ y1 + y2 + y3 + y4; S2 =~ y5 + y6 + y7 + y8 + y9; G =~ y1
+ y2 + y3 + y4 + y5 + y6 + y7 + y8 + y9
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{45}\sigma_{17} - \sigma_{47}\sigma_{15}$. In B the pairs (y_4, y_5), (y_1, y_7), (y_4, y_7), (y_1, y_5) are in different groups, so their covariances are $g_i g_j \psi_G$ and the tetrad vanishes. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = s_4 s_5 \psi_{S_2}\, g_1 g_7 \psi_G \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 1.426 \neq 0$.

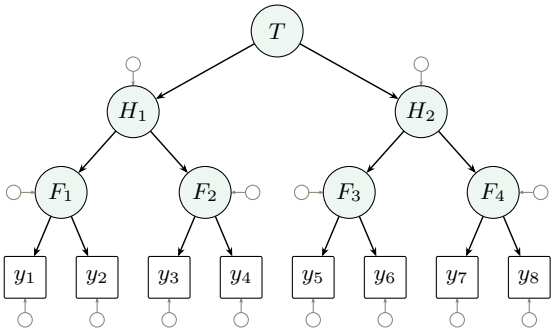
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{34}\sigma_{17} - \sigma_{37}\sigma_{14}$. In A the pairs (y_3, y_4), (y_1, y_7), (y_3, y_7), (y_1, y_4) are in different groups. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = 0.05760 \neq 0$. □

In the game: Both bifactor models have 27 parameters but group the items differently: y4 shares a group factor with y5 and y6 in A, with y1 to y3 in B. Each allows covariance that the other routes through G alone.

P01 A third-order factor on two second-order factors

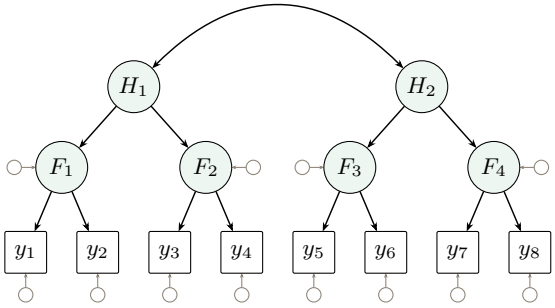
Level: Psychometrika. Model families: Higher-order. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 22$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; H1
=~ F1 + F2; H2 =~ F3 + F4; T =~ H1 + H2
```

Model B $q_B = 21$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; H1
=~ F1 + F2; H2 =~ F3 + F4; H1 ~~ H2
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Same first-order part; the observed covariances depend on H_1, H_2 only through their 2×2 covariance matrix, which is free in B. In A, $\mathrm{Cov}(H_1, H_2) = t_2 \psi_T$ and $\mathrm{Var}(H_k) = t_k^2 \psi_T + \zeta_{H_k}$ ($t_1 = 1$): with $t_2 = 1$, $\psi_T = \mathrm{Cov}(H_1, H_2)$, $\zeta_{H_k} = \mathrm{Var}(H_k) - \psi_T$ every 2×2 matrix is reached. Lemmas B(b) and G. A has $q_A = q_B + 1$ and is not identified.

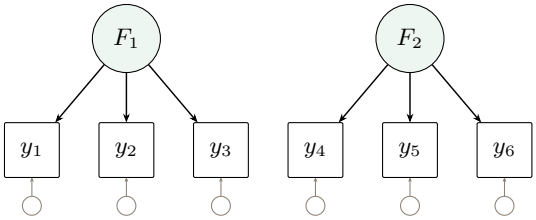
(ii) $B \preceq A$. See above. □

In the game: A third-order factor on only two second-order factors adds parameters but no restriction: like a factor with two indicators, it just produces one covariance between H1 and H2. A has one parameter more than B but is not identified.

P02 Orthogonal factors and network cliques

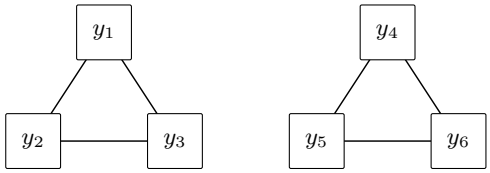
Level: Psychometrika. Model families: CFA, GGM. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 12$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6
```

Model B $q_B = 12$




```
network y1 y2 y3 y4 y5 y6; y1 - y2 + y3; y2 - y3; y4 - y5 + y6;
y5 - y6
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. $\mathcal{M}_A$: block-diagonal Σ with blocks $\{1,2,3\}, \{4,5,6\}$, each block a one-factor model with three indicators, which reproduces every positive definite block generically (Lemma B(a)). By Lemma I, Σ is block-diagonal iff Σ^{-1} is, which is $\mathcal{M}_B$ (two cliques, no edges between). Hence $\overline{\mathcal{M}_A} = \overline{\mathcal{M}_B}$.

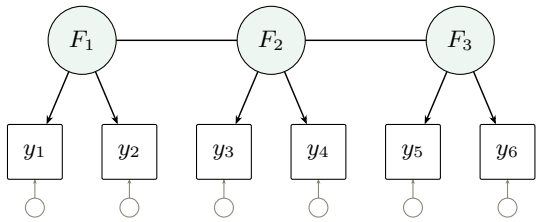
(ii) $B \preceq A$. See above. □

In the game: Orthogonal factors make the covariance matrix block-diagonal, and so is its inverse (two separate cliques in the GGM). With three indicators each factor is saturated within its block.

P03 Latent network chain equals latent chain

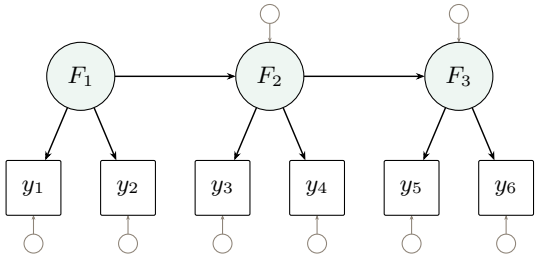
Level: Psychometrika. Model families: SEM, Latent / residual. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; network F1 F2 F3;
F1 - F2; F2 - F3
```

Model B $q_B = 14$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F2 ~ F1; F3 ~ F2
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Same measurement part. In A, $\Psi = K^{-1}$ with $k_{13} = 0$, which by Lemma I means $\psi_{12}\psi_{23} = \psi_{13}\psi_{22}$; by Lemma C that is exactly the set of Ψ of the chain $F_1 \to F_2 \to F_3$. Lemma B(b).

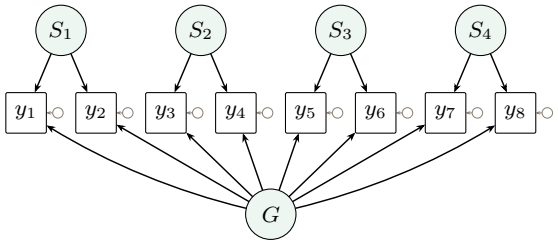
(ii) $B \preceq A$. See above. □

In the game: A latent network model (LNM) with edges F1 — F2 — F3 says F1 and F3 are independent given F2, exactly like the latent chain F1 → F2 → F3.

P04 Bifactor with two-indicator groups

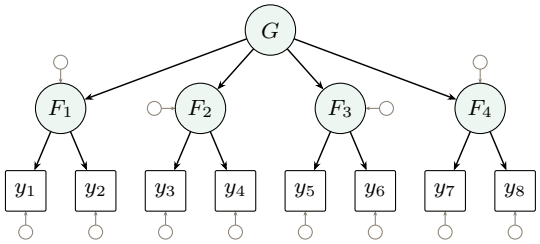
Level: Psychometrika. Model families: Bifactor. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 24$




```
S1 =~ y1 + y2; S2 =~ y3 + y4; S3 =~ y5 + y6; S4 =~ y7 + y8; G
=~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8
```

Model B $q_B = 20$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; G
=~ F1 + F2 + F3 + F4
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. A group factor with two indicators y_i, y_j only adds $s_i s_j \psi_S$ to σ_{ij} (and to the variances): it acts as a free residual covariance. So $A \equiv$ one factor (G) on $y_1, \ldots, y_8$ plus four within-pair residual covariances, which is equivalent to B by H04.

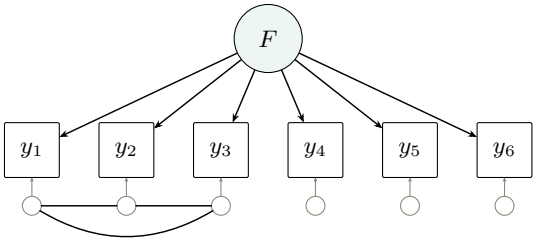
(ii) $B \preceq A$. See above. □

In the game: With two indicators per group factor, the bifactor model only frees the covariance within each pair on top of a general factor, and so does the higher-order model. Here the two models are equivalent, not merely nested!

P05 Residual clique equals residual covariance block

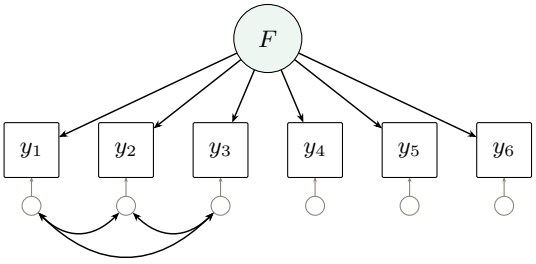
Level: Psychometrika. Model families: Latent / residual. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 15$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6; network y1 y2 y3; y1 - y2 + y3; y2 - y3
```

Model B $q_B = 15$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6; y1 ~~ y2 + y3; y2 ~~ y3
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Same factor part. The residual covariance matrix is $\Theta = K^{-1}$ in A and free on the block $\{1, 2, 3\}$ (diagonal elsewhere) in B; by Lemma I these are the same sets of Θ (block-diagonal inverse).

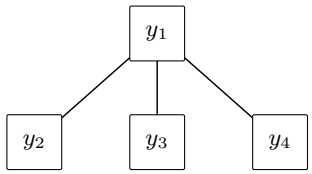
(ii) $B \preceq A$. See above. □

In the game: A residual network model (RNM): a fully connected block of residual partial correlations is the same as a fully free block of residual covariances (the inverse of a block-diagonal matrix is block-diagonal).

P06 Star network inside one factor

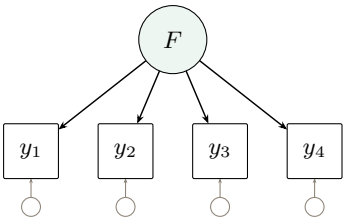
Level: Psychometrika. Model families: CFA, GGM. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 7$




```
network y1 y2 y3 y4; y1 - y2 + y3 + y4
```

Model B $q_B = 8$




```
F =~ y1 + y2 + y3 + y4
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. By Lemma C the star GGM equals the DAG $y_1 \to y_k$ ($k = 2, 3, 4$), which by Lemma T is the one-factor model with $\theta_1 = 0$.

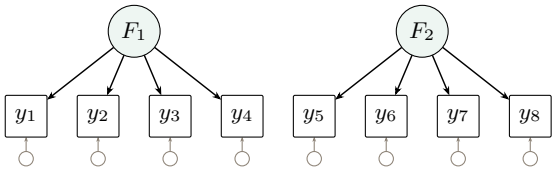
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 8 at the rational point listed in Appendix B, while model A has only $q_A = 7$ free parameters. □

In the game: In the star network y1 explains all associations. Make y1 a perfect indicator of F (no residual variance) and the factor model reproduces the star: A is nested in B.

P07 Orthogonal factors inside network cliques

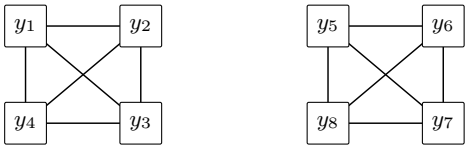
Level: Psychometrika. Model families: CFA, GGM. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 16$




```
F1 =~ y1 + y2 + y3 + y4; F2 =~ y5 + y6 + y7 + y8
```

Model B $q_B = 20$




```
network y1 y2 y3 y4 y5 y6 y7 y8; y1 - y2 + y3 + y4; y2 - y3 +
y4; y3 - y4; y5 - y6 + y7 + y8; y6 - y7 + y8; y7 - y8
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Orthogonal factors give a block-diagonal Σ, hence (Lemma I) a block-diagonal Σ^{-1}: no edges between the clusters, which is B.

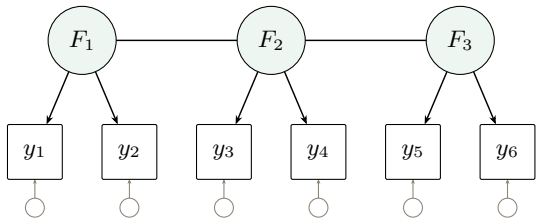
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 20 at the rational point listed in Appendix B, while model A has only $q_A = 16$ free parameters. □

In the game: Orthogonal factors give a block-diagonal covariance matrix, so its inverse is block-diagonal too: two fully connected clusters in the GGM. The GGM leaves each block free, the factors do not.

P08 Latent network chain inside a higher-order model

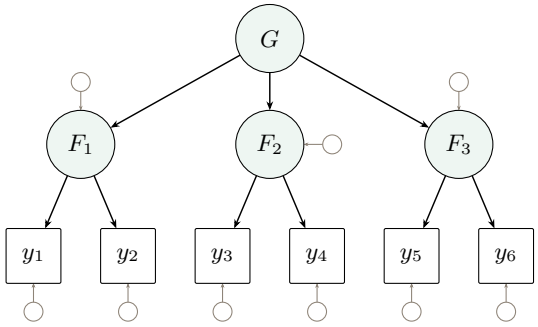
Level: Psychometrika. Model families: Higher-order, Latent / residual. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; network F1 F2 F3;
F1 - F2; F2 - F3
```

Model B $q_B = 15$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; G =~ F1 + F2 + F3
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. By P03, A is the latent chain; its Ψ is a particular covariance matrix of (F_1, F_2, F_3). The second-order factor of B reproduces every Ψ with nonzero covariances (as in M02). Lemma B(b).

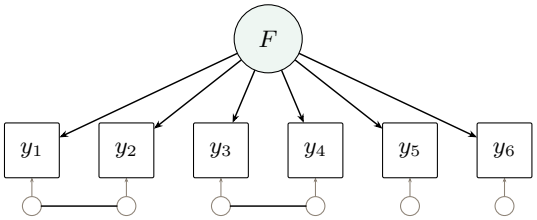
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 15 at the rational point listed in Appendix B, while model A has only $q_A = 14$ free parameters. □

In the game: The latent chain network removes the F1 — F3 edge (one restriction). A second-order factor on three factors is just-identified, so it reproduces any latent covariance matrix.

P09 Isolated residual edges

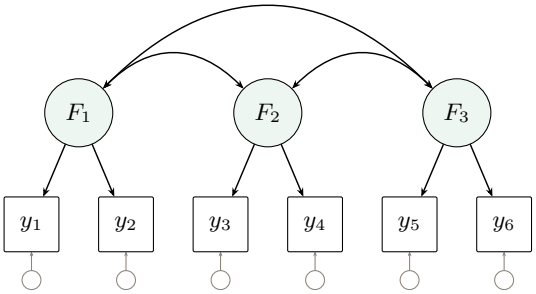
Level: Psychometrika. Model families: Latent / residual. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6; network y1 y2 y3 y4; y1 - y2;
y3 - y4
```

Model B $q_B = 15$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F1 ~~ F2 + F3; F2
~~ F3
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. By Lemma I, a residual network with the two separate edges $y_1 - y_2$ and $y_3 - y_4$ has residual covariance matrix with free θ_{12}, θ_{34} and nothing else, so A is the model of H10, which is nested in B.

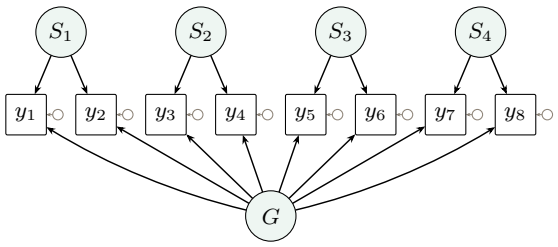
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 15 at the rational point listed in Appendix B, while model A has only $q_A = 14$ free parameters. □

In the game: Isolated residual edges in an RNM are just residual covariances. And a residual covariance within a pair acts like a two-indicator factor, so A is the three-factor model with one pair not separated.

P10 Bifactor with pairs inside correlated factors

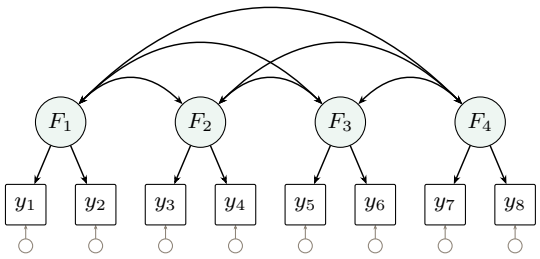
Level: Psychometrika. Model families: Bifactor. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 24$




```
S1 =~ y1 + y2; S2 =~ y3 + y4; S3 =~ y5 + y6; S4 =~ y7 + y8; G
=~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8
```

Model B $q_B = 22$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; F1
~~ F2 + F3 + F4; F2 ~~ F3 + F4; F3 ~~ F4
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. By P04, A is equivalent to the higher-order model over $F_1, \ldots, F_4$, which is nested in B (M06).

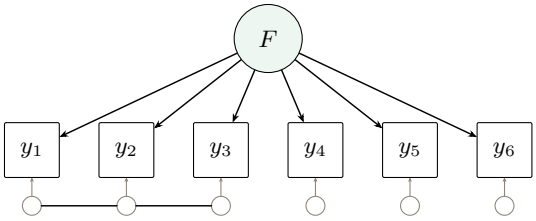
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{13}\sigma_{57} - \sigma_{15}\sigma_{37}$. In A items of different groups covary only through G: $\sigma_{ij} = g_i g_j \psi_G$, so this tetrad vanishes. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = -0.02500 \neq 0$. □

In the game: With two-indicator groups the bifactor model is equivalent to the higher-order model, and the higher-order model is nested in four freely correlated factors.

P11 One residual edge inside a residual chain

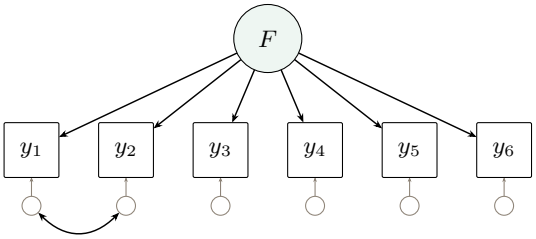
Level: Psychometrika. Model families: Latent / residual. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6; network y1 y2 y3; y1 - y2; y2
- y3
```

Model B $q_B = 13$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6; y1 ~~ y2
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 14 at the rational point listed in Appendix B, while model B has only $q_B = 13$ free parameters.

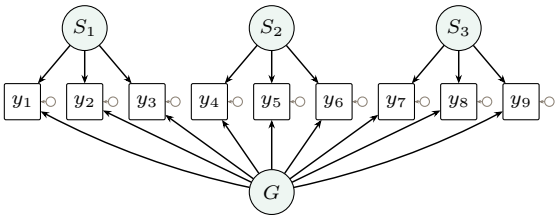
(ii) $B \preceq A$. By Lemma I the residual covariance θ_{12} alone is the residual network with only the edge $y_1 - y_2$; that is A with the edge $y_2 - y_3$ fixed to 0 (Lemma R). □

In the game: One residual covariance is one residual edge. A residual network with edges y1 — y2 — y3 contains the network with only y1 — y2, so B is nested in A.

P12 A two-indicator group factor

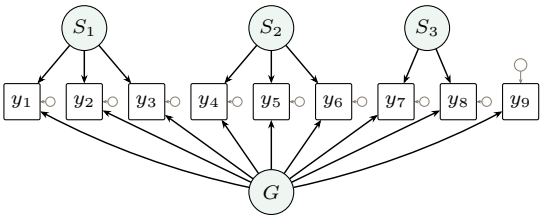
Level: Psychometrika. Model families: Bifactor. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9$ ($p = 9$, 45 moments).

Model A $q_A = 27$




```
S1 =~ y1 + y2 + y3; S2 =~ y4 + y5 + y6; S3 =~ y7 + y8 + y9; G
   =~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8 + y9
```

Model B $q_B = 26$




```
S1 =~ y1 + y2 + y3; S2 =~ y4 + y5 + y6; S3 =~ y7 + y8; G =~ y1
   + y2 + y3 + y4 + y5 + y6 + y7 + y8 + y9
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 27 at the rational point listed in Appendix B, while model B has only $q_B = 26$ free parameters.

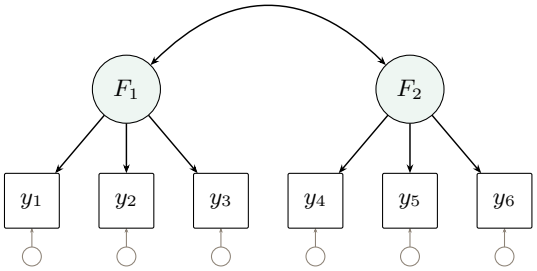
(ii) $B \preceq A$. A group factor with two indicators y_7, y_8 adds $s_7 s_8 \psi_{S_3}$ to σ_{78} (and to the variances) and nothing else: it acts as a free residual covariance θ_{78} (conversely $s_7 = 1$, $\psi_{S_3} = \theta_{78}/s_8$). By H03 the three-indicator group factors act as free residual blocks. So $B \equiv$ one factor plus free blocks $\{1,2,3\}$, $\{4,5,6\}$ and θ_{78}, and $A \equiv$ one factor plus free blocks $\{1,2,3\}$, $\{4,5,6\}$, $\{7,8,9\}$: B is A with $\theta_{79} = \theta_{89} = 0$ (Lemma R). □

In the game: A group factor with three indicators frees all covariances within its group; one with two indicators only frees one (here y7 ↔ y8). So B is A with the covariances of y9 with y7 and y8 fixed to zero.

P13 A tree network inside two factors

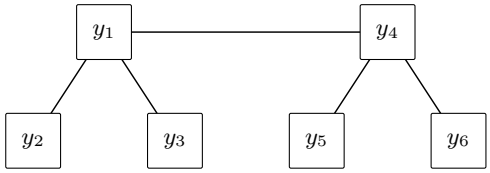
Level: Psychometrika. Model families: CFA, GGM. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 13$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2
```

Model B $q_B = 11$




```
network y1 y2 y3 y4 y5 y6; y1 - y2 + y3 + y4; y4 - y5 + y6
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 13 at the rational point listed in Appendix B, while model B has only $q_B = 11$ free parameters.

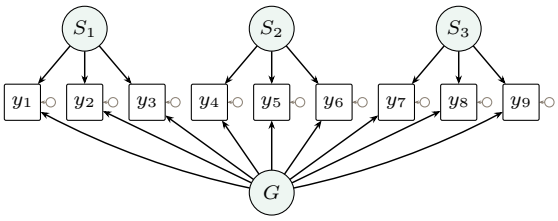
(ii) $B \preceq A$. By Lemma C, B equals the DAG $y_1 \to y_2, y_3, y_4$ and $y_4 \to y_5, y_6$. In A put $\theta_1 = \theta_4 = 0$, so $F_1 = y_1$ and $F_2 = y_4$ (Lemma T); then A reproduces this DAG ($\lambda_k = \beta_{k\cdot}$, $\psi_{12} = \mathrm{Cov}(y_1, y_4)$). □

In the game: Give y1 and y4 zero residual variance and they become F1 and F2 themselves. The two-factor model then becomes this tree-shaped network, so B is nested in A.

P14 A latent network inside a bifactor model

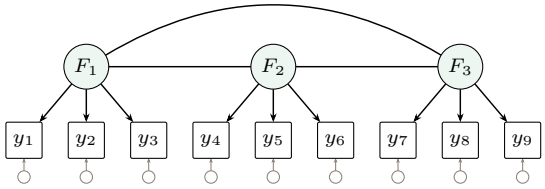
Level: Psychometrika. Model families: Bifactor, Latent / residual. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9$ ($p = 9$, 45 moments).

Model A $q_A = 27$




```
S1 =~ y1 + y2 + y3; S2 =~ y4 + y5 + y6; S3 =~ y7 + y8 + y9; G
  =~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8 + y9
```

Model B $q_B = 21$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F3 =~ y7 + y8 + y9;
network F1 F2 F3; F1 - F2 + F3; F2 - F3
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 27 at the rational point listed in Appendix B, while model B has only $q_B = 21$ free parameters.

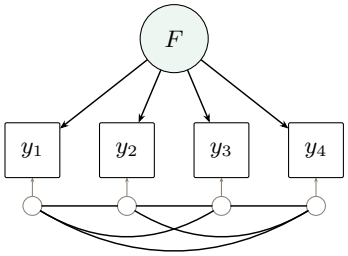
(ii) $B \preceq A$. A complete latent network has $\Psi = K^{-1}$ with K free, i.e. any $\Psi \succ 0$: B is the correlated three-factor model, which is nested in A by H07. □

In the game: A fully connected latent network is just three correlated factors, and those are nested in the bifactor model (via the higher-order model and Schmid–Leiman).

P15 Factor plus complete residual network is saturated

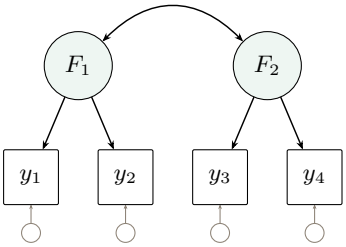
Level: Psychometrika. Model families: Latent / residual. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 14$




```
F =~ y1 + y2 + y3 + y4; network y1 y2 y3 y4; y1 - y2 + y3 + y4;
y2 - y3 + y4; y3 - y4
```

Model B $q_B = 9$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F1 ~~ F2
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 10 at the rational point listed in Appendix B, while model B has only $q_B = 9$ free parameters.

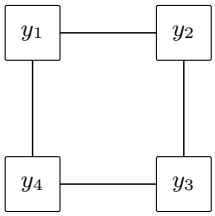
(ii) $B \preceq A$. Given any $\Sigma \succ 0$, choose λ and a small $\psi > 0$ with $\Sigma - \lambda\lambda^\top\psi \succ 0$, and $K = (\Sigma - \lambda\lambda^\top\psi)^{-1}$ (all edges free). So A is saturated and every model is nested in it. □

In the game: A factor model whose residuals form a complete network is saturated (and not identified): it reproduces any covariance matrix. Every model is nested in it.

P16 Cycle network versus one factor

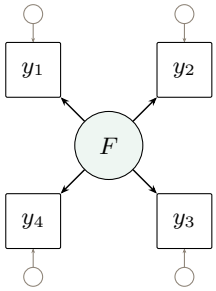
Level: Psychometrika. Model families: CFA, GGM. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 8$




```
network y1 y2 y3 y4; y1 - y2; y2 - y3; y3 - y4; y4 - y1
```

Model B $q_B = 8$




```
F =~ y1 + y2 + y3 + y4
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{12}\sigma_{34} - \sigma_{13}\sigma_{24}$. A tetrad: zero in every one-factor model. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A at the point of Appendix B, $f = 0.006844 \neq 0$.

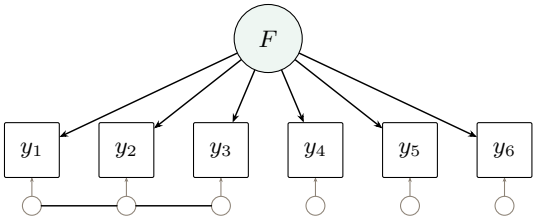
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \det \begin{pmatrix} \sigma_{12} & \sigma_{13} & \sigma_{14} \\ \sigma_{22} & \sigma_{23} & \sigma_{24} \\ \sigma_{42} & \sigma_{43} & \sigma_{44} \end{pmatrix}$. This minor is the cofactor of $(\Sigma^{-1})_{13}$; in A, $k_{13} = 0$. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = -0.8640 \neq 0$. □

In the game: The cycle network sets two partial correlations to zero (y1 — y3, y2 — y4); a factor model sets no partial correlation to zero but imposes tetrads. Both have 8 parameters, but the restrictions differ.

P17 Residual chain network versus residual covariance chain

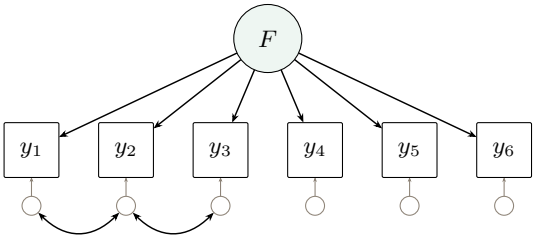
Level: Psychometrika. Model families: Latent / residual. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6; network y1 y2 y3; y1 - y2; y2 - y3
```

Model B $q_B = 14$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6; y1 ~~ y2; y2 ~~ y3
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{13}\sigma_{45} - \sigma_{14}\sigma_{35}$. In B, $\theta_{13} = 0$, so $\sigma_{13} = \lambda_1\lambda_3\psi$ and the tetrad vanishes. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \theta_{13}\lambda_4\lambda_5\psi \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.01863 \neq 0$.

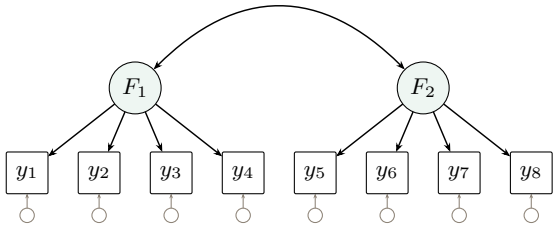
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = (\sigma_{13}\sigma_{45} - \sigma_{14}\sigma_{35})(\sigma_{22}\sigma_{45} - \sigma_{24}\sigma_{25}) - (\sigma_{12}\sigma_{45} - \sigma_{14}\sigma_{25})(\sigma_{23}\sigma_{45} - \sigma_{24}\sigma_{35})$. In A, $\sigma_{ij}\sigma_{45} - \sigma_{i4}\sigma_{j5} = \theta_{ij}\sigma_{45}$ for $i, j \in \{1, 2, 3\}$, so $f = \sigma_{45}^2(\theta_{13}\theta_{22} - \theta_{12}\theta_{23})$, and $k_{13} = 0$ means $\theta_{13}\theta_{22} = \theta_{12}\theta_{23}$ (Lemma I). So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = -0.05530 \neq 0$. □

In the game: A residual chain network sets a residual partial correlation to zero (y1, y3); the residual covariances in B set the residual covariance itself to zero. Same picture, different restriction.

P18 Correlated factors versus network cliques

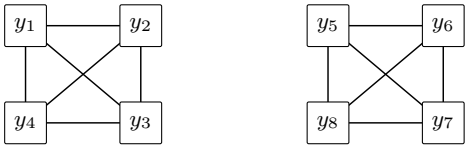
Level: Psychometrika. Model families: CFA, GGM. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 17$




```
F1 =~ y1 + y2 + y3 + y4; F2 =~ y5 + y6 + y7 + y8; F1 ~~ F2
```

Model B $q_B = 20$




```
network y1 y2 y3 y4 y5 y6 y7 y8; y1 - y2 + y3 + y4; y2 - y3 +
y4; y3 - y4; y5 - y6 + y7 + y8; y6 - y7 + y8; y7 - y8
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{15}$. In B there are no edges between the clusters, so Σ is block-diagonal (Lemma I). So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \lambda_1\lambda_5\psi_{12} \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.3000 \neq 0$.

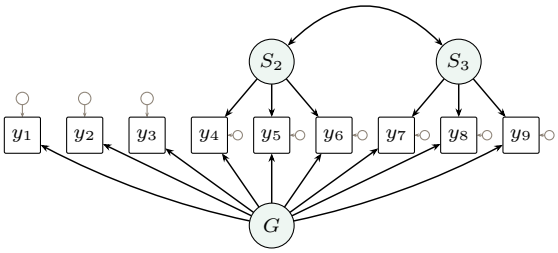
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 20 at the rational point listed in Appendix B, while model A has only $q_A = 17$ free parameters. □

In the game: Correlated factors connect the clusters (the GGM cannot); the GGM leaves each cluster saturated (the factors impose tetrads). Neither is nested.

P19 Bifactor-(S−1) versus bifactor

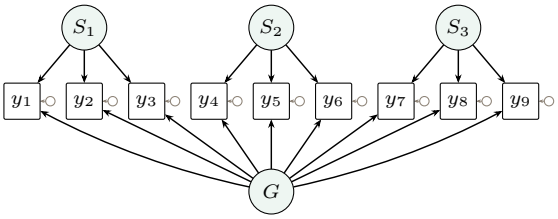
Level: Psychometrika. Model families: Bifactor. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9$ ($p = 9$, 45 moments).

Model A $q_A = 25$




```
G =~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8 + y9; S2 =~ y4 + y5
+ y6; S3 =~ y7 + y8 + y9; S2 ~~ S3
```

Model B $q_B = 27$




```
S1 =~ y1 + y2 + y3; S2 =~ y4 + y5 + y6; S3 =~ y7 + y8 + y9; G
=~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8 + y9
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{47}\sigma_{58} - \sigma_{48}\sigma_{57}$. In B (bifactor) items of groups 2 and 3 covary only through G: the block of their covariances has rank one. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \psi_G \psi_{S_2 S_3}(g_4 s_5 - s_4 g_5)(g_7 s_8 - s_7 g_8) \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = -0.03960 \neq 0$.

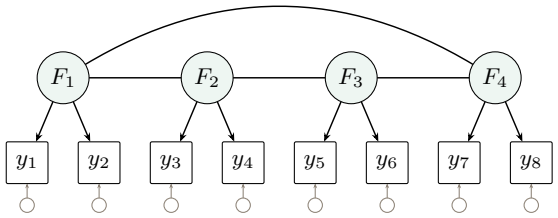
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{12}\sigma_{34} - \sigma_{14}\sigma_{23}$. In A the reference items y_1, y_2, y_3 load only on G, and y_4 covaries with them only through G. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B, $f = s_1 s_2 \psi_{S_1}\, g_3 g_4 \psi_G \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = -0.1403 \neq 0$. □

In the game: The bifactor-(S−1) model (A) has no group factor for the reference items y1–y3 and lets the other group factors correlate. That frees covariance between groups 2 and 3 (only through G in B), but restricts group 1 (free in B). Neither contains the other.

P20 Latent cycle network versus general factor

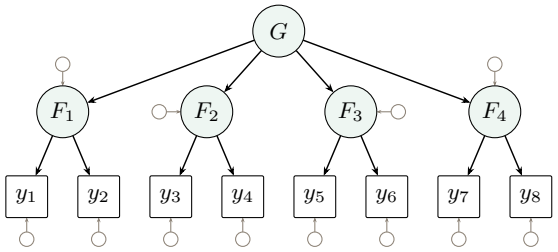
Level: Psychometrika. Model families: Higher-order, Latent / residual. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 20$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8;
network F1 F2 F3 F4; F1 - F2; F2 - F3; F3 - F4; F4 - F1
```

Model B $q_B = 20$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; G
=~ F1 + F2 + F3 + F4
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{13}\sigma_{57} - \sigma_{15}\sigma_{37}$. In B the latent covariances follow one second-order factor: $\psi_{12}\psi_{34} = \psi_{13}\psi_{24}$. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A at the point of Appendix B, $f = 0.006844 \neq 0$.

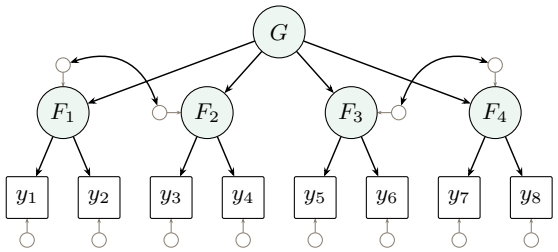
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \det \begin{pmatrix} \sigma_{14} & \sigma_{16} & \sigma_{18} \\ \sigma_{34} & \sigma_{36} & \sigma_{38} \\ \sigma_{74} & \sigma_{76} & \sigma_{78} \end{pmatrix}$. The matrix C of covariances between first indicators (y_1, y_3, y_5, y_7) and second indicators (y_2, y_4, y_6, y_8) equals $D_1 \Psi D_2$ (D_k diagonal loadings), so $(C^{-1})_{13} \propto (\Psi^{-1})_{13} = k_{13} = 0$ in A; f is the corresponding cofactor. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = -0.6159 \neq 0$. □

In the game: Both explain the latent correlations with 8 parameters, but the cycle network zeroes two latent partial correlations while the general factor imposes two tetrads. Different restrictions, so neither is nested.

P21 Disturbance covariances as second-order factors

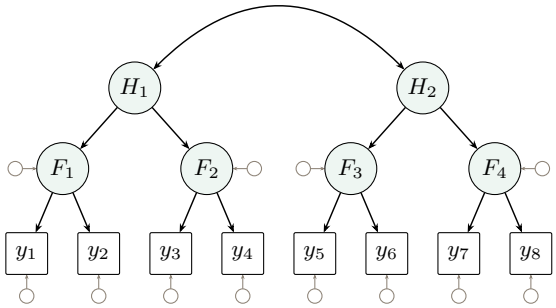
Level: Psychometrika. Model families: Higher-order. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 22$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; G
=~ F1 + F2 + F3 + F4; F1 ~~ F2; F3 ~~ F4
```

Model B $q_B = 21$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; H1
=~ F1 + F2; H2 =~ F3 + F4; H1 ~~ H2
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. The observed covariances depend on the first-order factors only through $\Psi = \mathrm{Cov}(F_1, \ldots, F_4)$ (Lemma B(b)). In A the F_k are indicators of G with disturbances as residuals, so Lemma P applies one level up: the disturbance covariance $F_1 \leftrightarrow F_2$ is equivalent to a factor H_1 for (F_1, F_2) regressed on G, and likewise H_2 for (F_3, F_4). Then G has no other children, and $H_1 \leftarrow G \rightarrow H_2$ produces every 2×2 covariance matrix of (H_1, H_2), as does $H_1 \leftrightarrow H_2$ (Lemma B(b)). Hence $A \equiv B$; as in H02, A is not identified ($q_A = q_B + 1$).

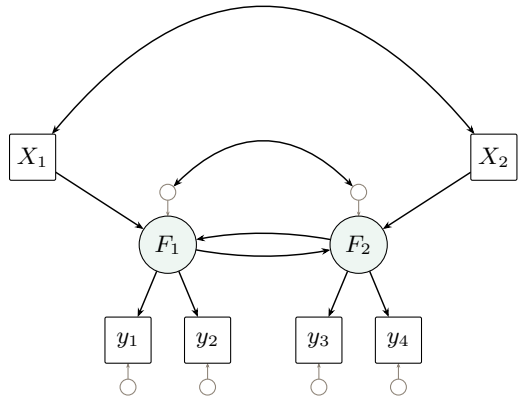
(ii) $B \preceq A$. See above. □

In the game: A disturbance covariance between two first-order factors acts like an extra second-order factor for that pair (just like a residual covariance acts like a factor, one level up). So A is G above two pair factors, and that is two correlated second-order factors. Counting trap: A has one parameter more and is not identified.

P22 A feedback loop with instruments is saturated

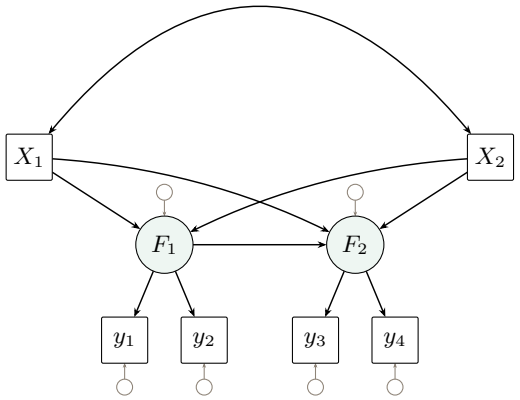
Level: Psychometrika. Model families: SEM. Observed variables: $X_1, X_2, y_1, y_2, y_3, y_4$ ($p = 6$, 21 moments).

Model A $q_A = 16$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F1 ~ X1 + F2; F2 ~ X2 + F1; X1 ~~
X2; F1 ~~ F2
```

Model B $q_B = 16$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F1 ~ X1 + X2; F2 ~ X1 + X2 + F1;
X1 ~~ X2
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Same measurement part; by Lemma B(b) it is enough that both structural parts produce every covariance matrix Φ of (X_1, X_2, F_1, F_2) (generically). B is a complete recursive model on these four variables (with $X_1 \leftrightarrow X_2$), saturated by Lemma B(a), so $A \preceq B$.

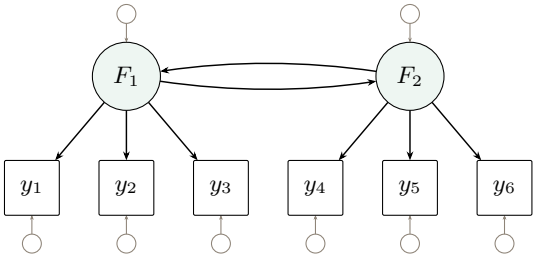
(ii) $B \preceq A$. In A, $F = \mathbf{B}F + \Gamma X + \zeta$ with $\mathbf{B} = \begin{pmatrix} 0 & b_{12} \\ b_{21} & 0 \end{pmatrix}$ and $\Gamma = \operatorname{diag}(g_1, g_2)$. Given Φ, let $\Pi = \Phi_{FX}\Phi_{XX}^{-1}$ (the regression of F on X); A needs $(I - \mathbf{B})\Pi = \Gamma$, solved by $b_{12} = \Pi_{12}/\Pi_{22}$, $b_{21} = \Pi_{21}/\Pi_{11}$, $g_1 = \Pi_{11} - b_{12}\Pi_{21}$, $g_2 = \Pi_{22} - b_{21}\Pi_{12}$; then $\operatorname{Cov}(\zeta) = (I - \mathbf{B})(\Phi_{FF} - \Pi\Phi_{XX}\Pi^\top)(I - \mathbf{B})^\top$ is a free 2×2 matrix in A and Φ_{XX} is free. This works whenever $\Pi_{11}\Pi_{22} \det \Pi \neq 0$ (then $\det(I - \mathbf{B}) = \det \Pi/(\Pi_{11}\Pi_{22}) \neq 0$); Lemma G. □

In the game: The feedback loop F1 ⇄ F2 is identified because each factor has its own instrument (X1, X2). With correlated disturbances the structural part has 10 parameters for the 10 (co)variances of X1, X2, F1 and F2: saturated, like the recursive model B.

P23 A feedback loop without instruments

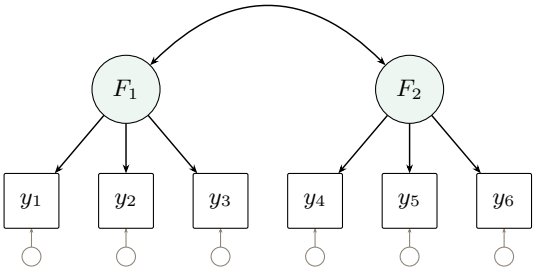
Level: Psychometrika. Model families: SEM. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~ F2; F2 ~ F1
```

Model B $q_B = 13$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Same measurement part, and in B the latent covariance matrix Ψ is free, so $A \preceq B$ (Lemma B(b)).

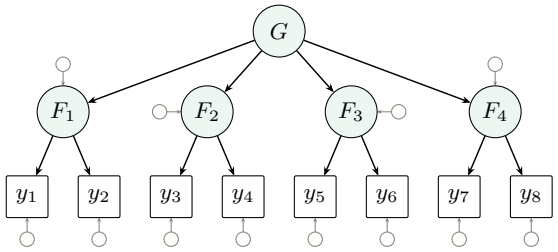
(ii) $B \preceq A$. In A fix $b_{12} = 0$: $F_2 = b_{21}F_1 + \zeta_2$ reproduces every 2×2 Ψ with $\psi_{11} \neq 0$ (E03). Lemmas B(b) and G. A has $q_A = q_B + 1$ and is not identified. □

In the game: A feedback loop without instruments: two effects for one covariance between the factors. A is not identified and produces exactly the same factor covariance matrices as a single correlation.

P24 Higher-order model inside paired residuals plus one

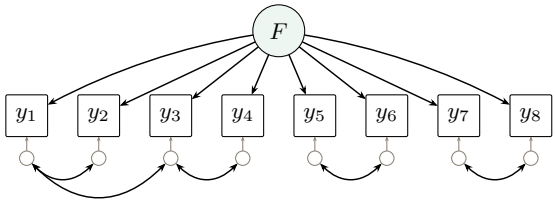
Level: Psychometrika. Model families: Higher-order. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 20$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; G
=~ F1 + F2 + F3 + F4
```

Model B $q_B = 21$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8; y1 ~~ y2; y3 ~~ y4;
y5 ~~ y6; y7 ~~ y8; y1 ~~ y3
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. By H04, A is equivalent to one factor plus the residual covariances $\theta_{12}, \theta_{34}, \theta_{56}, \theta_{78}$, which is B with $\theta_{13} = 0$ (Lemma R).

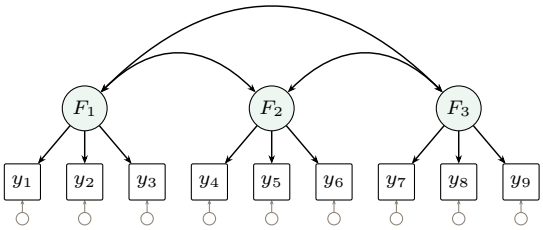
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 21 at the rational point listed in Appendix B, while model A has only $q_A = 20$ free parameters. □

In the game: The higher-order model over four two-indicator factors is equivalent to one factor plus a residual covariance within each pair. B has those four and one more (y1 ↔ y3).

P25 Correlated factors inside the bifactor-(S−1) model

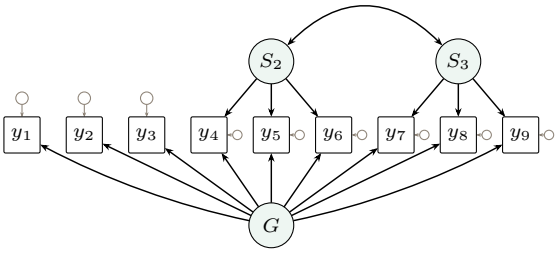
Level: Psychometrika. Model families: Bifactor. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9$ ($p = 9$, 45 moments).

Model A $q_A = 21$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F3 =~ y7 + y8 + y9; F1
~~ F2 + F3; F2 ~~ F3
```

Model B $q_B = 25$




```
G =~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8 + y9; S2 =~ y4 + y5
+ y6; S3 =~ y7 + y8 + y9; S2 ~~ S3
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Given A (factor covariances ψ_{kl}, loadings λ_i with markers $\lambda_1 = \lambda_4 = \lambda_7 = 1$): put $G = F_1$ and, for $k = 2, 3$, $S_k = F_k - \alpha_k F_1$ with $\alpha_k = \psi_{1k}/\psi_{11}$, so that S_k is uncorrelated with G. An item i of group $k \geq 2$ is $y_i = \lambda_i \alpha_k G + \lambda_i S_k + e_i$. This is B with G-loadings λ_i (group 1) and $\lambda_i \alpha_k$, S_k-loadings λ_i, $\mathrm{Var}(G) = \psi_{11}$, $\mathrm{Var}(S_k) = \psi_{kk} - \psi_{1k}^2/\psi_{11}$ and $\mathrm{Cov}(S_2, S_3) = \psi_{23} - \psi_{12}\psi_{13}/\psi_{11}$ (Lemma G, $\psi_{11} \neq 0$).

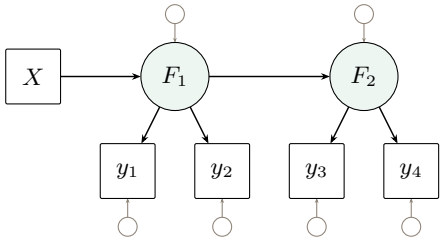
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 25 at the rational point listed in Appendix B, while model A has only $q_A = 21$ free parameters. □

In the game: The bifactor-(S−1) model (Eid et al.): G is the factor of the reference items y1–y3, S2 and S3 are what remains of the other factors after regressing them on G. Three correlated factors are the special case with proportional G and S loadings within each group.

P26 A covariate as an indicator

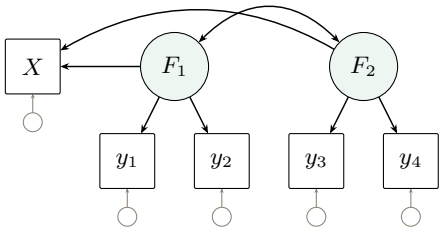
Level: Psychometrika. Model families: SEM. Observed variables: X, y_1, y_2, y_3, y_4 ($p = 5$, 15 moments).

Model A $q_A = 11$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F1 ~ X; F2 ~ F1
```

Model B $q_B = 12$




```
F1 =~ y1 + y2 + X; F2 =~ y3 + y4 + X; F1 ~~ F2
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. In A the variables (X, F_1, F_2) form the chain $X \to F_1 \to F_2$; the measurement parts attach to F_1 and F_2 only. By Lemma C the chain produces the same covariance matrices of (X, F_1, F_2) as the fork $X \leftarrow F_1 \to F_2$, i.e. X as one more indicator of F_1 (Lemma B(b)). With X an indicator, $F_1 \to F_2$ and $F_1 \leftrightarrow F_2$ both leave $\mathrm{Cov}(F_1, F_2)$ free (E03). So A is equivalent to the two-factor model with X loading on F_1 only, which is B with the loading of X on F_2 fixed to 0 (Lemma R).

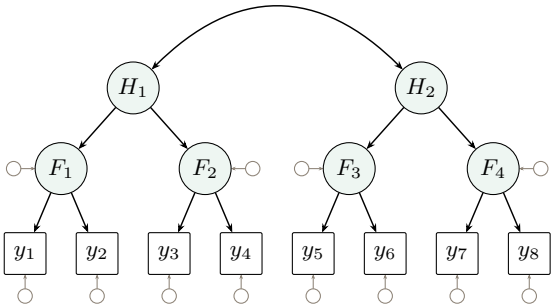
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 12 at the rational point listed in Appendix B, while model A has only $q_A = 11$ free parameters. □

In the game: A covariate that predicts F1 is the same as one more indicator of F1, and between two factors a regression is a covariance. So A is the two-factor model with X loading on F1 only; B lets X load on F2 as well.

P27 Paired residuals inside two second-order factors

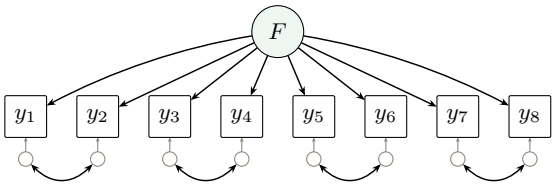
Level: Psychometrika. Model families: Higher-order. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 21$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; H1
=~ F1 + F2; H2 =~ F3 + F4; H1 ~~ H2
```

Model B $q_B = 20$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8; y1 ~~ y2; y3 ~~ y4;
y5 ~~ y6; y7 ~~ y8
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 21 at the rational point listed in Appendix B, while model B has only $q_B = 20$ free parameters.

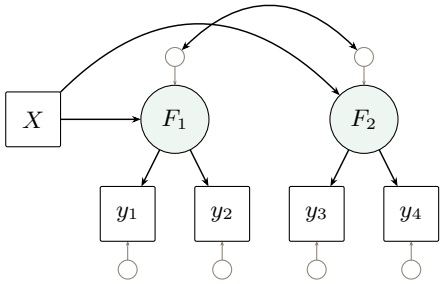
(ii) $B \preceq A$. By H04, B is equivalent to one second-order factor over the four pair factors $F_1, \ldots, F_4$, which is nested in A by M25 (merge H_1 and H_2). □

In the game: A residual covariance within a pair acts like a factor for that pair, so B is one second-order factor over four pair factors. And one second-order factor is two perfectly correlated ones.

P28 A saturated block of covariate and factors

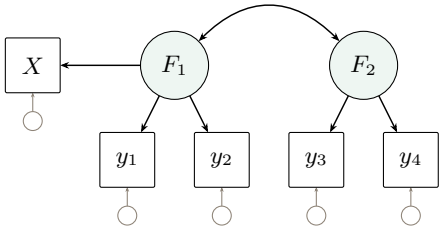
Level: Psychometrika. Model families: SEM. Observed variables: X, y_1, y_2, y_3, y_4 ($p = 5$, 15 moments).

Model A $q_A = 12$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F1 ~ X; F2 ~ X; F1 ~~ F2
```

Model B $q_B = 11$




```
F1 =~ y1 + y2 + X; F2 =~ y3 + y4; F1 ~~ F2
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 12 at the rational point listed in Appendix B, while model B has only $q_B = 11$ free parameters.

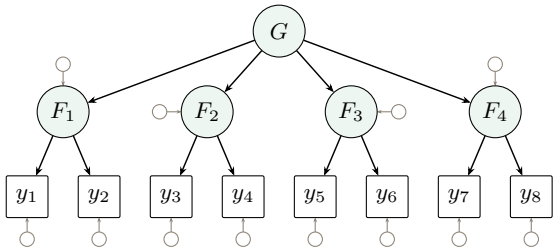
(ii) $B \preceq A$. By the argument of P26 (read backwards), B is equivalent to the chain $X \to F_1 \to F_2$ with the measurement parts of A. In A, $X \to F_1$, $X \to F_2$ and the disturbance covariance $F_1 \leftrightarrow F_2$ reproduce every covariance matrix of (X, F_1, F_2) with $\mathrm{Var}(X) \neq 0$, so $B \preceq A$ (Lemma B(b)). □

In the game: Making X an indicator of F1 is the same as X predicting F1, with F2 related to X only through F1. A leaves all covariances among X, F1 and F2 free, so it contains B.

P29 Higher-order model versus differently paired residuals

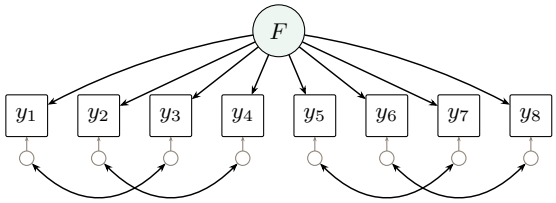
Level: Psychometrika. Model families: Higher-order. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 20$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; G
=~ F1 + F2 + F3 + F4
```

Model B $q_B = 20$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8; y1 ~~ y3; y2 ~~ y4;
y5 ~~ y7; y6 ~~ y8
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{12}\sigma_{36} - \sigma_{16}\sigma_{23}$. In B none of the pairs $(1,2), (3,6), (1,6), (2,3)$ has a residual covariance, so their covariances are $\lambda_i\lambda_j\psi$ and the tetrad vanishes. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \lambda_1\lambda_2\lambda_3\lambda_6\gamma_2\gamma_3\phi\,\zeta_1 \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.3240 \neq 0$.

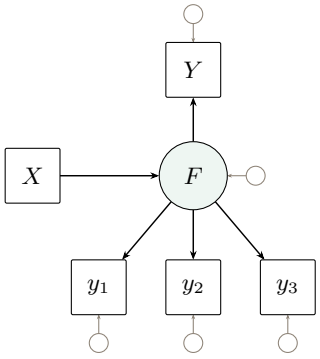
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{13}\sigma_{26} - \sigma_{16}\sigma_{23}$. In A, $\sigma_{ij} = \lambda_i\lambda_j\psi_{k(i)k(j)}$; here y_1, y_2 are indicators of F_1, so $f = \lambda_1\lambda_2\lambda_3\lambda_6(\psi_{12}\psi_{13} - \psi_{13}\psi_{12}) = 0$. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B, $f = \theta_{13}\lambda_2\lambda_6\psi \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.3300 \neq 0$. □

In the game: A is the same as one factor with residual covariances within the pairs (y1, y2), (y3, y4), ... B pairs the items differently. Same number of parameters, different restrictions.

P30 Cause of the factor or indicator with an effect

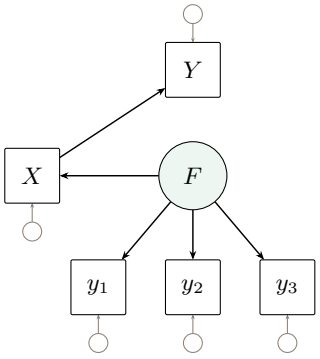
Level: Psychometrika. Model families: SEM. Observed variables: X, Y, y_1, y_2, y_3 ($p = 5$, 15 moments).

Model A $q_A = 10$




```
F =~ y1 + y2 + y3; F ~ X; Y ~ F
```

Model B $q_B = 10$




```
F =~ y1 + y2 + y3 + X; Y ~ X
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{Y1}\sigma_{XX} - \sigma_{XY}\sigma_{X1}$. In B, $Y = \beta X + e_Y$ with e_Y uncorrelated with everything else, so Y and y_1 are independent given X. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \lambda_Y \lambda_1 \mathrm{Var}(X)\, \psi_\zeta \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 1.200 \neq 0$.

(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{XY}\sigma_{12} - \sigma_{X1}\sigma_{Y2}$. In A, $\sigma_{XY} = \lambda_Y \gamma v$, $\sigma_{12} = \lambda_1 \lambda_2 \Phi$, $\sigma_{X1} = \lambda_1 \gamma v$, $\sigma_{Y2} = \lambda_Y \lambda_2 \Phi$: the tetrad vanishes. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B, $f = \beta \lambda_1 \lambda_2 \Phi\, \theta_X \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.5000 \neq 0$. □

In the game: Is X a cause of F, or an indicator of F that affects Y? A says Y relates to X only through F; B says Y relates to F only through X. Different conditional independencies: neither is nested.

Appendix B Rational points used in Lemmas S and D

For every non-nesting argument: the model, the rational parameter values (in the order of the model's free parameters; $X \to Y$ is a loading or regression, Var a (residual) variance, Cov a (residual) covariance, k an element of a precision matrix) and the exact value of f or the exact rank of the Jacobian at that point. The implied Σ is positive definite at each point.

E06, model B (for $B \not\preceq A$): rank of the Jacobian $= 14 > q_A = 13$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_1 \to y_4 = \frac{6}{5}$, $F_2 \to y_5 = \frac{4}{5}$, $F_2 \to y_6 = \frac{11}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$.

E07, model B (for $B \not\preceq A$): rank of the Jacobian $= 13 > q_A = 12$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_2 \to y_5 = \frac{6}{5}$, $F_2 \to y_6 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$.

E08, model B (for $B \not\preceq A$): rank of the Jacobian $= 6 > q_A = 5$. $A \to B = \frac{1}{2}$, $A \to C = \frac{3}{5}$, $B \to C = \frac{2}{5}$, $\mathrm{Var}(A) = 1$, $\mathrm{Var}(B) = \frac{6}{5}$, $\mathrm{Var}(C) = \frac{9}{10}$.

E09, model B (for $B \not\preceq A$): rank of the Jacobian $= 15 > q_A = 14$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(F_3, F_1) = \frac{1}{5}$, $\mathrm{Cov}(F_3, F_2) = \frac{2}{5}$.

E10, model B (for $B \not\preceq A$): rank of the Jacobian $= 13 > q_A = 12$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_2 \to y_5 = \frac{6}{5}$, $F_2 \to y_6 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$.

E11, model A (for $A \not\preceq B$): rank of the Jacobian $= 14 > q_B = 13$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_2 \to y_5 = \frac{6}{5}$, $F_2 \to y_6 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(y_4, y_3) = \frac{1}{5}$.

E12, model A (for $A \not\preceq B$): rank of the Jacobian $= 6 > q_B = 5$. $A \to C = \frac{1}{2}$, $B \to C = \frac{3}{5}$, $\mathrm{Var}(A) = 1$, $\mathrm{Var}(B) = \frac{6}{5}$, $\mathrm{Var}(C) = \frac{9}{10}$, $\mathrm{Cov}(B, A) = \frac{3}{10}$.

E13, model A (for $A \not\preceq B$): rank of the Jacobian $= 8 > q_B = 6$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(F) = 1$.

E14, model A (for $A \not\preceq B$): rank of the Jacobian $= 9 > q_B = 8$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $X \to F = \frac{1}{2}$, $X \to y_1 = \frac{3}{5}$, $\mathrm{Var}(X) = 1$, $\mathrm{Var}(y_1) = \frac{6}{5}$, $\mathrm{Var}(y_2) = \frac{9}{10}$, $\mathrm{Var}(y_3) = \frac{4}{5}$, $\mathrm{Var}(F) = 1$.

E15, model A (for $A \not\preceq B$): rank of the Jacobian $= 5 > q_B = 3$. $A \to B = \frac{1}{2}$, $A \to C = \frac{3}{5}$, $\mathrm{Var}(A) = 1$, $\mathrm{Var}(B) = \frac{6}{5}$, $\mathrm{Var}(C) = \frac{9}{10}$.

E16, model A (for $A \not\preceq B$): $f = 0.300000$. $A \to B = \frac{1}{2}$, $B \to C = \frac{3}{5}$, $\mathrm{Var}(A) = 1$, $\mathrm{Var}(B) = \frac{6}{5}$, $\mathrm{Var}(C) = \frac{9}{10}$.

E16, model B (for $B \not\preceq A$): $f = -0.270000$. $A \to B = \frac{1}{2}$, $C \to B = \frac{3}{5}$, $\mathrm{Var}(A) = 1$, $\mathrm{Var}(B) = \frac{6}{5}$, $\mathrm{Var}(C) = \frac{9}{10}$.

E17, model A (for $A \not\preceq B$): $f = 0.0648000$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_2 \to y_5 = \frac{6}{5}$, $F_2 \to y_6 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(y_2, y_1) = \frac{1}{5}$.

E17, model B (for $B \not\preceq A$): $f = 0.0600000$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_2 \to y_5 = \frac{6}{5}$, $F_2 \to y_6 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(y_6, y_5) = \frac{1}{5}$.

E18, model A (for $A \not\preceq B$): $f = 1.44000$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_2 \to y_5 = \frac{6}{5}$, $F_2 \to y_6 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$.

E18, model B (for $B \not\preceq A$): $f = 1.20000$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $F \to y_5 = \frac{4}{5}$, $F \to y_6 = \frac{11}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F) = 1$.

E19, model A (for $A \not\preceq B$): $f = 0.450000$. $A \to B = \frac{1}{2}$, $B \to C = \frac{3}{5}$, $\mathrm{Var}(A) = 1$, $\mathrm{Var}(B) = \frac{6}{5}$, $\mathrm{Var}(C) = \frac{9}{10}$.

E19, model B (for $B \not\preceq A$): $f = 0.600000$. $A \to C = \frac{1}{2}$, $C \to B = \frac{3}{5}$, $\mathrm{Var}(A) = 1$, $\mathrm{Var}(B) = \frac{6}{5}$, $\mathrm{Var}(C) = \frac{9}{10}$.

E20, model A (for $A \not\preceq B$): $f = 0.500000$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_1 \to F_2 = \frac{1}{2}$, $F_2 \to F_3 = \frac{3}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$.

E20, model B (for $B \not\preceq A$): $f = 0.540000$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_1 \to F_3 = \frac{1}{2}$, $F_2 \to F_3 = \frac{3}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$.

E22, model B (for $B \not\preceq A$): rank of the Jacobian $= 6 > q_A = 5$. $k_{y_1,y_1} = 2$, $k_{y_2,y_2} = \frac{9}{4}$, $k_{y_3,y_3} = \frac{5}{2}$, $k_{y_2,y_1} = -\frac{1}{2}$, $k_{y_3,y_2} = -\frac{2}{5}$, $k_{y_3,y_1} = -\frac{3}{10}$.

E23, model A (for $A \not\preceq B$): rank of the Jacobian $= 7 > q_B = 6$. $k_{y_1,y_1} = 2$, $k_{y_2,y_2} = \frac{9}{4}$, $k_{y_3,y_3} = \frac{5}{2}$, $k_{y_4,y_4} = \frac{11}{5}$, $k_{y_2,y_1} = -\frac{1}{2}$, $k_{y_3,y_2} = -\frac{2}{5}$, $k_{y_4,y_3} = -\frac{3}{10}$.

E24, model A (for $A \not\preceq B$): $f = 0.0485201$. $k_{y_1,y_1} = 2$, $k_{y_2,y_2} = \frac{9}{4}$, $k_{y_3,y_3} = \frac{5}{2}$, $k_{y_2,y_1} = -\frac{1}{2}$, $k_{y_3,y_2} = -\frac{2}{5}$.

E24, model B (for $B \not\preceq A$): $f = 0.0482276$. $k_{y_1,y_1} = 2$, $k_{y_2,y_2} = \frac{9}{4}$, $k_{y_3,y_3} = \frac{5}{2}$, $k_{y_3,y_1} = -\frac{1}{2}$, $k_{y_3,y_2} = -\frac{2}{5}$.

E25, model A (for $A \not\preceq B$): rank of the Jacobian $= 7 > q_B = 4$. $k_{y_1,y_1} = 2$, $k_{y_2,y_2} = \frac{9}{4}$, $k_{y_3,y_3} = \frac{5}{2}$, $k_{y_4,y_4} = \frac{11}{5}$, $k_{y_2,y_1} = -\frac{1}{2}$, $k_{y_3,y_1} = -\frac{2}{5}$, $k_{y_4,y_1} = -\frac{3}{10}$.

E27, model A (for $A \not\preceq B$): rank of the Jacobian $= 15 > q_B = 14$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_1 \to F_3 = \frac{1}{2}$, $F_2 \to F_3 = \frac{3}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$.

M06, model B (for $B \not\preceq A$): rank of the Jacobian $= 22 > q_A = 20$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_4 \to y_8 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(F_4) = \frac{11}{10}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(F_3, F_1) = \frac{1}{5}$, $\mathrm{Cov}(F_4, F_1) = \frac{2}{5}$, $\mathrm{Cov}(F_3, F_2) = \frac{1}{4}$, $\mathrm{Cov}(F_4, F_2) = \frac{7}{20}$, $\mathrm{Cov}(F_4, F_3) = \frac{3}{20}$.

M07, model B (for $B \not\preceq A$): rank of the Jacobian $= 15 > q_A = 13$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $F \to y_5 = \frac{4}{5}$, $F \to y_6 = \frac{11}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F) = 1$, $\mathrm{Cov}(y_5, y_4) = \frac{3}{10}$, $\mathrm{Cov}(y_6, y_4) = \frac{1}{5}$, $\mathrm{Cov}(y_6, y_5) = \frac{2}{5}$.

M08, model B (for $B \not\preceq A$): rank of the Jacobian $= 15 > q_A = 13$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(F_3, F_1) = \frac{1}{5}$, $\mathrm{Cov}(F_3, F_2) = \frac{2}{5}$.

M09, model B (for $B \not\preceq A$): rank of the Jacobian $= 10 > q_A = 7$. $A \to B = \frac{1}{2}$, $A \to C = \frac{3}{5}$, $B \to C = \frac{2}{5}$, $A \to D = \frac{7}{10}$, $B \to D = \frac{3}{10}$, $C \to D = \frac{9}{20}$, $\mathrm{Var}(A) = 1$, $\mathrm{Var}(B) = \frac{6}{5}$, $\mathrm{Var}(C) = \frac{9}{10}$, $\mathrm{Var}(D) = \frac{4}{5}$.

M10, model B (for $B \not\preceq A$): rank of the Jacobian $= 13 > q_A = 12$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_2 \to y_5 = \frac{6}{5}$, $F_2 \to y_6 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$.

M11, model A (for $A \not\preceq B$): rank of the Jacobian $= 10 > q_B = 8$. $y_1 \to y_2 = \frac{1}{2}$, $y_1 \to y_3 = \frac{3}{5}$, $y_2 \to y_3 = \frac{2}{5}$, $y_1 \to y_4 = \frac{7}{10}$, $y_2 \to y_4 = \frac{3}{10}$, $y_3 \to y_4 = \frac{9}{20}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$.

M12, model A (for $A \not\preceq B$): rank of the Jacobian $= 11 > q_B = 10$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $X \to F = \frac{1}{2}$, $F \to Y = \frac{6}{5}$, $X \to Y = \frac{3}{5}$, $\mathrm{Var}(X) = 1$, $\mathrm{Var}(y_1) = \frac{6}{5}$, $\mathrm{Var}(y_2) = \frac{9}{10}$, $\mathrm{Var}(y_3) = \frac{4}{5}$, $\mathrm{Var}(Y) = \frac{11}{10}$, $\mathrm{Var}(F) = 1$.

M13, model A (for $A \not\preceq B$): rank of the Jacobian $= 15 > q_B = 13$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(F_3, F_1) = \frac{1}{5}$, $\mathrm{Cov}(F_3, F_2) = \frac{2}{5}$.

M14, model A (for $A \not\preceq B$): rank of the Jacobian $= 11 > q_B = 10$. $A \to C = \frac{1}{2}$, $B \to C = \frac{3}{5}$, $C \to D = \frac{2}{5}$, $C \to E = \frac{7}{10}$, $D \to E = \frac{3}{10}$, $\mathrm{Var}(A) = 1$, $\mathrm{Var}(B) = \frac{6}{5}$, $\mathrm{Var}(C) = \frac{9}{10}$, $\mathrm{Var}(D) = \frac{4}{5}$, $\mathrm{Var}(E) = \frac{11}{10}$, $\mathrm{Cov}(B, A) = \frac{3}{10}$.

M15, model A (for $A \not\preceq B$): rank of the Jacobian $= 6 > q_B = 5$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(F) = 1$.

M16, model A (for $A \not\preceq B$): $f = 1.17216$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_1 \to y_4 = \frac{6}{5}$, $F_2 \to y_5 = \frac{4}{5}$, $F_2 \to y_6 = \frac{11}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$.

M16, model B (for $B \not\preceq A$): $f = 0.888000$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_2 \to y_4 = \frac{6}{5}$, $F_2 \to y_5 = \frac{4}{5}$, $F_2 \to y_6 = \frac{11}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$.

M17, model A (for $A \not\preceq B$): $f = 1.19880$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_2 \to y_5 = \frac{6}{5}$, $F_2 \to y_6 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$.

M17, model B (for $B \not\preceq A$): $f = 0.324000$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $F \to y_5 = \frac{4}{5}$, $F \to y_6 = \frac{11}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F) = 1$, $\mathrm{Cov}(y_2, y_1) = \frac{3}{10}$.

M18, model A (for $A \not\preceq B$): $f = 0.0720000$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_2 \to y_5 = \frac{6}{5}$, $F_2 \to y_6 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(y_4, y_3) = \frac{1}{5}$.

M18, model B (for $B \not\preceq A$): $f = 1.17216$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_1 \to y_4 = \frac{6}{5}$, $F_2 \to y_5 = \frac{4}{5}$, $F_2 \to y_6 = \frac{11}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$.

M19, model A (for $A \not\preceq B$): rank of the Jacobian $= 8 > q_B = 7$. $X \to M_1 = \frac{1}{2}$, $X \to M_2 = \frac{3}{5}$, $M_1 \to Y = \frac{2}{5}$, $M_2 \to Y = \frac{7}{10}$, $\mathrm{Var}(X) = 1$, $\mathrm{Var}(M_1) = \frac{6}{5}$, $\mathrm{Var}(M_2) = \frac{9}{10}$, $\mathrm{Var}(Y) = \frac{4}{5}$.

M19, model B (for $B \not\preceq A$): $f = 0.720000$. $X \to M_1 = \frac{1}{2}$, $M_1 \to M_2 = \frac{3}{5}$, $M_2 \to Y = \frac{2}{5}$, $\mathrm{Var}(X) = 1$, $\mathrm{Var}(M_1) = \frac{6}{5}$, $\mathrm{Var}(M_2) = \frac{9}{10}$, $\mathrm{Var}(Y) = \frac{4}{5}$.

M20, model A (for $A \not\preceq B$): $f = 0.324000$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(F) = 1$, $\mathrm{Cov}(y_2, y_1) = \frac{3}{10}$.

M20, model B (for $B \not\preceq A$): $f = 0.360000$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(F) = 1$, $\mathrm{Cov}(y_3, y_1) = \frac{3}{10}$.

M22, model A (for $A \not\preceq B$): $f = 0.600000$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $X \to F_1 = \frac{1}{2}$, $F_1 \to F_2 = \frac{3}{5}$, $\mathrm{Var}(X) = 1$, $\mathrm{Var}(y_1) = \frac{6}{5}$, $\mathrm{Var}(y_2) = \frac{9}{10}$, $\mathrm{Var}(y_3) = \frac{4}{5}$, $\mathrm{Var}(y_4) = \frac{11}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$.

M22, model B (for $B \not\preceq A$): $f = 0.600000$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $X \to F_1 = \frac{1}{2}$, $X \to F_2 = \frac{3}{5}$, $\mathrm{Var}(X) = 1$, $\mathrm{Var}(y_1) = \frac{6}{5}$, $\mathrm{Var}(y_2) = \frac{9}{10}$, $\mathrm{Var}(y_3) = \frac{4}{5}$, $\mathrm{Var}(y_4) = \frac{11}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$.

M23, model B (for $B \not\preceq A$): rank of the Jacobian $= 12 > q_A = 11$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $X \to F_1 = \frac{1}{2}$, $X \to F_2 = \frac{3}{5}$, $\mathrm{Var}(X) = 1$, $\mathrm{Var}(y_1) = \frac{6}{5}$, $\mathrm{Var}(y_2) = \frac{9}{10}$, $\mathrm{Var}(y_3) = \frac{4}{5}$, $\mathrm{Var}(y_4) = \frac{11}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$.

M25, model A (for $A \not\preceq B$): rank of the Jacobian $= 21 > q_B = 20$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_4 \to y_8 = \frac{4}{5}$, $H_1 \to F_2 = \frac{1}{2}$, $H_2 \to F_4 = \frac{3}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(F_4) = \frac{11}{10}$, $\mathrm{Var}(H_1) = \frac{9}{10}$, $\mathrm{Var}(H_2) = 1$, $\mathrm{Cov}(H_2, H_1) = \frac{3}{10}$.

M26, model A (for $A \not\preceq B$): $f = 0.243000$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_4 \to y_8 = \frac{4}{5}$, $H_1 \to F_2 = \frac{1}{2}$, $H_2 \to F_4 = \frac{3}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(F_4) = \frac{11}{10}$, $\mathrm{Var}(H_1) = \frac{9}{10}$, $\mathrm{Var}(H_2) = 1$, $\mathrm{Cov}(H_2, H_1) = \frac{3}{10}$.

M26, model B (for $B \not\preceq A$): $f = 0.243000$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_4 \to y_8 = \frac{4}{5}$, $H_1 \to F_3 = \frac{1}{2}$, $H_2 \to F_4 = \frac{3}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(F_4) = \frac{11}{10}$, $\mathrm{Var}(H_1) = \frac{9}{10}$, $\mathrm{Var}(H_2) = 1$, $\mathrm{Cov}(H_2, H_1) = \frac{3}{10}$.

M27, model B (for $B \not\preceq A$): rank of the Jacobian $= 21 > q_A = 20$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_4 \to y_8 = \frac{4}{5}$, $G \to F_2 = \frac{1}{2}$, $G \to F_3 = \frac{3}{5}$, $G \to F_4 = \frac{2}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(F_4) = \frac{11}{10}$, $\mathrm{Var}(G) = \frac{9}{10}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$.

M28, model A (for $A \not\preceq B$): $f = 0.0134599$. $k_{y_1,y_1} = 2$, $k_{y_2,y_2} = \frac{9}{4}$, $k_{y_3,y_3} = \frac{5}{2}$, $k_{y_4,y_4} = \frac{11}{5}$, $k_{y_2,y_1} = -\frac{1}{2}$, $k_{y_3,y_2} = -\frac{2}{5}$, $k_{y_4,y_3} = -\frac{3}{10}$.

M28, model B (for $B \not\preceq A$): $f = -0.0135007$. $k_{y_1,y_1} = 2$, $k_{y_2,y_2} = \frac{9}{4}$, $k_{y_3,y_3} = \frac{5}{2}$, $k_{y_4,y_4} = \frac{11}{5}$, $k_{y_2,y_1} = -\frac{1}{2}$, $k_{y_3,y_2} = -\frac{2}{5}$, $k_{y_4,y_2} = -\frac{3}{10}$.

M30, model B (for $B \not\preceq A$): rank of the Jacobian $= 8 > q_A = 7$. $k_{A,A} = 2$, $k_{B,B} = \frac{9}{4}$, $k_{C,C} = \frac{5}{2}$, $k_{D,D} = \frac{11}{5}$, $k_{B,A} = -\frac{1}{2}$, $k_{C,A} = -\frac{2}{5}$, $k_{C,B} = -\frac{3}{10}$, $k_{D,C} = -\frac{1}{4}$.

M31, model A (for $A \not\preceq B$): rank of the Jacobian $= 8 > q_B = 7$. $k_{y_1,y_1} = 2$, $k_{y_2,y_2} = \frac{9}{4}$, $k_{y_3,y_3} = \frac{5}{2}$, $k_{y_4,y_4} = \frac{11}{5}$, $k_{y_2,y_1} = -\frac{1}{2}$, $k_{y_3,y_2} = -\frac{2}{5}$, $k_{y_4,y_3} = -\frac{3}{10}$, $k_{y_4,y_1} = -\frac{1}{4}$.

M32, model A (for $A \not\preceq B$): $f = 0.0192911$. $k_{y_1,y_1} = 2$, $k_{y_2,y_2} = \frac{9}{4}$, $k_{y_3,y_3} = \frac{5}{2}$, $k_{y_3,y_1} = -\frac{1}{2}$, $k_{y_3,y_2} = -\frac{2}{5}$.

M32, model B (for $B \not\preceq A$): $f = -0.360000$. $y_1 \to y_3 = \frac{1}{2}$, $y_2 \to y_3 = \frac{3}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$.

H06, model B (for $B \not\preceq A$): rank of the Jacobian $= 27 > q_A = 21$. $S_1 \to y_2 = 1$, $S_1 \to y_3 = \frac{9}{10}$, $S_2 \to y_5 = \frac{6}{5}$, $S_2 \to y_6 = \frac{4}{5}$, $S_3 \to y_8 = \frac{11}{10}$, $S_3 \to y_9 = \frac{7}{10}$, $G \to y_2 = \frac{13}{10}$, $G \to y_3 = \frac{3}{4}$, $G \to y_4 = \frac{17}{20}$, $G \to y_5 = \frac{21}{20}$, $G \to y_6 = 1$, $G \to y_7 = \frac{9}{10}$, $G \to y_8 = \frac{6}{5}$, $G \to y_9 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(y_9) = \frac{19}{20}$, $\mathrm{Var}(S_1) = 1$, $\mathrm{Var}(S_2) = \frac{6}{5}$, $\mathrm{Var}(S_3) = \frac{4}{5}$, $\mathrm{Var}(G) = \frac{11}{10}$.

H07, model B (for $B \not\preceq A$): rank of the Jacobian $= 27 > q_A = 21$. $S_1 \to y_2 = 1$, $S_1 \to y_3 = \frac{9}{10}$, $S_2 \to y_5 = \frac{6}{5}$, $S_2 \to y_6 = \frac{4}{5}$, $S_3 \to y_8 = \frac{11}{10}$, $S_3 \to y_9 = \frac{7}{10}$, $G \to y_2 = \frac{13}{10}$, $G \to y_3 = \frac{3}{4}$, $G \to y_4 = \frac{17}{20}$, $G \to y_5 = \frac{21}{20}$, $G \to y_6 = 1$, $G \to y_7 = \frac{9}{10}$, $G \to y_8 = \frac{6}{5}$, $G \to y_9 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(y_9) = \frac{19}{20}$, $\mathrm{Var}(S_1) = 1$, $\mathrm{Var}(S_2) = \frac{6}{5}$, $\mathrm{Var}(S_3) = \frac{4}{5}$, $\mathrm{Var}(G) = \frac{11}{10}$.

H08, model B (for $B \not\preceq A$): rank of the Jacobian $= 15 > q_A = 14$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_1 \to F_2 = \frac{1}{2}$, $F_1 \to F_3 = \frac{3}{5}$, $F_2 \to F_3 = \frac{2}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$.

H09, model B (for $B \not\preceq A$): rank of the Jacobian $= 8 > q_A = 7$. $F \to B = 1$, $F \to C = \frac{9}{10}$, $F \to D = \frac{6}{5}$, $\mathrm{Var}(A) = 1$, $\mathrm{Var}(B) = \frac{6}{5}$, $\mathrm{Var}(C) = \frac{9}{10}$, $\mathrm{Var}(D) = \frac{4}{5}$, $\mathrm{Var}(F) = 1$.

H10, model B (for $B \not\preceq A$): rank of the Jacobian $= 15 > q_A = 14$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(F_3, F_1) = \frac{1}{5}$, $\mathrm{Cov}(F_3, F_2) = \frac{2}{5}$.

H11, model A (for $A \not\preceq B$): rank of the Jacobian $= 22 > q_B = 20$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_4 \to y_8 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(F_4) = \frac{11}{10}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(F_3, F_1) = \frac{1}{5}$, $\mathrm{Cov}(F_4, F_1) = \frac{2}{5}$, $\mathrm{Cov}(F_3, F_2) = \frac{1}{4}$, $\mathrm{Cov}(F_4, F_2) = \frac{7}{20}$, $\mathrm{Cov}(F_4, F_3) = \frac{3}{20}$.

H12, model A (for $A \not\preceq B$): rank of the Jacobian $= 9 > q_B = 8$. $F \to y_1 = 1$, $F \to y_2 = \frac{9}{10}$, $F \to y_3 = \frac{6}{5}$, $\mathrm{Var}(X) = 1$, $\mathrm{Var}(y_1) = \frac{6}{5}$, $\mathrm{Var}(y_2) = \frac{9}{10}$, $\mathrm{Var}(y_3) = \frac{4}{5}$, $\mathrm{Var}(F) = 1$, $\mathrm{Cov}(y_1, X) = \frac{3}{10}$.

H13, model A (for $A \not\preceq B$): rank of the Jacobian $= 13 > q_B = 12$. $X_1 \to X_3 = \frac{1}{2}$, $X_2 \to X_3 = \frac{3}{5}$, $X_1 \to X_4 = \frac{2}{5}$, $X_2 \to X_4 = \frac{7}{10}$, $X_3 \to X_4 = \frac{3}{10}$, $X_4 \to X_5 = \frac{9}{20}$, $X_2 \to X_5 = \frac{1}{2}$, $\mathrm{Var}(X_1) = 1$, $\mathrm{Var}(X_2) = \frac{6}{5}$, $\mathrm{Var}(X_3) = \frac{9}{10}$, $\mathrm{Var}(X_4) = \frac{4}{5}$, $\mathrm{Var}(X_5) = \frac{11}{10}$, $\mathrm{Cov}(X_2, X_1) = \frac{3}{10}$.

H14, model A (for $A \not\preceq B$): rank of the Jacobian $= 15 > q_B = 13$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $F \to y_5 = \frac{4}{5}$, $F \to y_6 = \frac{11}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F) = 1$, $\mathrm{Cov}(y_2, y_1) = \frac{3}{10}$, $\mathrm{Cov}(y_4, y_3) = \frac{1}{5}$, $\mathrm{Cov}(y_6, y_5) = \frac{2}{5}$.

H15, model A (for $A \not\preceq B$): rank of the Jacobian $= 22 > q_B = 17$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_4 \to y_8 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(F_4) = \frac{11}{10}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(F_3, F_1) = \frac{1}{5}$, $\mathrm{Cov}(F_4, F_1) = \frac{2}{5}$, $\mathrm{Cov}(F_3, F_2) = \frac{1}{4}$, $\mathrm{Cov}(F_4, F_2) = \frac{7}{20}$, $\mathrm{Cov}(F_4, F_3) = \frac{3}{20}$.

H16, model A (for $A \not\preceq B$): $f = 0.180000$. $y_1 \to y_2 = \frac{1}{2}$, $y_2 \to y_3 = \frac{3}{5}$, $y_3 \to y_4 = \frac{2}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$.

H16, model B (for $B \not\preceq A$): rank of the Jacobian $= 8 > q_A = 7$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(F) = 1$.

H17, model A (for $A \not\preceq B$): $f = 0.274725$. $X_1 \to X_3 = \frac{1}{2}$, $X_2 \to X_3 = \frac{3}{5}$, $X_1 \to X_4 = \frac{2}{5}$, $X_2 \to X_4 = \frac{7}{10}$, $X_3 \to X_4 = \frac{3}{10}$, $X_4 \to X_5 = \frac{9}{20}$, $X_2 \to X_5 = \frac{1}{2}$, $\mathrm{Var}(X_1) = 1$, $\mathrm{Var}(X_2) = \frac{6}{5}$, $\mathrm{Var}(X_3) = \frac{9}{10}$, $\mathrm{Var}(X_4) = \frac{4}{5}$, $\mathrm{Var}(X_5) = \frac{11}{10}$, $\mathrm{Cov}(X_2, X_1) = \frac{3}{10}$.

H17, model B (for $B \not\preceq A$): $f = -0.122100$. $X_1 \to X_3 = \frac{1}{2}$, $X_2 \to X_3 = \frac{3}{5}$, $X_1 \to X_4 = \frac{2}{5}$, $X_2 \to X_4 = \frac{7}{10}$, $X_3 \to X_4 = \frac{3}{10}$, $X_2 \to X_5 = \frac{9}{20}$, $\mathrm{Var}(X_1) = 1$, $\mathrm{Var}(X_2) = \frac{6}{5}$, $\mathrm{Var}(X_3) = \frac{9}{10}$, $\mathrm{Var}(X_4) = \frac{4}{5}$, $\mathrm{Var}(X_5) = \frac{11}{10}$, $\mathrm{Cov}(X_2, X_1) = \frac{3}{10}$, $\mathrm{Cov}(X_5, X_4) = \frac{1}{5}$.

H18, model A (for $A \not\preceq B$): $f = 0.950400$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_2 \to y_5 = \frac{6}{5}$, $F_2 \to y_6 = \frac{4}{5}$, $F_3 \to y_8 = \frac{11}{10}$, $F_3 \to y_9 = \frac{7}{10}$, $G \to F_2 = \frac{1}{2}$, $G \to F_3 = \frac{3}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(y_9) = \frac{19}{20}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(G) = \frac{11}{10}$.

H18, model B (for $B \not\preceq A$): $f = 0.0840000$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $F \to y_5 = \frac{4}{5}$, $F \to y_6 = \frac{11}{10}$, $F \to y_7 = \frac{7}{10}$, $F \to y_8 = \frac{13}{10}$, $F \to y_9 = \frac{3}{4}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(y_9) = \frac{19}{20}$, $\mathrm{Var}(F) = 1$, $\mathrm{Cov}(y_2, y_1) = \frac{3}{10}$, $\mathrm{Cov}(y_3, y_1) = \frac{1}{5}$, $\mathrm{Cov}(y_3, y_2) = \frac{2}{5}$.

H19, model A (for $A \not\preceq B$): $f = 0.540000$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $X \to F_1 = \frac{1}{2}$, $F_1 \to F_2 = \frac{3}{5}$, $\mathrm{Var}(X) = 1$, $\mathrm{Var}(y_1) = \frac{6}{5}$, $\mathrm{Var}(y_2) = \frac{9}{10}$, $\mathrm{Var}(y_3) = \frac{4}{5}$, $\mathrm{Var}(y_4) = \frac{11}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$.

H19, model B (for $B \not\preceq A$): $f = 0.500000$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $X \to F_2 = \frac{1}{2}$, $F_2 \to F_1 = \frac{3}{5}$, $\mathrm{Var}(X) = 1$, $\mathrm{Var}(y_1) = \frac{6}{5}$, $\mathrm{Var}(y_2) = \frac{9}{10}$, $\mathrm{Var}(y_3) = \frac{4}{5}$, $\mathrm{Var}(y_4) = \frac{11}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$.

H20, model A (for $A \not\preceq B$): rank of the Jacobian $= 20 > q_B = 19$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_4 \to y_8 = \frac{4}{5}$, $G \to F_2 = \frac{1}{2}$, $G \to F_3 = \frac{3}{5}$, $G \to F_4 = \frac{2}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(F_4) = \frac{11}{10}$, $\mathrm{Var}(G) = \frac{9}{10}$.

H20, model B (for $B \not\preceq A$): $f = 0.160000$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_4 \to y_8 = \frac{4}{5}$, $F_1 \to F_2 = \frac{1}{2}$, $F_2 \to F_3 = \frac{3}{5}$, $F_3 \to F_4 = \frac{2}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(F_4) = \frac{11}{10}$.

H23, model B (for $B \not\preceq A$): rank of the Jacobian $= 6 > q_A = 5$. $X \to M = \frac{1}{2}$, $X \to Y = \frac{3}{5}$, $\mathrm{Var}(X) = 1$, $\mathrm{Var}(M) = \frac{6}{5}$, $\mathrm{Var}(Y) = \frac{9}{10}$, $\mathrm{Cov}(Y, M) = \frac{3}{10}$.

H25, model B (for $B \not\preceq A$): rank of the Jacobian $= 22 > q_A = 21$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_1 \to y_4 = \frac{6}{5}$, $F_2 \to y_5 = \frac{4}{5}$, $F_2 \to y_6 = \frac{11}{10}$, $F_3 \to y_8 = \frac{7}{10}$, $F_3 \to y_9 = \frac{13}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(y_9) = \frac{19}{20}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(F_3, F_1) = \frac{1}{5}$, $\mathrm{Cov}(F_3, F_2) = \frac{2}{5}$.

H26, model A (for $A \not\preceq B$): rank of the Jacobian $= 21 > q_B = 18$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_2 \to y_5 = \frac{6}{5}$, $F_2 \to y_6 = \frac{4}{5}$, $F_3 \to y_8 = \frac{11}{10}$, $F_3 \to y_9 = \frac{7}{10}$, $G \to F_2 = \frac{1}{2}$, $G \to F_3 = \frac{3}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(y_9) = \frac{19}{20}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(G) = \frac{11}{10}$.

H27, model A (for $A \not\preceq B$): $f = 0.450000$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_4 \to y_8 = \frac{4}{5}$, $G \to F_2 = \frac{1}{2}$, $G \to F_3 = \frac{3}{5}$, $G \to F_4 = \frac{2}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(F_4) = \frac{11}{10}$, $\mathrm{Var}(G) = \frac{9}{10}$.

H27, model B (for $B \not\preceq A$): rank of the Jacobian $= 21 > q_A = 20$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_4 \to y_8 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(F_4) = \frac{11}{10}$, $\mathrm{Cov}(F_3, F_1) = \frac{3}{10}$, $\mathrm{Cov}(F_4, F_1) = \frac{1}{5}$, $\mathrm{Cov}(F_3, F_2) = \frac{2}{5}$, $\mathrm{Cov}(F_4, F_2) = \frac{1}{4}$, $\mathrm{Cov}(F_4, F_3) = \frac{7}{20}$.

H28, model A (for $A \not\preceq B$): rank of the Jacobian $= 27 > q_B = 21$. $S_1 \to y_2 = 1$, $S_1 \to y_3 = \frac{9}{10}$, $S_2 \to y_5 = \frac{6}{5}$, $S_2 \to y_6 = \frac{4}{5}$, $S_3 \to y_8 = \frac{11}{10}$, $S_3 \to y_9 = \frac{7}{10}$, $G \to y_2 = \frac{13}{10}$, $G \to y_3 = \frac{3}{4}$, $G \to y_4 = \frac{17}{20}$, $G \to y_5 = \frac{21}{20}$, $G \to y_6 = 1$, $G \to y_7 = \frac{9}{10}$, $G \to y_8 = \frac{6}{5}$, $G \to y_9 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(y_9) = \frac{19}{20}$, $\mathrm{Var}(S_1) = 1$, $\mathrm{Var}(S_2) = \frac{6}{5}$, $\mathrm{Var}(S_3) = \frac{4}{5}$, $\mathrm{Var}(G) = \frac{11}{10}$.

H29, model A (for $A \not\preceq B$): $f = 1.42560$. $S_1 \to y_2 = 1$, $S_1 \to y_3 = \frac{9}{10}$, $S_2 \to y_5 = \frac{6}{5}$, $S_2 \to y_6 = \frac{4}{5}$, $S_3 \to y_8 = \frac{11}{10}$, $S_3 \to y_9 = \frac{7}{10}$, $G \to y_2 = \frac{13}{10}$, $G \to y_3 = \frac{3}{4}$, $G \to y_4 = \frac{17}{20}$, $G \to y_5 = \frac{21}{20}$, $G \to y_6 = 1$, $G \to y_7 = \frac{9}{10}$, $G \to y_8 = \frac{6}{5}$, $G \to y_9 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(y_9) = \frac{19}{20}$, $\mathrm{Var}(S_1) = 1$, $\mathrm{Var}(S_2) = \frac{6}{5}$, $\mathrm{Var}(S_3) = \frac{4}{5}$, $\mathrm{Var}(G) = \frac{11}{10}$.

H29, model B (for $B \not\preceq A$): $f = 0.0576000$. $S_1 \to y_2 = 1$, $S_1 \to y_3 = \frac{9}{10}$, $S_1 \to y_4 = \frac{6}{5}$, $S_2 \to y_6 = \frac{4}{5}$, $S_2 \to y_7 = \frac{11}{10}$, $S_2 \to y_8 = \frac{7}{10}$, $S_2 \to y_9 = \frac{13}{10}$, $G \to y_2 = \frac{3}{4}$, $G \to y_3 = \frac{17}{20}$, $G \to y_4 = \frac{21}{20}$, $G \to y_5 = 1$, $G \to y_6 = \frac{9}{10}$, $G \to y_7 = \frac{6}{5}$, $G \to y_8 = \frac{4}{5}$, $G \to y_9 = \frac{11}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(y_9) = \frac{19}{20}$, $\mathrm{Var}(S_1) = 1$, $\mathrm{Var}(S_2) = \frac{6}{5}$, $\mathrm{Var}(G) = \frac{4}{5}$.

P06, model B (for $B \not\preceq A$): rank of the Jacobian $= 8 > q_A = 7$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(F) = 1$.

P07, model B (for $B \not\preceq A$): rank of the Jacobian $= 20 > q_A = 16$. $k_{y_1,y_1} = 2$, $k_{y_2,y_2} = \frac{9}{4}$, $k_{y_3,y_3} = \frac{5}{2}$, $k_{y_4,y_4} = \frac{11}{5}$, $k_{y_5,y_5} = \frac{12}{5}$, $k_{y_6,y_6} = \frac{21}{10}$, $k_{y_7,y_7} = 2$, $k_{y_8,y_8} = \frac{9}{4}$, $k_{y_2,y_1} = -\frac{1}{2}$, $k_{y_3,y_1} = -\frac{2}{5}$, $k_{y_4,y_1} = -\frac{3}{10}$, $k_{y_3,y_2} = -\frac{1}{4}$, $k_{y_4,y_2} = -\frac{7}{20}$, $k_{y_4,y_3} = -\frac{9}{20}$, $k_{y_6,y_5} = -\frac{1}{2}$, $k_{y_7,y_5} = -\frac{2}{5}$, $k_{y_8,y_5} = -\frac{3}{10}$, $k_{y_7,y_6} = -\frac{1}{4}$, $k_{y_8,y_6} = -\frac{7}{20}$, $k_{y_8,y_7} = -\frac{9}{20}$.

P08, model B (for $B \not\preceq A$): rank of the Jacobian $= 15 > q_A = 14$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $G \to F_2 = \frac{1}{2}$, $G \to F_3 = \frac{3}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(G) = \frac{11}{10}$.

P09, model B (for $B \not\preceq A$): rank of the Jacobian $= 15 > q_A = 14$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(F_3, F_1) = \frac{1}{5}$, $\mathrm{Cov}(F_3, F_2) = \frac{2}{5}$.

P10, model B (for $B \not\preceq A$): $f = -0.0250000$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_4 \to y_8 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(F_4) = \frac{11}{10}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(F_3, F_1) = \frac{1}{5}$, $\mathrm{Cov}(F_4, F_1) = \frac{2}{5}$, $\mathrm{Cov}(F_3, F_2) = \frac{1}{4}$, $\mathrm{Cov}(F_4, F_2) = \frac{7}{20}$, $\mathrm{Cov}(F_4, F_3) = \frac{3}{20}$.

P11, model A (for $A \not\preceq B$): rank of the Jacobian $= 14 > q_B = 13$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $F \to y_5 = \frac{4}{5}$, $F \to y_6 = \frac{11}{10}$, $k_{y_1,y_1} = 2$, $k_{y_2,y_2} = \frac{9}{4}$, $k_{y_3,y_3} = \frac{5}{2}$, $k_{y_2,y_1} = -\frac{1}{2}$, $k_{y_3,y_2} = -\frac{2}{5}$, $\mathrm{Var}(y_4) = 1$, $\mathrm{Var}(y_5) = \frac{6}{5}$, $\mathrm{Var}(y_6) = \frac{9}{10}$, $\mathrm{Var}(F) = 1$.

P12, model A (for $A \not\preceq B$): rank of the Jacobian $= 27 > q_B = 26$. $S_1 \to y_2 = 1$, $S_1 \to y_3 = \frac{9}{10}$, $S_2 \to y_5 = \frac{6}{5}$, $S_2 \to y_6 = \frac{4}{5}$, $S_3 \to y_8 = \frac{11}{10}$, $S_3 \to y_9 = \frac{7}{10}$, $G \to y_2 = \frac{13}{10}$, $G \to y_3 = \frac{3}{4}$, $G \to y_4 = \frac{17}{20}$, $G \to y_5 = \frac{21}{20}$, $G \to y_6 = 1$, $G \to y_7 = \frac{9}{10}$, $G \to y_8 = \frac{6}{5}$, $G \to y_9 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(y_9) = \frac{19}{20}$, $\mathrm{Var}(S_1) = 1$, $\mathrm{Var}(S_2) = \frac{6}{5}$, $\mathrm{Var}(S_3) = \frac{4}{5}$, $\mathrm{Var}(G) = \frac{11}{10}$.

P13, model A (for $A \not\preceq B$): rank of the Jacobian $= 13 > q_B = 11$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_2 \to y_5 = \frac{6}{5}$, $F_2 \to y_6 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$.

P14, model A (for $A \not\preceq B$): rank of the Jacobian $= 27 > q_B = 21$. $S_1 \to y_2 = 1$, $S_1 \to y_3 = \frac{9}{10}$, $S_2 \to y_5 = \frac{6}{5}$, $S_2 \to y_6 = \frac{4}{5}$, $S_3 \to y_8 = \frac{11}{10}$, $S_3 \to y_9 = \frac{7}{10}$, $G \to y_2 = \frac{13}{10}$, $G \to y_3 = \frac{3}{4}$, $G \to y_4 = \frac{17}{20}$, $G \to y_5 = \frac{21}{20}$, $G \to y_6 = 1$, $G \to y_7 = \frac{9}{10}$, $G \to y_8 = \frac{6}{5}$, $G \to y_9 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(y_9) = \frac{19}{20}$, $\mathrm{Var}(S_1) = 1$, $\mathrm{Var}(S_2) = \frac{6}{5}$, $\mathrm{Var}(S_3) = \frac{4}{5}$, $\mathrm{Var}(G) = \frac{11}{10}$.

P15, model A (for $A \not\preceq B$): rank of the Jacobian $= 10 > q_B = 9$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $k_{y_1,y_1} = 2$, $k_{y_2,y_2} = \frac{9}{4}$, $k_{y_3,y_3} = \frac{5}{2}$, $k_{y_4,y_4} = \frac{11}{5}$, $k_{y_2,y_1} = -\frac{1}{2}$, $k_{y_3,y_1} = -\frac{2}{5}$, $k_{y_4,y_1} = -\frac{3}{10}$, $k_{y_3,y_2} = -\frac{1}{4}$, $k_{y_4,y_2} = -\frac{7}{20}$, $k_{y_4,y_3} = -\frac{9}{20}$, $\mathrm{Var}(F) = 1$.

P16, model A (for $A \not\preceq B$): $f = 0.00684402$. $k_{y_1,y_1} = 2$, $k_{y_2,y_2} = \frac{9}{4}$, $k_{y_3,y_3} = \frac{5}{2}$, $k_{y_4,y_4} = \frac{11}{5}$, $k_{y_2,y_1} = -\frac{1}{2}$, $k_{y_3,y_2} = -\frac{2}{5}$, $k_{y_4,y_3} = -\frac{3}{10}$, $k_{y_4,y_1} = -\frac{1}{4}$.

P16, model B (for $B \not\preceq A$): $f = -0.864000$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(F) = 1$.

P17, model A (for $A \not\preceq B$): $f = 0.0186317$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $F \to y_5 = \frac{4}{5}$, $F \to y_6 = \frac{11}{10}$, $k_{y_1,y_1} = 2$, $k_{y_2,y_2} = \frac{9}{4}$, $k_{y_3,y_3} = \frac{5}{2}$, $k_{y_2,y_1} = -\frac{1}{2}$, $k_{y_3,y_2} = -\frac{2}{5}$, $\mathrm{Var}(y_4) = 1$, $\mathrm{Var}(y_5) = \frac{6}{5}$, $\mathrm{Var}(y_6) = \frac{9}{10}$, $\mathrm{Var}(F) = 1$.

P17, model B (for $B \not\preceq A$): $f = -0.0552960$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $F \to y_5 = \frac{4}{5}$, $F \to y_6 = \frac{11}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F) = 1$, $\mathrm{Cov}(y_2, y_1) = \frac{3}{10}$, $\mathrm{Cov}(y_3, y_2) = \frac{1}{5}$.

P18, model A (for $A \not\preceq B$): $f = 0.300000$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_1 \to y_4 = \frac{6}{5}$, $F_2 \to y_6 = \frac{4}{5}$, $F_2 \to y_7 = \frac{11}{10}$, $F_2 \to y_8 = \frac{7}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$.

P18, model B (for $B \not\preceq A$): rank of the Jacobian $= 20 > q_A = 17$. $k_{y_1,y_1} = 2$, $k_{y_2,y_2} = \frac{9}{4}$, $k_{y_3,y_3} = \frac{5}{2}$, $k_{y_4,y_4} = \frac{11}{5}$, $k_{y_5,y_5} = \frac{12}{5}$, $k_{y_6,y_6} = \frac{21}{10}$, $k_{y_7,y_7} = 2$, $k_{y_8,y_8} = \frac{9}{4}$, $k_{y_2,y_1} = -\frac{1}{2}$, $k_{y_3,y_1} = -\frac{2}{5}$, $k_{y_4,y_1} = -\frac{3}{10}$, $k_{y_3,y_2} = -\frac{1}{4}$, $k_{y_4,y_2} = -\frac{7}{20}$, $k_{y_4,y_3} = -\frac{9}{20}$, $k_{y_6,y_5} = -\frac{1}{2}$, $k_{y_7,y_5} = -\frac{2}{5}$, $k_{y_8,y_5} = -\frac{3}{10}$, $k_{y_7,y_6} = -\frac{1}{4}$, $k_{y_8,y_6} = -\frac{7}{20}$, $k_{y_8,y_7} = -\frac{9}{20}$.

P19, model A (for $A \not\preceq B$): $f = -0.0396000$. $G \to y_2 = 1$, $G \to y_3 = \frac{9}{10}$, $G \to y_4 = \frac{6}{5}$, $G \to y_5 = \frac{4}{5}$, $G \to y_6 = \frac{11}{10}$, $G \to y_7 = \frac{7}{10}$, $G \to y_8 = \frac{13}{10}$, $G \to y_9 = \frac{3}{4}$, $S_2 \to y_5 = \frac{17}{20}$, $S_2 \to y_6 = \frac{21}{20}$, $S_3 \to y_8 = 1$, $S_3 \to y_9 = \frac{9}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(y_9) = \frac{19}{20}$, $\mathrm{Var}(G) = 1$, $\mathrm{Var}(S_2) = \frac{6}{5}$, $\mathrm{Var}(S_3) = \frac{4}{5}$, $\mathrm{Cov}(S_3, S_2) = \frac{3}{10}$.

P19, model B (for $B \not\preceq A$): $f = -0.140250$. $S_1 \to y_2 = 1$, $S_1 \to y_3 = \frac{9}{10}$, $S_2 \to y_5 = \frac{6}{5}$, $S_2 \to y_6 = \frac{4}{5}$, $S_3 \to y_8 = \frac{11}{10}$, $S_3 \to y_9 = \frac{7}{10}$, $G \to y_2 = \frac{13}{10}$, $G \to y_3 = \frac{3}{4}$, $G \to y_4 = \frac{17}{20}$, $G \to y_5 = \frac{21}{20}$, $G \to y_6 = 1$, $G \to y_7 = \frac{9}{10}$, $G \to y_8 = \frac{6}{5}$, $G \to y_9 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(y_9) = \frac{19}{20}$, $\mathrm{Var}(S_1) = 1$, $\mathrm{Var}(S_2) = \frac{6}{5}$, $\mathrm{Var}(S_3) = \frac{4}{5}$, $\mathrm{Var}(G) = \frac{11}{10}$.

P20, model A (for $A \not\preceq B$): $f = 0.00684402$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_4 \to y_8 = \frac{4}{5}$, $k_{F_1,F_1} = 2$, $k_{F_2,F_2} = \frac{9}{4}$, $k_{F_3,F_3} = \frac{5}{2}$, $k_{F_4,F_4} = \frac{11}{5}$, $k_{F_2,F_1} = -\frac{1}{2}$, $k_{F_3,F_2} = -\frac{2}{5}$, $k_{F_4,F_3} = -\frac{3}{10}$, $k_{F_4,F_1} = -\frac{1}{4}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$.

P20, model B (for $B \not\preceq A$): $f = -0.615859$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_4 \to y_8 = \frac{4}{5}$, $G \to F_2 = \frac{1}{2}$, $G \to F_3 = \frac{3}{5}$, $G \to F_4 = \frac{2}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(F_4) = \frac{11}{10}$, $\mathrm{Var}(G) = \frac{9}{10}$.

P24, model B (for $B \not\preceq A$): rank of the Jacobian $= 21 > q_A = 20$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $F \to y_5 = \frac{4}{5}$, $F \to y_6 = \frac{11}{10}$, $F \to y_7 = \frac{7}{10}$, $F \to y_8 = \frac{13}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(F) = 1$, $\mathrm{Cov}(y_2, y_1) = \frac{3}{10}$, $\mathrm{Cov}(y_4, y_3) = \frac{1}{5}$, $\mathrm{Cov}(y_6, y_5) = \frac{2}{5}$, $\mathrm{Cov}(y_8, y_7) = \frac{1}{4}$, $\mathrm{Cov}(y_3, y_1) = \frac{7}{20}$.

P25, model B (for $B \not\preceq A$): rank of the Jacobian $= 25 > q_A = 21$. $G \to y_2 = 1$, $G \to y_3 = \frac{9}{10}$, $G \to y_4 = \frac{6}{5}$, $G \to y_5 = \frac{4}{5}$, $G \to y_6 = \frac{11}{10}$, $G \to y_7 = \frac{7}{10}$, $G \to y_8 = \frac{13}{10}$, $G \to y_9 = \frac{3}{4}$, $S_2 \to y_5 = \frac{17}{20}$, $S_2 \to y_6 = \frac{21}{20}$, $S_3 \to y_8 = 1$, $S_3 \to y_9 = \frac{9}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(y_9) = \frac{19}{20}$, $\mathrm{Var}(G) = 1$, $\mathrm{Var}(S_2) = \frac{6}{5}$, $\mathrm{Var}(S_3) = \frac{4}{5}$, $\mathrm{Cov}(S_3, S_2) = \frac{3}{10}$.

P26, model B (for $B \not\preceq A$): rank of the Jacobian $= 12 > q_A = 11$. $F_1 \to y_2 = 1$, $F_1 \to X = \frac{9}{10}$, $F_2 \to y_4 = \frac{6}{5}$, $F_2 \to X = \frac{4}{5}$, $\mathrm{Var}(X) = 1$, $\mathrm{Var}(y_1) = \frac{6}{5}$, $\mathrm{Var}(y_2) = \frac{9}{10}$, $\mathrm{Var}(y_3) = \frac{4}{5}$, $\mathrm{Var}(y_4) = \frac{11}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$.

P27, model A (for $A \not\preceq B$): rank of the Jacobian $= 21 > q_B = 20$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_4 \to y_8 = \frac{4}{5}$, $H_1 \to F_2 = \frac{1}{2}$, $H_2 \to F_4 = \frac{3}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(F_4) = \frac{11}{10}$, $\mathrm{Var}(H_1) = \frac{9}{10}$, $\mathrm{Var}(H_2) = 1$, $\mathrm{Cov}(H_2, H_1) = \frac{3}{10}$.

P28, model A (for $A \not\preceq B$): rank of the Jacobian $= 12 > q_B = 11$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $X \to F_1 = \frac{1}{2}$, $X \to F_2 = \frac{3}{5}$, $\mathrm{Var}(X) = 1$, $\mathrm{Var}(y_1) = \frac{6}{5}$, $\mathrm{Var}(y_2) = \frac{9}{10}$, $\mathrm{Var}(y_3) = \frac{4}{5}$, $\mathrm{Var}(y_4) = \frac{11}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$.

P29, model A (for $A \not\preceq B$): $f = 0.324000$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_4 \to y_8 = \frac{4}{5}$, $G \to F_2 = \frac{1}{2}$, $G \to F_3 = \frac{3}{5}$, $G \to F_4 = \frac{2}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(F_4) = \frac{11}{10}$, $\mathrm{Var}(G) = \frac{9}{10}$.

P29, model B (for $B \not\preceq A$): $f = 0.330000$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $F \to y_5 = \frac{4}{5}$, $F \to y_6 = \frac{11}{10}$, $F \to y_7 = \frac{7}{10}$, $F \to y_8 = \frac{13}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(F) = 1$, $\mathrm{Cov}(y_3, y_1) = \frac{3}{10}$, $\mathrm{Cov}(y_4, y_2) = \frac{1}{5}$, $\mathrm{Cov}(y_7, y_5) = \frac{2}{5}$, $\mathrm{Cov}(y_8, y_6) = \frac{1}{4}$.

P30, model A (for $A \not\preceq B$): $f = 1.20000$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $X \to F = \frac{1}{2}$, $F \to Y = \frac{6}{5}$, $\mathrm{Var}(X) = 1$, $\mathrm{Var}(Y) = \frac{6}{5}$, $\mathrm{Var}(y_1) = \frac{9}{10}$, $\mathrm{Var}(y_2) = \frac{4}{5}$, $\mathrm{Var}(y_3) = \frac{11}{10}$, $\mathrm{Var}(F) = 1$.

P30, model B (for $B \not\preceq A$): $f = 0.500000$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to X = \frac{6}{5}$, $X \to Y = \frac{1}{2}$, $\mathrm{Var}(X) = 1$, $\mathrm{Var}(Y) = \frac{6}{5}$, $\mathrm{Var}(y_1) = \frac{9}{10}$, $\mathrm{Var}(y_2) = \frac{4}{5}$, $\mathrm{Var}(y_3) = \frac{11}{10}$, $\mathrm{Var}(F) = 1$.

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